

GE Healthcare

LightSpeed 7.X

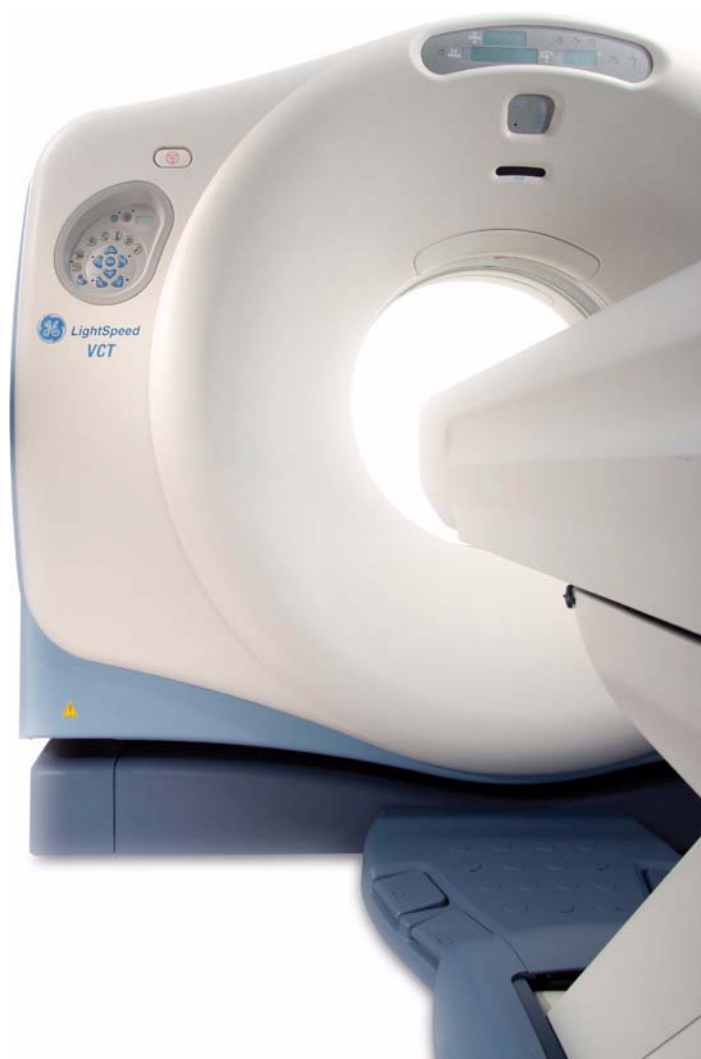
Pre-Installation Manual

LightSpeed VCT XT

LightSpeed VCT

LightSpeed VCT Select

LightSpeed Pro32



OPERATING DOCUMENTATION



5116410-100
Rev 12

The information in this manual applies to the following GE Healthcare LightSpeed 7.X CT Scanners:

- LightSpeed VCT XT
- LightSpeed VCT
- LightSpeed VCT Select
- LightSpeed Pro32

The information in this manual does NOT apply to non-fixed (mobile) installations.

IMPORTANT PRECAUTIONS

LANGUAGE

Предупреждение

(BG)

- ТОВА УПЪТВАНЕ ЗА РАБОТА Е НАЛИЧНО САМО НА АНГЛИЙСКИ ЕЗИК.
- АКО ДОСТАВЧИКЪТ НА УСЛУГАТА НА КЛИЕНТА ИЗИСКА ЕЗИК, РАЗЛИЧЕН ОТ АНГЛИЙСКИ, ЗАДЪЛЖЕНИЕ НА КЛИЕНТА Е ДА ОСИГУРИ ПРЕВОД.
- НЕ ИЗПОЛЗВАЙТЕ ОБОРУДВАНЕТО ПРЕДИ ДА СТЕ СЕ КОНСУЛТИРАЛИ И РАЗБРАЛИ УПЪТВАНЕТО ЗА РАБОТА.
- НЕСПАЗВАНЕТО НА ТОВА ПРЕДУПРЕЖДЕНИЕ МОЖЕ ДА ДОВЕДЕ ДО НАРАНЯВАНЕ НА ДОСТАВЧИКА НА УСЛУГАТА, ОПЕРАТОРА ИЛИ ПАЦИЕНТ В РЕЗУЛТАТ НА ТОКОВ УДАР ИЛИ МЕХАНИЧНА ИЛИ ДРУГА ОПАСНОСТ.

警告

(ZH-CN)

- 本维修手册仅提供英文版本。
- 如果维修服务提供商需要非英文版本，客户需自行提供翻译服务。
- 未详细阅读和完全理解本维修手册之前，不得进行维修。
- 忽略本警告可能对维修人员，操作员或患者造成触电、机械伤害或其他形式的伤害。

VÝSTRAHA

(CS)

- Tento provozní návod existuje pouze v anglickém jazyce.
- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL

(DA)

- Denne servicemanual findes kun på engelsk.
- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret medmindre denne servicemanual har været konsulteret og er forstået.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING

(NL)

- Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.
- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden voordat deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING

(EN)

- This Service Manual is available in English only.
- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator, or patient, from electric shock or from mechanical or other hazards.

HOIATUS

(ET)

- Käesolev teenindusjuhend on saadaval ainult inglise keeles.
- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS

(FI)

- Tämä huolto-ohje on saatavilla vain englanniksi.
- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION

(FR)

- Ce manuel de service n'est disponible qu'en anglais.
- Si le technicien du client a besoin de ce manuel dans une autre langue que l'anglais, c'est au client qu'il incombe de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que le manuel service n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG

(DE)

- Diese Serviceanleitung existiert nur in Englischer Sprache.
- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine Entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, Mechanische oder Sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ

(EL)

- Το παρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο.
- Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Εάν δε λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS

(HU)

- Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.
- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészítése.
- Ne próbálja elkezdni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN

(IS)

- Þessi þjónustuhandbók er eingöngu fáanleg á ensku.
- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA

(IT)

- Il presente manuale di manutenzione è disponibile soltanto in inglese.
- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Si proceda alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il non rispetto della presente avvertenza potrebbe far compiere operazioni da cui derivino lesioni all'addetto, alla manutenzione, all'utilizzatore ed al paziente per folgorazione elettrica, per urti meccanici od altri rischi.

警告

(JA)

- このサービスマニュアルには英語版しかありません。
- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고

(KO)

- 본 서비스 지침서는 영어로만 이용하실 수 있습니다 .
- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우 , 번역 서비스를 제공하는 것은 고객의 책임입니다 .
- 본 서비스 지침서를 참고했고 이해하지 않는 한은 해당 장비를 수리하려고 시도하지 마십시오 .
- 이 경고에 유의하지 않으면 전기 쇼크 , 기계상의 혹은 다른 위험으로부터 서비스 제공자 , 운영자 혹은 환자에게 위해를 가할 수 있습니다 .

BRDINJUMS

(LV)

- Šī apkalpes rokasgrāmata ir pieejama tikai anglu valodā.
- Ja klienta apkalpes sniedzējam nepieciešama informācija citā valodā, nevis anglu, klienta pienākums ir nodrošināt tulkošanu.
- Neveiciet aprīkojuma apkalpi bez apkalpes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošana var radīt elektriskās strāvas trieciena, mehānisku vai citu risku izraisītu traumu apkalpes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS

(LT)

- Šis eksploatavimo vadovas yra prieinamas tik anglų kalba.
- Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, numatyti vertimo paslaugas yra kliento atsakomybė.
- Nemėginkite atlikti įrangos techninės priežiūros, nebent atsižvelgėte į šį eksploatavimo vadovą ir jį supratote.
- Jei neatkreipsite dėmesio į šį perspėjimą, galimi sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų paslaugų tiekėjui, operatoriui ar pacientui.

ADVARSEL

(NO)

- Denne servicehåndboken finnes bare på engelsk.
- Hvis kundens serviceleverandør trenger et annet språk, er det kundens ansvar å sørge for oversettelse.
- Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått.
- Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.

OSTRZEŻENIE

(PL)

- Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim.
- Jeśli dostawca usług klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta.
- Nie próbować serwisować wyposażenia bez zapoznania się i zrozumienia niniejszego podręcznika serwisowego.
- Niezastosowanie się do tego ostrzeżenia może spowodować urazy dostawcy usług, operatora lub pacjenta w wyniku porażenia elektrycznego, zagrożenia mechanicznego bądź innego.

ATENÇÃO

(PT-BR)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode por em perigo a segurança do técnico, operador ou paciente devido a choques elétricos, mecânicos ou outros

ATENÇÃO

(PT-PT)

- Este manual de assistência técnica só se encontra disponível em inglês.
- Se qualquer outro serviço de assistência técnica, que não a gems, solicitar estes manuais noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução.
- Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica.
- O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE

(RO)

- Acest manual de service este disponibil numai în limba engleză.
- Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere.
- Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service.
- Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.

ОСТОРОЖНО!

(RU)

- Данное руководство по обслуживанию предлагается только на английском языке.
- Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод.
- Перед обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения.
- Несоблюдение требований данного предупреждения может привести к тому, что специалист по обслуживанию, оператор или пациент получат удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE

(SR)

- Ovo servisno uputstvo je dostupno samo na engleskom jeziku.
- Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilacke usluge.
- Ne pokušavajte da opravite uredaj ako niste procitali i razumeli ovo servisno uputstvo.
- Zanemarivanje ovog upozorenja može dovesti do povređivanja serviser, rukovaoca ili pacijenta usled strujnog udara ili mehanickih i drugih opasnosti

UPOZORNENIE

(SK)

- Tento návod na obsluhu je k dispozícii len v angličtine.
- Ak zákazníkovi poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka.
- Nepokúšajte sa o obsluhu zariadenia skôr, ako si neprečítate návod na obsluhu a neporozumiete mu.
- Zanedbanie tohto upozornenia môže vyústiť do zranenia poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, do mechanického alebo iného nebezpečenstva.

ATENCION

(ES)

- Este manual de servicio sólo existe en inglés.
- Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual.
- No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio.
- La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING

(SV)

- Den här servicehandboken finns bara tillgänglig på engelska.
- Om en kunds servicetekniker har behov av ett annat språk än engelska ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

DIKKAT

(TR)

- Bu servis kilavuzunun sadece ingilizcesi mevcuttur.
- Eğer müşteri teknisyeni bu kilavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kilavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm>
- Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is strongly recommended. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS

CAUTION



Risk of Explosion.

Danger of explosion if battery is incorrectly replaced.

Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.

ATTENTION



Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie.

Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives.

GE personnel, please use the Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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Revision History

Revision	Date	Reason for change
12	10/24/08	Updated Important Precautions, Language section to add new languages. PQR 13168182: Updated Chapter 9, Section 5.0, Table 9-6 to add A1 information for Europe. SPR FCTge 29404: Removed Figures 9-1 and 9-2 from manual. They are incorrect and not needed for siting. SPR FCTge 36007: Updates to Chapters 1 - 4: Finished floor and wall requirements Construction site requirements Relocatable building installations System siting requirements Update of system PMI tasks Site-ready visit requirements SPR FCTge36010: Chapter 4, Section 5.3: removed bullet point allowing for remote mount of the console. Chapter 4: Section 8 added subsection 8.4: Ceiling Requirements (including Boom in a Room) Chapter 5: Sections 4.9, 4.10, 4.11, 4.12, and 4.13 (EMC Compliance) replaced to match information in DVCT Pre-Installation manual. Chapter 7, Section 7.0: update the URL for the umi-dolly shop; remove information about tilting dollies for GT1700 table in Table 7-3. Chapter 7, updated Section 4.0 "Storage Requirements" and Section 5.0 Extreme Temperature Transportation and Deliveries.
11	9/21/07	Added Bulgarian to Important Precautions PQR 13108575: Chapter 7:Corrected weight in Table 7-1.
10	2/23/07 2/9/07	Updated measurements to metric to meet European standards and compliance. General revision of manual. Updated the Warnings section to include new languages. PQR 13092823,13093839: Chapter 8: Updated Section 2.1 - Power Source Configuration
9	12/7/06	Chapter 7: Added weights to Table 7-1. (PQR 13079923/FCTge243738)
8	11/16/06	Title Page: Added product names Chapter 4: - Updated Section 3.1 - Minimum Room Size - Updated Section 3.2 - System Specifications and Section 3.4 - Minimum Room Size & Requirement Layouts (added metric units) - Updated Section 10 - Radiation Protection (scatter plots)

Revision	Date	Reason for change
7	10/2/06	Chapter 4: Added Section 3.0 - Additional Regulatory Clearance Information; other updates
6	7/17/06	Updates for Two-Step Installations; "Limited Access Rooms"
5	1/27/06	Chapter 1: - Updated Section 1.0 - Site Readiness - Updated Section 2.0 - Section 2.0 - Deleted old Section 3.0 - Site Preparation Prior to Equipment Delivery - Added/Updated new Section 3.0 - Customer Installation Checklists (moved from Chapter 2) - Added "Pre-Installation Block Diagram" (moved from Chapter 2) Chapter 2: - Moved Checklist and Block Diagram to Chapter 1. - Added Notice to review DVD prior to beginning Pre-Install work. Chapter 3: - Added Section 1.0 - Option Catalog Numbers - Updated photo of base scanner Chapter 4: Major updates, including (but not limited to) definition of <i>Regulatory</i> vs. <i>Service</i> Clearances. Chapter 5: - Updated <i>Cooling Requirements Worksheet</i> - Updated PDU EMI envelope dimensions Chapter 7: - Edited order information for zero-clearance dollies. - Updated Section 4.0 - Storage Requirements - Added Section 8.0 - Tilt Dollies Chapter 8: Updated Section 2.1 - Power Source Configuration Chapter 9: Updated Section 5.0 - UPS Interconnect Appendix: Updated <i>VCT & FCT Installation Site Ready Form</i>
4	7/12/05	Chapter 2: Added "Floor Levelness Specification" to page 37 Chapter 4: - Updated Minimum & Recommended Scan Room sizes - Corrected Long Cable Set catalog number - Updated "Typical Room Layout" and "Service Clearance" drawings - Corrected table height in Figure 4-8 - Corrected Floor Levelness Specification - Corrected "Typical Scatter Survey" drawings Chapter 6: - Corrected "Redundant Anchor Locations" in Figure 6-3 - Updated Floor Template drawing - Added Seismic Floor Mounting Hole Locations to PDU drawing Chapter 7: Added Table Delivery dimensions information Chapter 9: Updated System Interconnect Diagram

Revision	Date	Reason for change
3	6/17/05	Chapter 5: Added note to Section 1.1. Chapter 9: Added CCC Compliance Note, to Figure 9-1. Removed n/a HP Table information.
2	03/02/05	Chapter 2: - Updated Pre-Installation Block Diagram. - Updated Pre-Installation Checklist Chapter 4: Corrected Typographical Error in Figure 4-17. Chapter 5: Corrected Typographical Errors, in “EMC Edition 2 Compliance” statement. Chapter 6: - Corrected GT 2000 Table weight, in Table 6-1. - Updated Figure 6-4 Floor Template - Gantry and (GT) Table Base
1	12/17/04	Initial Release, for NPI.

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Appendix:

VCT & FCT Installation Site Ready Form 121

Chapter 1

Introduction

This direction contains physical and electrical data necessary for planning and preparing a site. Pre-installation work is defined as site preparation for installation of the GE CT scanner. It is the responsibility of the purchaser to arrange and pay for this work. Pre-installation work includes:

- Installation of electrical conduit, junction boxes, ducts, outlets, and line safety switches.
- Installation of interconnection wiring that is AWG stranded copper. The electrical contractor shall ring out and tag all wires at both ends. Color-coded wires are recommended for easier identification. Wires shall be continuous without splices. Ground wires must conform to local codes.
- Any site renovation.
- Alterations and modifications to products not specifically included in the sales contract.

All work must conform to local building and safety codes. Unless specifically mentioned, GE Healthcare does not provide or install wires, conduits, junction boxes, and ducts as illustrated in this publication.

All CT site plans, preliminary concepts and final working drawings must be reviewed by General Electric Headquarters Architectural Planning prior to construction or approval.

Contact your local General Electric sales representative for complete information regarding your site-specific room layout.

Section 1.0

Site Readiness

Site ready is a requirement that must be achieved to install a CT product. For your convenience, a site ready visit inspection shall be performed at least three (3) days prior to the installation date. The site inspection must conclude with a minimum of a conditional pass status to be ready on the requested installation delivery date. Site ready inspections on the delivery date are not acceptable unless prior arrangements are made.

Pre-Installation and Site Ready Tools:

- VCT Pre-Installation Quick Start Kit & Video
- VCT Floor template
- Pre-Installation Check List
- Pre-Installation Block Diagram
- Site Room Layouts
- Power and Grounding Inspection
- Pre-Installation Support

Section 2.0

Responsibility of Purchaser

2.1 Customer Room Prep Items

The CT air intake is near the bottom of the gantry and draws air in through a filter in the gantry heater assembly. Fine dust, as listed below, clogs the filter and is deposited throughout the gantry, table, operator console and PDU electronics. This fine dust cannot be completely removed and can damage electronic components.

For these reasons, the scanner should be the last item installed in your CT suite area.

“Pre-installation” is work necessary to plan and prepare a site for installation of equipment.

Pre-installation work helps the user (customer) avoid:

- Application delay and scheduling
- Surprise siting discoveries
- Installation confusion
- Waste of manpower

The following **MUST** be completed before installation work can begin for a GE CT scanner:

- Completely finished:
 - Wall painted or have final wall covering
 - Ceiling tiles installed and no remaining ceiling work is strongly recommended
 - Final floor covering installed with no remaining dust-causing floor work is strongly recommended
 - All room millwork installed as shown on the site print
 - All plumbing work in the CT suite is completed
 - No construction in or around the scan suite AREA that will produce:
 - * Concrete dust
 - * Drywall dust
 - * Ceiling tile dust
 - * Wood sawdust or shavings
 - * Dust tracked into the CT suite area
- Active Broad Band connection
 - A completed network connection is strongly recommended for ALL CT installations.
 - A GE Healthcare network specialist may be needed to complete the VCN connection. This may take a week or longer to schedule.
- Power available to A1, with provision for Lockout/Tagout at the A1 disconnect.
If a UPS is needed, a GE A1 breaker* is needed to complete this installation. Refer to the electrical section for more details.



NOTICE

SERVICE NOTICE: An improperly prepared site (i.e., one that is in a state of construction) can result in increased installation time.

A CT scanner installed in a dirty environment is more prone to contamination, which can result in decreased reliability and increased scanner downtime.

2.2 Purchaser Site Preparation Work

The following list describes many of the items to consider when planning for a system replacement or designing a room for new equipment:

- Determine room dimensions and verify that doorways are large enough for the scanner system.
- Install appropriate conduits and duct work for system cables. If additional components are needed in the CT suite, their connection consideration must be determined and completed.
- Install junction boxes of correct size with covers at locations shown in installation plan.
- Installation of A1 main disconnect.
- Install power supply of correct voltage output and adequate KVA rating.
- Install local disconnects, including proper over-current protection.
- Install “steelwork” or other suitable support work for mounting equipment on walls or from ceiling.
- Camera should be on-site at the time of installation.
- Complete all suite and room alterations and modifications.
- Verify that room shielding is adequate for the system being installed.
- Review structural requirements - including floor vibration, levelness, and thickness. (See [Structural Requirements](#), on page 63.)
- Review all ceiling-mounted options requirements. (See [Ceiling Requirements](#), on page 65.)
- Review HVAC requirements including system regulation and patient comfort.
- Review operational clearances to see if your daily used items fit, such as beds and carts.
- Consider emergency medical equipment.
- Storage cabinets and sink (if needed) must be shown on the site print.
- The following contractors, and others, may be needed to help confirm that the site meets all installation requirements:
 - Structural Engineer and /or Architect
 - HVAC contractor
 - Electrical contractor
 - Qualified radiological health physicist
- The project manager or sales will deliver a copy of the pre-installation quick start kit.

The above items can be found in chapter 2 through 9 in this manual.

It is suggested that this work be completed at least three days prior to delivery.

2.3 Manufacturer’s System Level Siting Requirements

The following siting requirements are the minimum that must be met in order to install a new or replacement system:

- Network Communication in place and active
- Meets all scan room regulatory and service requirements
- Meets all minimum scan room structural requirements
- Meets minimum scan room HVAC requirements
- Meets minimum scan room electrical requirements
- Reviewed radiation protection section in the Pre-Installation manual
- All in-room items shown on the final GE Healthcare site print and the final print are on site
- No construction in the scan room or neighboring suite areas

- Includes all finished doors, floors, windows, ceilings, and walls, with all plumbing and cabinets already installed. (*Finished Floor Exception 1*, on page 65 and *Finished Floor Exception 2*, on page 65 may apply. *Finished Walls Requirements*, on page 65 may apply.)
 - All necessary parties have received the VCT Quick Start Kit and reviewed the requirements
- It is suggested that this work be completed at least three days prior to delivery.

2.3.1 Meeting Site-Ready Requirements

To ensure timely delivery and installation, GE recommends that the customer complete all necessary work and schedule a site-ready visit prior to the delivery date. This visit verifies that the site meets all system siting recommendations and results in a call or report to the Project Manager of Installation, indicating a site-ready condition, and confirming that the installation can proceed. The site-ready visit reviews and verifies all of the installation recommendations. The customer and/or the customer's contractors need to confirm that all work was completed, including floor levelness, PRIOR to delivery.

The site ready visit results in a report to the project manager indicating one of the following:

Pass - All strongly recommended items are present, completed, and the site is ready for installation.

Conditional Pass - Eighty percent (80%) of all of the tasks are completed and all parties agree that the remaining 20% will be completed by the installation delivery date.

If a "Conditional Pass" is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are completed and that the site is ready for delivery one business day prior to delivery.

Fail - Less than 80% of the tasks are completed and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites will be rescheduled when all items are completed.

2.3.2 Quick Installs

Quick installations are described as sites where minimum room improvements are needed. These include, but are not limited to, the following items:

- Existing electrical disconnect device, wire size, and grounds meet all of the above requirements.
- Existing structural items, including floor thickness, meet all of the above requirements.
- Existing HVAC capacity and regulation meet all of the above requirements.
- Existing CT suite meets all of the above regulatory and minimum size requirements.
- Existing facility can accommodate the delivery and meet all of the above delivery requirements.

Quick Installs are subject to the following restrictions:

- Quick installs must have a new room print that accurately reflects the rooms to be upgraded.
- Existing floor anchors from a non-VCT system CANNOT be reused.
- New floor anchors must be a minimum of 10.16 cm (4 in.) from any existing floor penetrations.

Quick Installs typically involve a weekend de-install and room prep completion, with a next business day delivery and install.

2.3.3 Two-Step and Upgrade Installs

A Two-Step installation is the practice of temporarily installing one CT system in a site with the intention of upgrading the site to a different CT system at a later date.

- For a two-step installation to be considered, the room must meet the minimum room requirements for the upgraded/final project.

- As with any upgrade installation, “two-steps” are subject to ALL of the siting requirements imposed by the upgraded/final system. This includes the recommended room size as well as electrical, structural, and HVAC requirements.
- “Two-steps” and other upgrades may be done as “Quick” Installs. In this case, all requirements described in Section 2.3.2 (above) also apply.
- It is the customer’s responsibility to check that all requirements are met.
- Sites that do not meet the minimum room requirements for the final product must either upgrade (or enlarge) their room, or consider the “Left-Side Limited Access” option.

2.3.4 Construction Site Installations

A *construction installation* describes installations at sites without an occupancy permit, often with ongoing construction. In general, construction sites fail to meet the recommended specifications for delivery of the system. GE Healthcare does not recommend construction installations, as they can result in delays, increased costs, and possible damage to the system. When construction-site delivery proves unavoidable, the installation falls into one of two categories:

- Full construction site with completed radiology area
- Full construction site with limited delivery access

Review the following categories to determine which most closely matches the condition of the planned installation site.

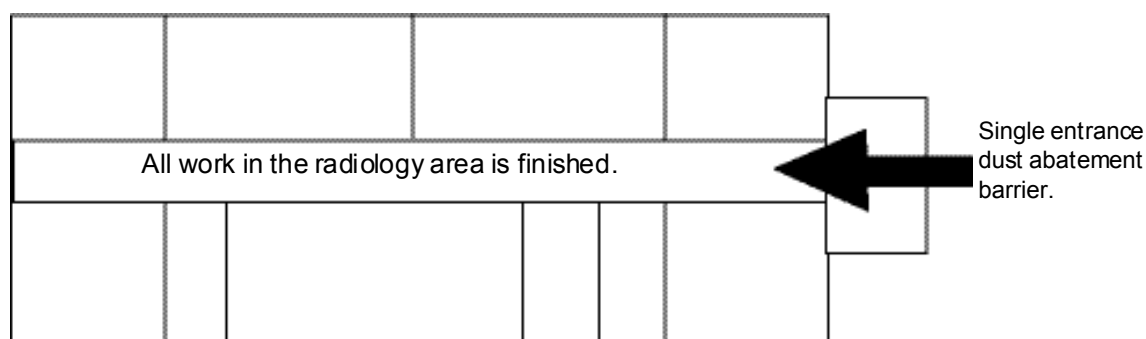


Figure 1-1 Full construction site with completed radiology area

2.3.4.1 Full Construction Site with Completed Radiology Area

This type of site consists of a finished, dust-free, occupancy-ready radiology suite. While there is no remaining construction in or around the scan suite area, there may be ongoing construction in other areas. At the time of delivery such sites feature:

- Dust control measures deployed in the radiology suite area.
- Scan suite access limited to a single entrance (see [Figure 1-1](#)).
- Radiology suite sealed off from the remaining construction area.
- Operational HVAC, with a positive air pressure within the radiology suite.

In addition, the radiology suite at such a site REMAINS in a dust-free, occupancy-ready state after delivery and throughout the remaining construction phase.

2.3.4.2 Full Construction Site with Limited Delivery Access

This type of site allows delivery during ongoing construction of the radiology suite area. At such sites, delivery occurs prior to site completion, but the product remains stored until a finished, dust-free, occupancy-ready radiology suite area is ready. This type of site requires the LightSpeed 7.X to be delivered in a sealed package with dollies. Delivery to the storage area may require a lift truck or riggers. Installation work begins only when the site reaches the completed, dust-free, occupancy-ready radiology suite requirement.

Note: If delivery requires vertical or horizontal lifting, the PMI adds the necessary identifier to the order.

2.3.5 Relocatable Building Installations

A relocatable building is made in a factory and delivered to the site of its permanent location. Relocatable buildings qualify as fixed sites and must satisfy all of the requirements of a fixed site. The PET scanner and table must be mounted on a solid concrete floor. Any other floor type installations must be designed by the customer's structural engineer and meet all GE Healthcare's specifications listed in this manual.

Refer to [Table and Gantry Mounting Requirements](#), on page 63 of this manual for further information.

2.4 Site Ready Inspection Visit

The GE Healthcare Project Manager of Installation (GE PMI) will review the site delivery process with you to determine how to best transfer the equipment from the transportation truck to your room. At the time of the site ready visit, the site must meet all requirements in sections 1 and 2, plus the following items.

- Delivery information
 - Determine the delivery type: ground, dock, or tilt-bed truck
 - Determine if delivery requires tilt dollies or riggers; order dollies and lifting crates, as needed
 - Determine delivery route into the scan room
 - Determine if elevators, doorways, and hallways are adequate for delivery
 - Determine if floor protection is needed
 - Determine if a tilt-bed truck is needed for ground delivery, and inform GE Transportation of the need for a tilt-bed truck
- Regulatory Requirement
 - Room size meets the minimum requirements
 - Site print is present and accurately reflects the room size and layout
 - No grounded walls are present in the regulatory clearance areas
 - All regulatory clearances space is met
 - Room meets all local codes
- Manufacturer Requirements: As listed in section 2, all requirements are met
- Purchaser's Site Preparation Work: As listed in section 1, all actions are completed

A site-ready visit should occur prior to the delivery date. This visit verifies that the site meets all system siting recommendations and confirms that installation can proceed. During the site-ready visit, a GE representative confirms that the site meets all of the recommended site-ready conditions, including floor levelness and delivery route readiness. Lifting options and construction site packaging must be ordered prior to delivery and cannot be added on site.

Section 3.0

Customer Installation Checklists

Information for the Site

Must be completed before the scheduled delivery date

Hospital Name as it appears on the system screens:

Network ID numbers / IP addresses

PACS: _____

Camera: _____

AW: _____

Other - Specify type & ID: _____

Other - Specify type & ID: _____

Camera setup information: _____

AW Direct Connect address: _____

Do you want HIPAA enabled? No___ Yes___

Do you want automatic downloads enabled? No___ Yes___

:

Table 1-1 Schedule Date Commitments

GE	Cust	Dates
Y N	Y N	
	☐ ☐	Has the project schedule been verified with facilities department, contractor, and GE?
	☐ ☐	Will the committed site-ready date be met?
☐ ☐	☐ ☐	Does the completion date for any/all construction meet or precede the delivery date?
☐ ☐	☐ ☐	Is the Power & Ground survey complete? Date: _____ Hospital contact: _____
	☐ ☐	Site-Ready visit is scheduled. Date: _____
	☐ ☐	Delivery date is scheduled. Date: _____
	☐ ☐	Installation date is scheduled. Date: _____
☐ ☐	☐ ☐	Installation timing: A: Weekdays___ B: Weekend___ C: Quick Install___ If B or C, have all sub-contractors been notified? No___ Yes___
☐ ☐	☐ ☐	Does the delivery and/or installation date need to be adjusted?
	☐ ☐	First-Use date is scheduled. Date: _____
☐ ☐	☐ ☐	Applications/Training dates: On-Site Training Date: _____ Healthcare Institute Training Date: _____

Table 1-2 General Site Planning

GE	CUST	General / Site Requirements
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Have final drawings been approved and distributed to the contractors?
☐ ☐	☐ ☐	Are final drawings “signed off” to approve equipment layout / orientation?
☐ ☐	☐ ☐	Do the actual room dimensions match those on the final drawings?
	☐ ☐	Has the radiologist health physician reviewed and approved the room layout and shielding requirements?
	☐ ☐	Have any additional requirements or questions about the installation been discussed with GE? List: _____ _____ _____ _____
	☐ ☐	Is there a person assigned to review and verify that all installation requirements are met? Name: _____
	☐ ☐	Have the specific site requirements been discussed with the contractor? Refer to the GE final drawings specifications. (See Table 1-3 below.)
	☐ ☐	Has the responsibility of cabling, installing, and interfacing accessories not on the order been discussed?
	☐ ☐	Are all third-party vendors identified, notified and scheduled? (Examples: Netcom, Medrad, etc.)
	☐ ☐	Have all regulatory requirements been met per Regulatory and Service Clearances , on page 42?
	☐ ☐	Will existing network, broadband, and camera cable drops reach new locations and will they meet the requirements and function with LightSpeed 7.X? If not, what are the requirements? List: _____ _____ _____

Table 1-3 References for Specific Site Requirements

Sections for Specific Requirements	
Room Planning - Chapter 4 (page 41)	Floor Loads & Weights - Chapter 6 (page 87)
Radiation Protection - page 72	Delivery - Chapter 7 (page 97)
Environment - Chapter 5 (page 75)	Power - Chapter 8 (page 105)
<u>All work</u> by contractors must be completed before the scheduled delivery date.	

Table 1-4 Equipment Compatibility

GE	Cust	Equipment
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Has the order been reviewed for completeness and compatibility with existing equipment? Typical equipment: Remote monitors ____ AW relocation ____ Cardiac option ____ Injectors ____
☐ ☐		Are interfaces to existing and/or new accessories ordered and planned for accordingly?
☐ ☐	☐ ☐	Have the following peripheral locations been included in the site drawings? EKG monitor ____ Injector control ____ Laser camera ____ UPS ____ 2 nd Monitor ____
☐ ☐	☐ ☐	Will GE Healthcare provide additional services per contract negotiations?
☐ ☐		Are correct length cables on order?

Table 1-5 Network Connections

GE	Cust	Network Installation and Setup
Y N	Y N	<i>Must be completed 5 weeks before scheduled delivery date</i>
☐ ☐	☐ ☐	Have IP addresses and Host Names been obtained? No ____ Yes ____
☐ ☐	☐ ☐	Will a network camera be used? No ____ Yes ____
☐ ☐	☐ ☐	Mandatory: Is the network installed, are network jacks installed, and is the entire network tested?
☐ ☐	☐ ☐	Mandatory: Broadband VPN installed/setup?
☐ ☐	☐ ☐	Mandatory: Are network software options ordered ____ HIS RIS option ____ DICOM print ____ AW ____
	☐ ☐	Optional: Has modem option ordered? ____ (Requires a site escalation)
☐ ☐	☐ ☐	Optional: Is the LightSpeed 7.X service telephone line identified and installed for InSite? (Electrical, mechanical, etc.)

Table 1-6 Miscellaneous Tasks

GE	Cust	Other
Y N	Y N	<i>Must be completed before the scheduled delivery date</i>
	☐ ☐	Arrangements made in the schedule to allow adequate time for remodeling, if needed (such as wall, floor, or ceiling repair work, painting, other cosmetic finishes)
	☐ ☐	Have arrangements been made to clean the floor <i>after</i> equipment removal and <i>prior</i> to the installation of the new equipment?
☐ ☐	☐ ☐	Is de-installation of existing equipment needed? No ____ Yes ____ Removal date _____

Table 1-6 Miscellaneous Tasks (Continued)

GE	Cust	Other
Y N	Y N	<i>Must be completed before the scheduled delivery date</i>
☐ ☐	☐ ☐	Is there a trade-in of existing equipment? No __ Yes __ GoldSeal _____
☐ ☐	☐ ☐	Delivery route identified and verified with the proper hospital personnel? No__ Yes __ Elevators and doors checked for size and weight constraints? No__ Yes __
☐ ☐	☐ ☐	Have appropriate arrangements been made with traffic for delivery?No__ Yes __
☐ ☐	☐ ☐	Will acceptance/performance testing or bio-medical testing be needed?No__ Yes __ Date: _____
☐ ☐	☐ ☐	Are trash and/or recycling bins available for the removal of papers, boxes, etc. during the installation? No__ Yes __

Chapter 2

Pre-Installation Overview



NOTICE Prior to beginning the pre-installation process for a LightSpeed 7.X CT system, it is highly recommended that the user review the Pre-Installation Video (DVD) provided in the VCT Quick Start Kit.

Before a LightSpeed 7.X system can be installed, all pre-installation requirements must be complete.

- Chapter 4, Sections [8.0](#) and [9.0](#) Structural Requirements
- Chapter 4, Section [10.0](#) Radiation Protection
- Broadband Standard
- Site Ready Visit
- [Chapter 5, Environmental Conditions](#), Sections [1.0](#) & [2.0](#) HVAC Requirements
- [Chapter 8, Power Requirements](#) (Site Power Audit Required)
- [Chapter 9, Interconnection Data](#)

Site-specific items must be verified before the installation can begin.

Section 1.0

Dust/Dirt Contamination

The LightSpeed 7.X system (consisting of: console, PDU, table, and gantry) is highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site.



NOTICE Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.

Section 2.0

Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed Series components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration's should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 3.0

Walls, Ceiling, and Floor

All walls, ceiling, and flooring must be completed before installation can begin.
LightSpeed 7.X scanners can only be installed on a 10.16 cm (4 in.) concrete floor.

Section 4.0 Broad-band

For information on broad-band requirements, refer to Chapter 4, [Section 9.0, on page 68](#).

Section 5.0 Phone Line (for optional modem)

Two phone lines must be installed at or near the operator console and be operational prior to installation.

- One analog line (for modem use)
- one voice line

Section 6.0 Review

The LightSpeed Series systems use adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads. Refer to [Table 2-1](#), below, to determine the number of leveling pads, by table style.

TABLE	# OF PADS	TEMPLATE
GT 1700	4	5111499
GT 2000	4	5111499

Table 2-1 Number of Leveling Pads and Floor Template P/N, by Table Type

Using the GE print to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the template (see [Table 2-1](#)) shipped with the system, locate the anchor holes. Make sure they clear structural interferences in the floor.

Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the two (2) floor templates.
- 2.) Start with the gantry template—align it per the GE print.
- 3.) Place the table template over the top of the gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Check the level of the floor (See [Figure 2-1](#)) across the templates.

Note: Tiles (or other resilient flooring) around all holes will be cut during the installation process.

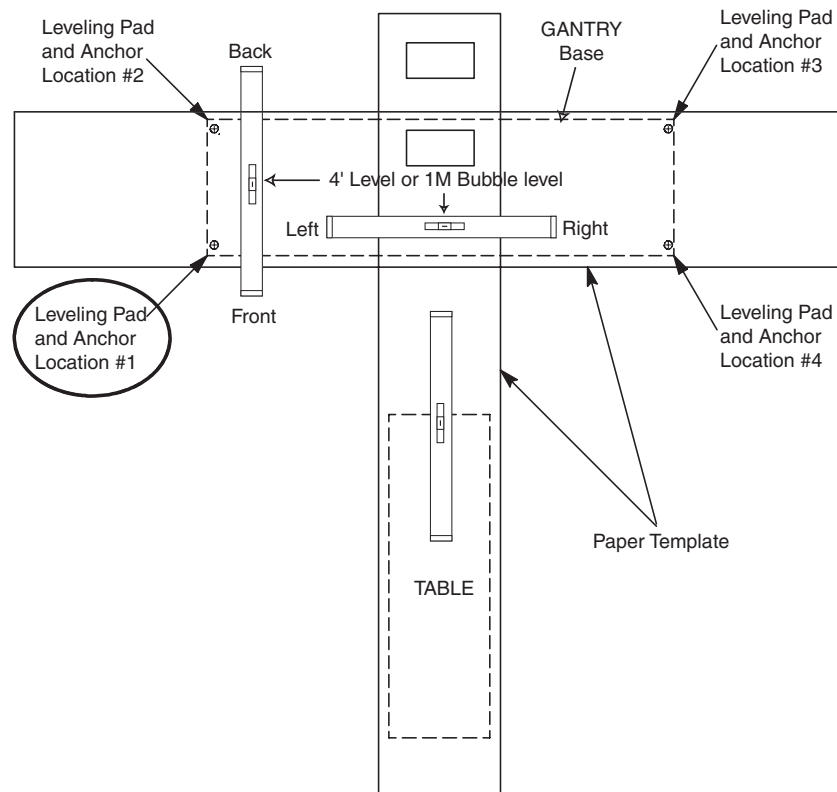


Figure 2-1 Hole Locations

FLOOR LEVELNESS SPECIFICATION

- 3 mm (1/8 in.) over 3048 mm (10 ft.)

This should be measured on the template over the table/gantry area, as shown in [Figure 2-1](#), above.

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Chapter 3

System Catalog

Section 1.0

Option Catalog Numbers

The following is a list of system options requiring site planning work for the LightSpeed 7.X system. Contact your local GE Healthcare Sales representative for a complete list of all system options or visit us at <http://www.gehealthcare.com>. Refer to the instruction manuals supplied with specific options for respective details.

CATALOG NUMBER	OPTION DESCRIPTION (COMMENTS)
B7850GD	MKE Gantry Dolly Only
B7850RC	Rear Cable Cover (Shipped with every MKE sourced system)
E7014LA	GE Color Printer
E8203JA	GE Color Printer includes Cassette
B7710LN	Boom-in-Room
B7864CB	SmartScore EKG Monitor and Recording Device (contains more than the monitor)
E8007RC	SmartScore EKG Monitor only
K9000L	6 Node, 10/100 Mbit Auto Sensing Switched Network Kit
B7540RB	Bar Code Reader
R4390JC	Seismic Kit (Earthquake zones and IBC Installations)
B7864JB	Long Cable Set
B7700MG	Global Modem Kit

Table 3-1 VCT Installed Options and Kits

Section 2.0

Base Scanner System

2.1 Application

The CT scanner system includes hardware and software to support patient data acquisition and image analysis for whole-body computed tomography.

2.2 Configuration

The base scanner system is configured as shown. All scan and analysis functions are controlled from the Operator Console (not shown).



Figure 3-1 Base Scanner System

Chapter 4

Room Planning

Section 1.0

Strongly Recommended Systems Clearances



NOTICE If your system was installed before December 2006, Limited Access is not available for your system. You must use the Preferred Room Size to reinstall it.

Consult your local GE Sales and Service Representative about your specific needs.

Some possible room size dimensions are shown in the table below. These room size dimensions are table dependent.

System Configuration	Preferred Scan Room Size	Typical Scan Room Size	Recommended Ctrl Room Size	Clearance Requirements
GT 1700 mm Table	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 6096 mm (13 ft. 6 in. x 20 ft.)	2743 mm x 4420 mm (9 ft. x 14 ft. 6 in.)	see Figure 4-1
GT 2000 mm Table	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft. 6 in.)	4115 mm x 6706 mm (13 ft. 6 in. x 22 ft.)	2743 mm x 4420 mm (9 ft. x 14 ft. 6 in.)	see Figure 4-1

Table 4-1 List of Room Layouts, by System Configuration

Limited Access Rooms (width only):

Minimum Width: 3556 mm (11 ft. 8 in.) 356 mm (14 in.) cover to wall

Average Width: 3912 mm (12 ft. 10 in.) 711 mm (28 in.) cover to wall

Additional component dimensions are available in [Figure 4-8](#) through [Figure 4-12](#) of this document. Consult your local General Electric Project Manager of Installation (GE PMI) for your appropriate room specifications.

For equipment clearance requirements, refer to [Section 2.0](#). Remember, sufficient Regulatory and Service clearances must be maintained around equipment for full operation, service, and safety.

Cable length is an important consideration in room layout. The LightSpeed 7.X system is shipped with standard short length cables, with a set of longer cables (B7864JB) available as an option. See the electrical page of the GE print for your specific requirements.

Note also, that where possible, the cables should enter the gantry from the rear, utilizing the rear cable cover assembly. Alternate cable entry is possible at the center of the gantry (refer to the Installation template).

- Excess cable length cannot be stored behind the operator console or PDU.
- A long cable must not be cut or shortened.
- Excess cable may be stored in the cable wall or floor duct, provided sufficient space is available.
- All NEC 70-E Electrical Regulations regarding conduit or duct fill must be observed.

Section 2.0

Regulatory and Service Clearances

2.1 Regulatory Clearances

MINIMUM CLEARANCES UNDER U.S. FEDERAL REGULATIONS AND NATIONAL STANDARDS:

29 CFR 1910 (OSHA),

NFPA 70E (STANDARD FOR ELECTRICAL SAFETY IN THE WORKPLACE)

NFPA 101 (LIFE SAFETY CODE):

Figure 4-1 is a map of clearance requirements for U.S. regulatory compliance. See clearance tables on the following pages for detailed dimensional clearances. Please note: all systems installed in the United States must comply with all Federal and local regulations. For installations outside the United States, country-specific or other local regulatory clearance requirements must be met.

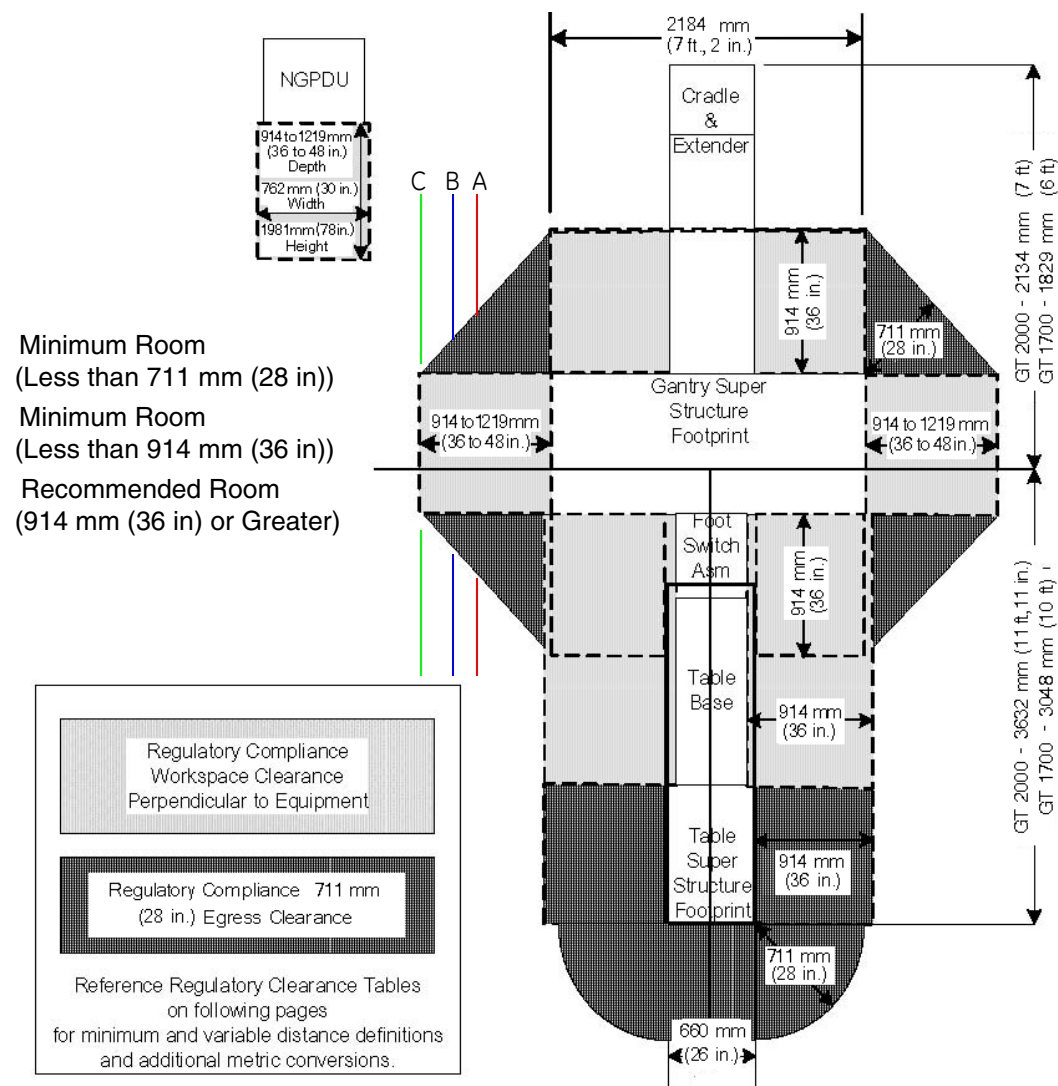


Figure 4-1 Regulatory Clearance Requirements for LightSpeed VCT System Configurations

Note: See [Section 4.0](#) for Service Clearances

2.1.1 Regulated Minimum Working Clearance by Major Subsystem

- Requirements apply to equipment operating at 600 V or less, where examination, adjustment, servicing, or maintenance is likely to be performed while live parts are exposed.
- Direction of Service Access is defined as perpendicular to the surface of the equipment being serviced.
- Strongly recommended regulatory clearance distances must be maintained and may not be used for storage. This includes normal system operation as well as service inspection or maintenance.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access	N/A No exposed live part hazards	

Table 4-2 Console

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (Front of PDU)	914 mm (36 in.)	1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1067 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater
Head Clearance	1981 mm (78 in.)	The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s). 1981 mm (78 in.) or height of equipment, whichever is greater

Table 4-3 NGPDU

- For the gantry and table, distances are measured from the enclosure, not the finish covers.

Work Space Requirement	Minimum Clear Space	Additional Conditions
Direction of Service Access (All Sides)	914 mm (36 in.)	1219 mm (48 in.), if exposed live parts of 151 - 600 volts are present on both sides of workspace with the operator between. 1067 mm (42 in.), if opposite wall is grounded and exposed live parts of 151 - 600 volts are present.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater.

Table 4-4 Gantry

Work Space Requirement	Minimum Clear Space in Inches	Additional Conditions
Direction of Service Access (Table Head)	N/A	
Direction of Service Access (Table Sides)	914 mm (36 in.)	*Can be reduced to 711 mm (28 in.) provided a written and signed approval is obtained by the local team from the local AHJ (Authority Having Jurisdiction). Signed document must be on file within GE.
Direction of Service Access (Table Foot)	711 mm (28 in.)	457 mm (18 in.) minimum for Front gantry Cover removal only if unobstructed egress space of 711 mm (28 in.) is maintained around the equipment for room exit. This also means no trip hazards exist along the path of egress.
Service Access Width (Left-Right of workspace)	762 mm (30 in.)	This is the width of the working space in front of the equipment. 762 mm (30 in.) min or the width of the equipment, whichever is greater.

Table 4-5 Table

2.1.2 Terms and Definitions

EGRESS

The path of exit from within any room. U.S. regulatory requires a minimum of 711 mm (28 in.) of continuous and unobstructed space including trip hazards along the path of exit.

WORK SPACE

This is the dimensional box needed for safe inspection or service of energized equipment. It consists of depth, width, and height. The depth dimension is measured perpendicular to the direction of access. U.S regulation is minimum of 914 mm (36 in.). Additional conditions can increase the minimum requirement. FCT defines this as the envelope of the component superstructure. For the NGPDU it is with the front panel removed. For the gantry and table, it is with the patient or external covers removed.

SERVICE ACCESS WIDTH

This is the width of the working space in front of the equipment, a minimum of 762 mm (30 in.), or the width of the equipment, whichever is greater.

HEAD CLEARANCE

This is the height dimension of "Work Space". The height of the workspace measured from floor at the front edge of equipment to ceiling or overhead obstruction(s), 1981 mm (78 in.) or height of equipment, which ever is greater.

GROUNDING WALL

Any wall that can be electrically conductive to earth ground. Masonry, concrete, or tile, are considered conductive. Additional commonly found aspects of a wall should also be considered as grounded. This is not an all-inclusive list:

- Medical Gas ports
- Metal door and window frames
- Water sources and metallic sink structures
- Metallic wall-mounted cabinets
- A1 disconnect panel
- Equipment Emergency Off panels
- Industrial equipment, such as air conditioners and vents
- Expansion joints

The following are not considered as grounded elements of a common wall:

- Standard wall outlet
- Light switches
- Telephones
- Communication wall jacks

MINIMUM

The lowest limit permitted by law or other authority.

DIMENSIONS AND CLEARANCES

Consisting of, or representing, the lowest possible amount of degree for freedom permissible for equipment siting. This relationship must meet all safety, service, and regulatory requirements to be acceptable.

PRE-INSTALLATION ESCALATION

Process to consult with CT Engineering, the Design Center or EHS regarding pre-installation issues related to your siting concerns.

Section 3.0

Additional Regulatory Clearance Information

3.1 Minimum Room Size (Limited Access)

The CT Gantry Left Side Limited Access Initiative provides the capability to reduce the minimum room size for CT Systems while still meeting all installation requirements and specifications. This adds left side flexibility, allowing the CT system to be sited in rooms with widths 559 mm (22 in.) smaller than the current minimum room width. Left-side access and egress may be restricted. Refer to your site's installation print for your room's detail.

If you are using the square meters (square feet) to determine compliance, remember that the front cover clearance is wider than the regulatory clearance along the table length, and that the cover park position is behind the table in the home position. The cover is removed on tilting dollies and can be moved side-to-side to reach the park position of 457 mm (18 in.), if this is not an egress route.

Wall duct and conduit on walls within the regulatory clearance areas shall be 1067mm (42 in.), measured from the covers to the obstruction. Servicing of the CT System can be safely performed within the regulatory envelopes, however sufficient space must be maintained to remove system covers, and replace large system components. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. Surface floor raceway cannot be used in the egress route areas. OSHA ramps are available. The FE lifting the rear or front cover to avoid floor obstructions is not an EHS-approved service procedure.

One Service Engineer shall be able to accomplish all service component replace tasks listed without the need for special tools or equipment, such as a tube change, detector change, and HV tank.

3.1.1 Regulatory Caution

Site prints are needed for all system installations including relocation and moves. CT room layout, as shown on your site print, shall meet all regulatory requirements as described in the installation manual. Additional room components, such as cabinets, reduce room size. Equipment not shown on the site print may void the caution statement, making the room non-compliant. Actual site measurements before installation will be taken to determine room size and compliance.

3.1.2 Egress Clearance

Egress requires a clear, unobstructed route out of the room, either around the back of the gantry or around the back of the table. If your egress route is not around the back of the table, maintain 457 mm (18 in.) of clearance between the back of the table, with a continuous width of 3200 mm (126 in.), 1600 mm (63 in.) on each side of the table center line, on each side to any obstruction so that the front cover can be removed. Refer to the Pre-Installation manual for more details on service clearances.

Exceptions

Rooms smaller than 3556 mm x 6096 mm (11 ft. 8 in. x 20 ft.), or 3556 mm x 6706 mm (11 ft. 8 in. x 22 ft.), require construction to meet the minimum requirements. The design center or your GE PMI may have additional recommendations for your room size.

3.1.3 Operational Caution

In a minimum room layout 356 mm - 686 mm (14 in. - 27 in.), the customer should consider workflow, customer access for patient care, and critical-care operations space requirements. Additionally, there may be limited equipment access on the gantry left side when loading patients or when positioning patient equipment in the room between the gantry and the wall. Detailed customer installation tasks are detailed in the product Pre-Installation manual, Chapters 1-4.

3.1.4 System Specifications (Pro32 and VCT 64 Systems)

System Configuration	Preferred Room Size	Typical Room Size	Minimum Room Size
VCT with GT 1700	4420 mm x 7772 mm ¹ (14 ft. 6 in. x 25 ft. 6 in.)	3688 mm x 6096 mm ¹ (13 ft. 6 in. x 20 ft.)	3556 mm x 6096 mm ² (11 ft. 8 in. x 20 ft.)
VCT with GT 2000	4420 mm x 7772 mm (14 ft. 6 in. x 25 ft. 6 in.)	3688 mm x 6706 mm (13 ft. 6 in. x 22 ft.)	3556 mm x 6706 mm (11 ft. 8 in. x 22 ft.)

Table 4-6 System Specifications (Pro32 and VCT 64 Systems)

¹Same Regulatory requirements apply

²Same Regulatory requirements apply, with the addition of no energized left side service.

3.1.4.1 Preferred Room Size

The “preferred room size” configuration offers the most flexibility for future upgrades. It has sufficient workspace and space to add millwork, while meeting all regulatory requirements. This room is compatible with most two-step future installations. For Upgrades to CT-PET refer to [Section 3.1.5](#).

3.1.4.2 Typical Room Size

The “typical room size” configuration allows for some future upgrades. It has sufficient workspace, but limited space to add millwork and meet all regulatory requirements. This room may be compatible with some two-step future installations.

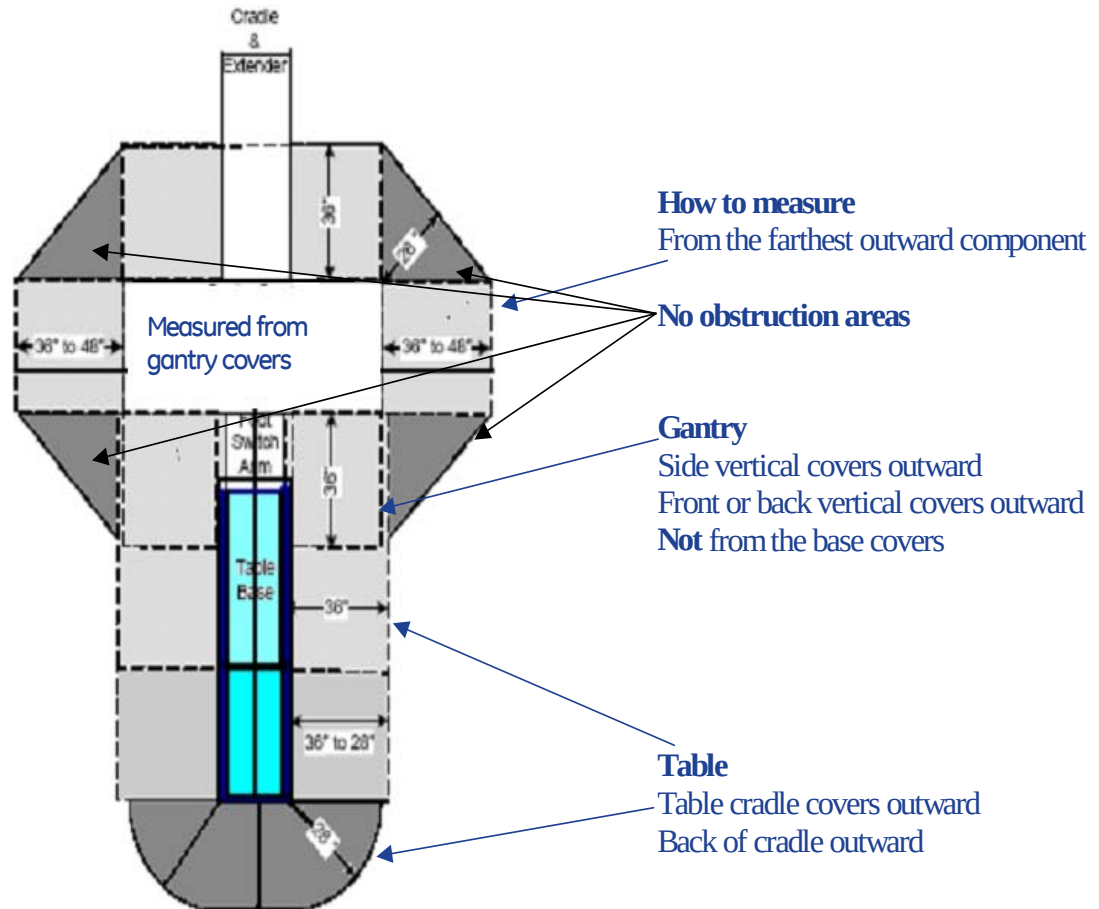
3.1.4.3 Minimum Room Size

The “minimum room size” configuration allows for no future upgrades. It has limited workspace and no in-room millwork, but meets all regulatory requirements. This room is not compatible with two-step future installations.

3.1.5 Room Size for PET Upgrades

To complete a future upgrade to PET, you need to meet the requirements, including room size, to support the CT PET configuration. To upgrade to Discovery VCT, you need a room size 7772 x 4420 mm (25 ft. 6 in. x 14 ft. 6 in.), which, depending on your layout, allows space to add millwork while meeting all regulatory requirements. This room would be compatible with most PET two-step future installations.

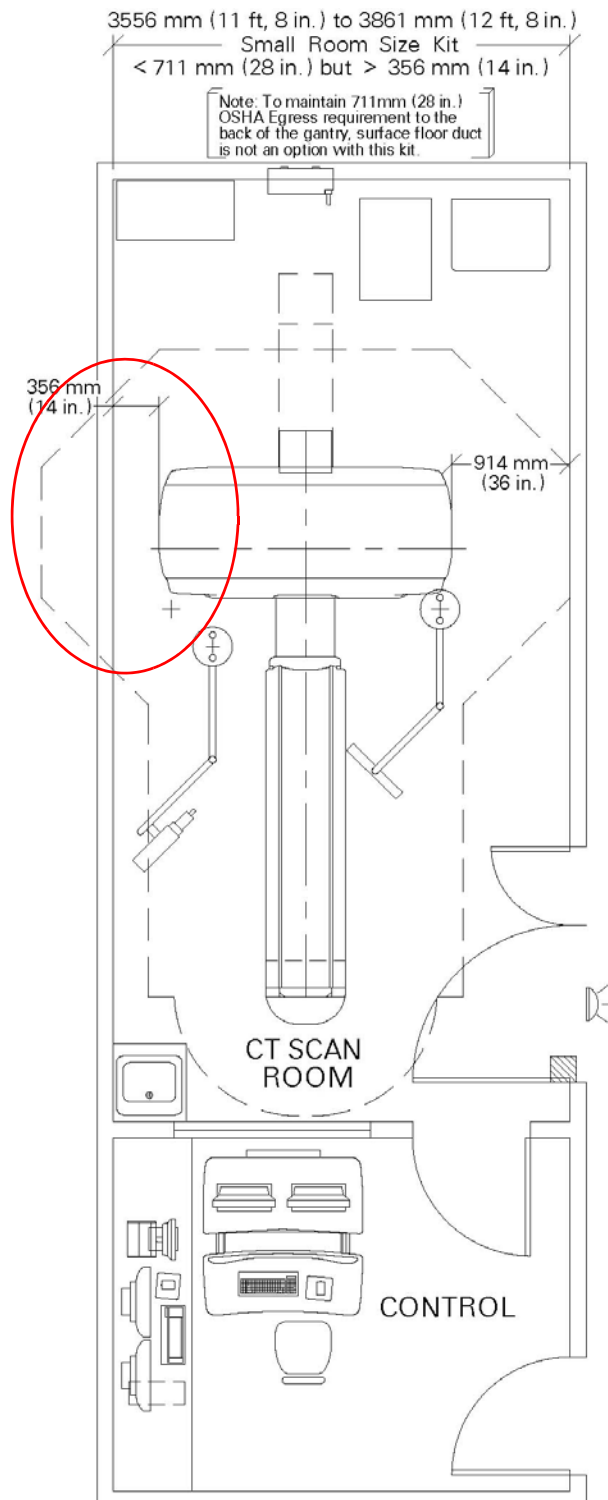
3.2 How to Measure



3.3 Minimum Room Size & Requirement Layouts

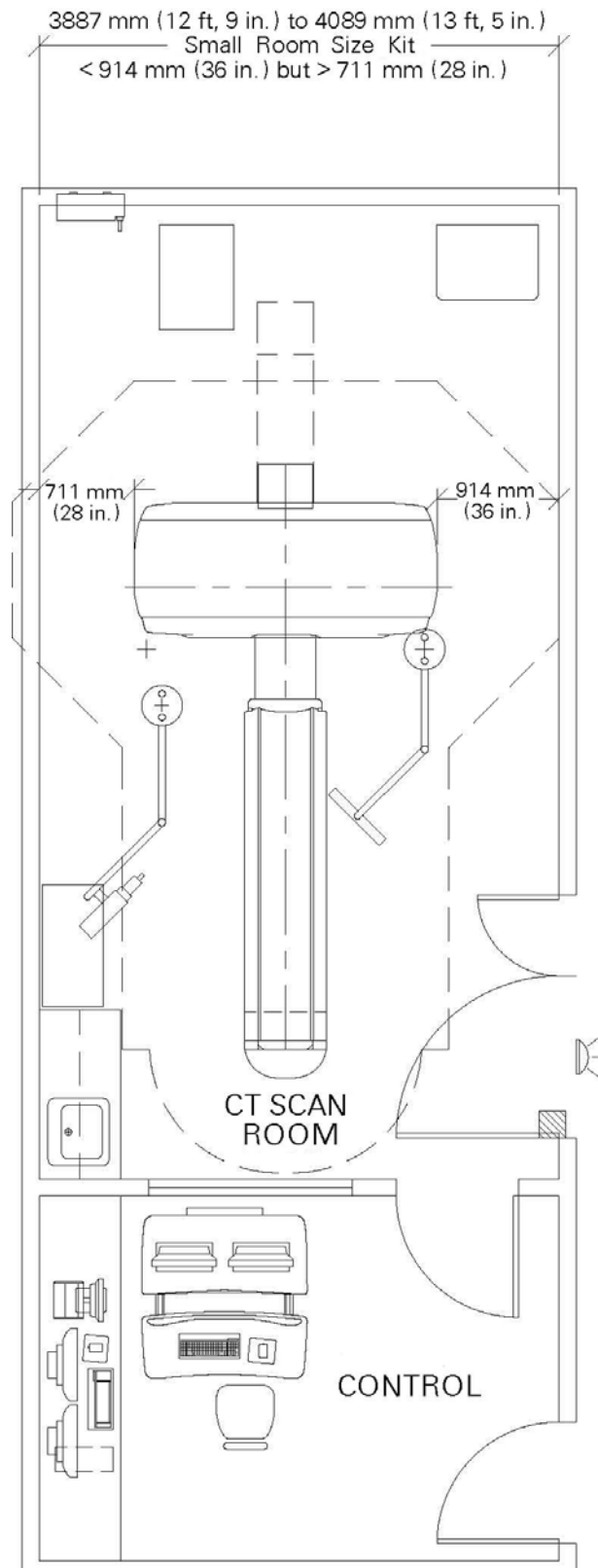
Room A - Less than 711 mm (28 in.), but greater than 256 mm (14 in.), measured from the covers to the left sidewall.

In this configuration service, egress and workspace are compromised around the gantry's left side.



Room B - Less than 914 mm (36 in.) but greater than 711 mm (28 in.) measured from the covers to the left sidewall.

In this configuration service, egress, and workspace are acceptable around the gantry.



3.4 Preferred and Typical Room Size & Requirement Layouts

Figure 4-2 and Figure 4-3 show preferred and typical room layouts, with and without in-room cabinets. You need to be aware of locations for me gas, surface ductwork, or other items that make a grounded wall.

Note: Your room layout may meet the preferred or typical room requirements but look different than Figure 4-2 or Figure 4-3. Contact your sales person to have a detail room layout completed for your site.

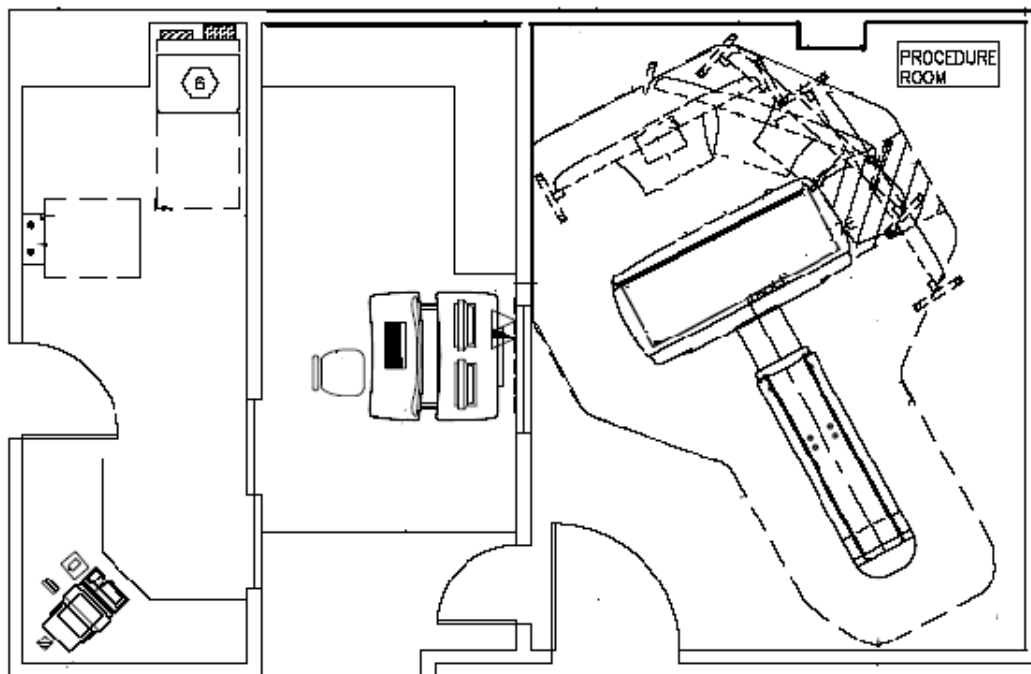


Figure 4-2 Room Layout with cabinets

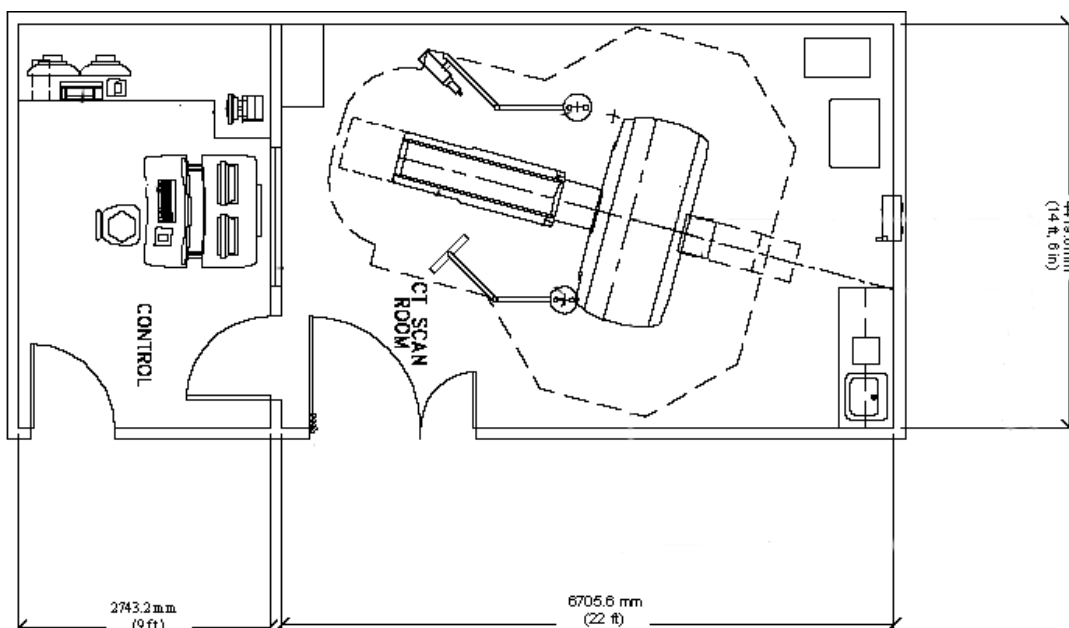


Figure 4-3 Room Layout without cabinets

Section 4.0 Service Clearances

Servicing the CT System can be safely performed within the regulatory envelopes defined in [Section 2.1](#), however sufficient space must be maintained to remove the covers from the system. To achieve this clearance for the gantry, clear space must be available to maneuver the gantry covers mounted on the service dollies. One Service Engineer can accomplish this. If such space is not available, then two Service Engineers are strongly recommended for gantry cover handling. This must be clearly communicated with the Customer.

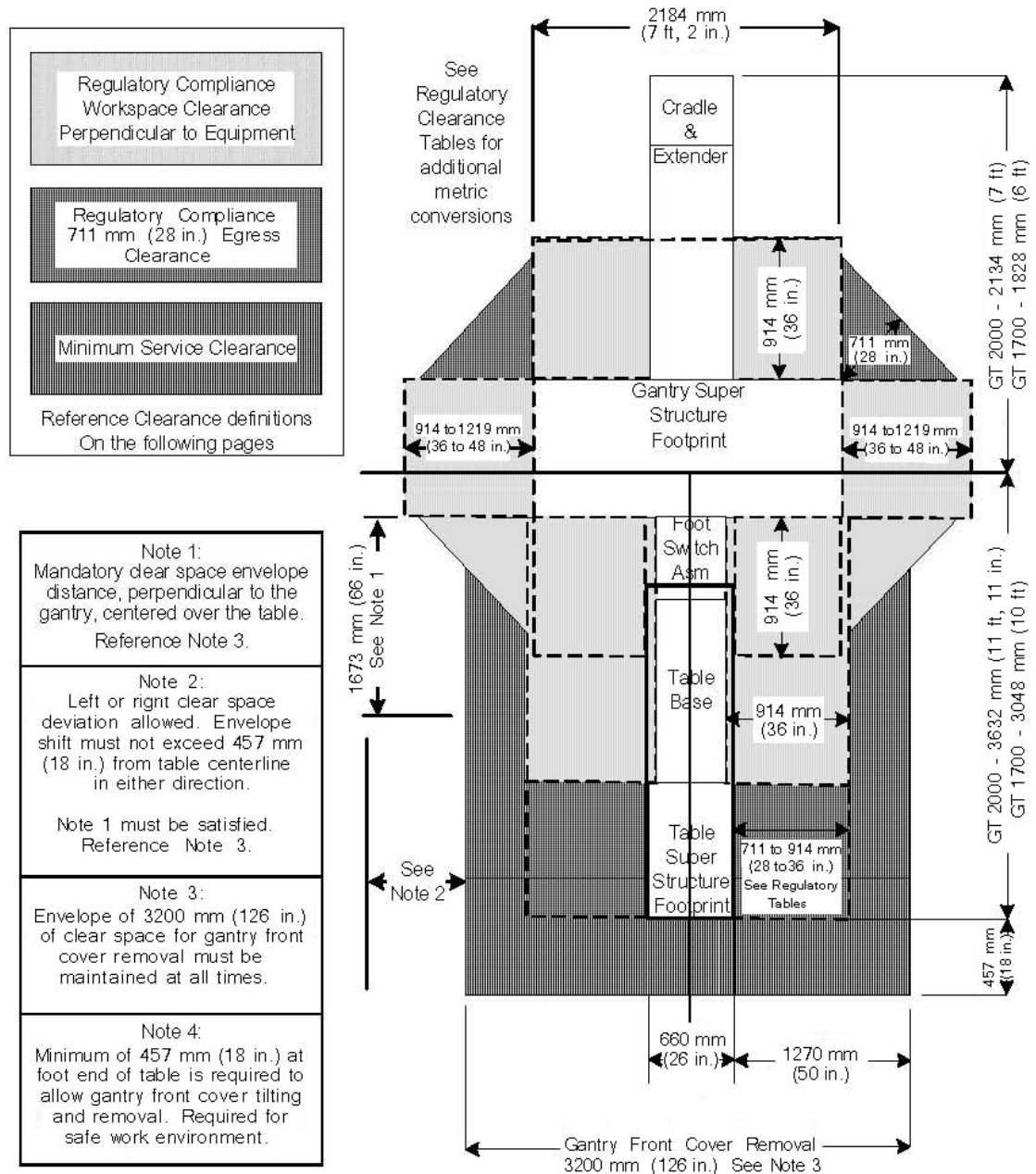


Figure 4-4 Minimum Service Clearance

4.1 Service Clearances for Single Service Engineer

- Gantry front cover removal requires the use of the Tilting Cover Dollies. These dollies allow the Service Engineer to separate the cover from the gantry, tilt the cover 90 degrees, roll the cover to the foot end of the table, and then tilt the cover an additional 90 degrees, such that the front cover is now upside down relative to the normal system-mounted condition. [Figure 4-4](#) illustrates the minimum clear space strongly recommended to achieve this operation of 3200 mm (126 in.). The gantry front cover must be removed to a position that satisfies the minimum regulatory clearances.
- The gantry rear cover with service dollies installed, requires a width of 2388 mm (94 in.) and a depth of 584 mm (23 in.) of clearance for removal as shown in [Section 6.0](#), [Figure 4-6](#). Sufficient space must be calculated to move the cover either straight back or to one side of the table to satisfy the minimum service clearance shown in [Figure 4-4](#). This means the rear cover cannot violate the workspace on the rear, or on either side of the gantry.
- If gantry service requires both the front and rear covers be removed, then these covers must be positioned within the room in such a manner as to not violate the egress clearances on any side of the gantry. This may necessitate removing the covers from within the suite. This should be discussed with the customer and provisions made to accommodate this potential event.
- A single Service Engineer can safely perform servicing of the table. Sufficient clear space must be available to maintain egress clearances when the table covers or cradle are removed.

4.2 Power Distribution Unit (NGPDU) Service Clearance

Positioning of this component must be considered for regulatory compliance as defined in [Section 2.1](#), Regulated Minimum Working Clearance by Major Subsystem.
See Regulatory Tables.

4.3 Console Service Clearance

The operator console does not present an exposed live parts hazard. However, a minimum working space depth of 1219 mm (48 in.) and full width of the operator console shall be maintained at all times for service activity. Additionally, sufficient space needs to be provided for repositioning of the operator console, and side clearance for rear service access. See [Figure 4-5](#) for a typical control room layout.

Section 5.0 Recommended Layouts

5.1 Storage Cabinet

A storage cabinet is provided by GE Healthcare to store all supplied service equipment (see [Table 4-7](#) for equipment list). This storage cabinet (457 mm D x 914 mm W x 1067 mm H) (18" D x 36" W x 42" H; ~90 lbs) should be located in the scan room suite area for easy service access.

ITEM	SIZE	WEIGHT (TOTAL)	
QA Phantom (water filled)	20 cm x 15 cm (7.9 in. x 5.9 in.)	5.5 kg	12 lb
35CM Phantom	35 cm x 7 cm (13.8 in. x 2.8 in.)	8.2 kg	18 lb
Phantom Holder	25 cm x 25 cm (9.8 in. x 9.8 in.)	3.6 kg	8 lb
FE Box (Purple)	30 cm x 38 cm x 30 cm (11.8 in. x 15 in. x 11.8 in.)	6.8 kg	15 lb
Rear Cover Dollies	158 cm x 82 cm (62.2 in. x 32.3 in.)	11.4 kg	25 lb
Front Cover Dollies	85 cm x 20 cm and 85 cm x 15 cm (33.5 in. x 7.9 in. and 33.5 in. x 5.9 in.)	15.9	35 lb
Install Support Kit (box)	30 cm x 30 cm x 38 cm (11.8 in. x 11.8 in. x 15 in.)	9.1 kg	20 lb
Tube Hoist	77 cm x 8 cm and 38 cm x 15 cm (30.3 in. x 3.1 in. and 15 in. x 5.9 in.)	9.1 kg	20 lb
Balance Weight Kit		33 kg	73 lb

Table 4-7 Equipment to be stored in storage cabinet

5.2 Advantage Workstation (AW)

Refer to Pre-Installation Manual 2111833 and Installation/Service Manual 2111831. For document access, please refer to GE Healthcare's Common Documentation Library web site.

5.3 Control Room Considerations

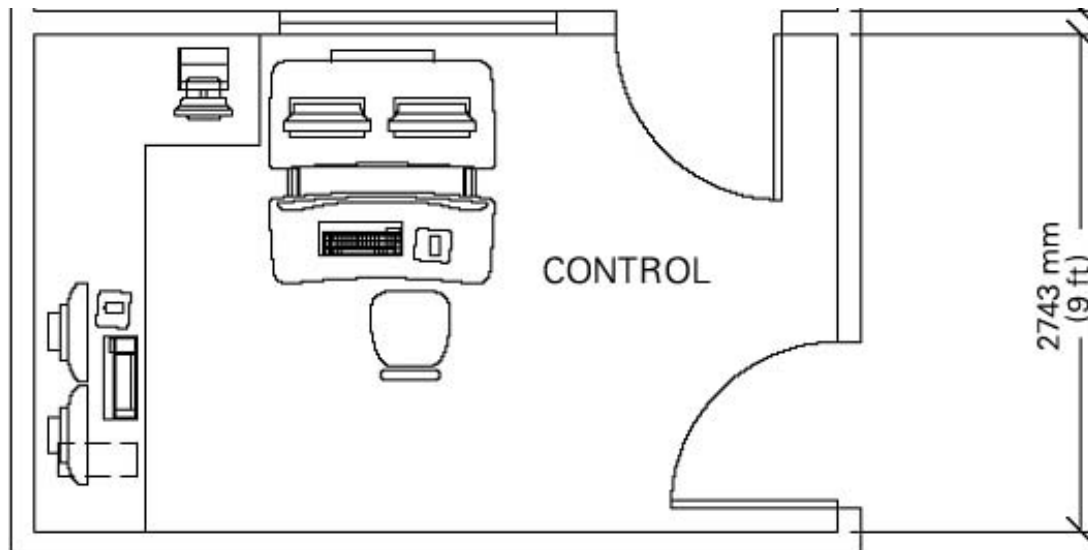


Figure 4-5 Typical Control Room Layout

- The control room must provide a suitable operating environment for the operator console electronic, and operator working comfort.
- The operator console cannot be dismantled or have components removed or rearranged in configurations other than as shipped.
- A suitable work area, which is within reach of the operator console, should be provided for placement of the injector control. Injector controls differ in dimensions depending on the brand selected.
- A PACS, workstation, image printer, or filming device are often placed in the operator console control room area, and sometimes may be directly linked to the operator console.
- Additional components although linked via network or ethernet cable, are not powered from the CT operator console.
- Additional room power and network connection should be considered when reviewing the operator console work space.

Section 6.0

Component Dimensions

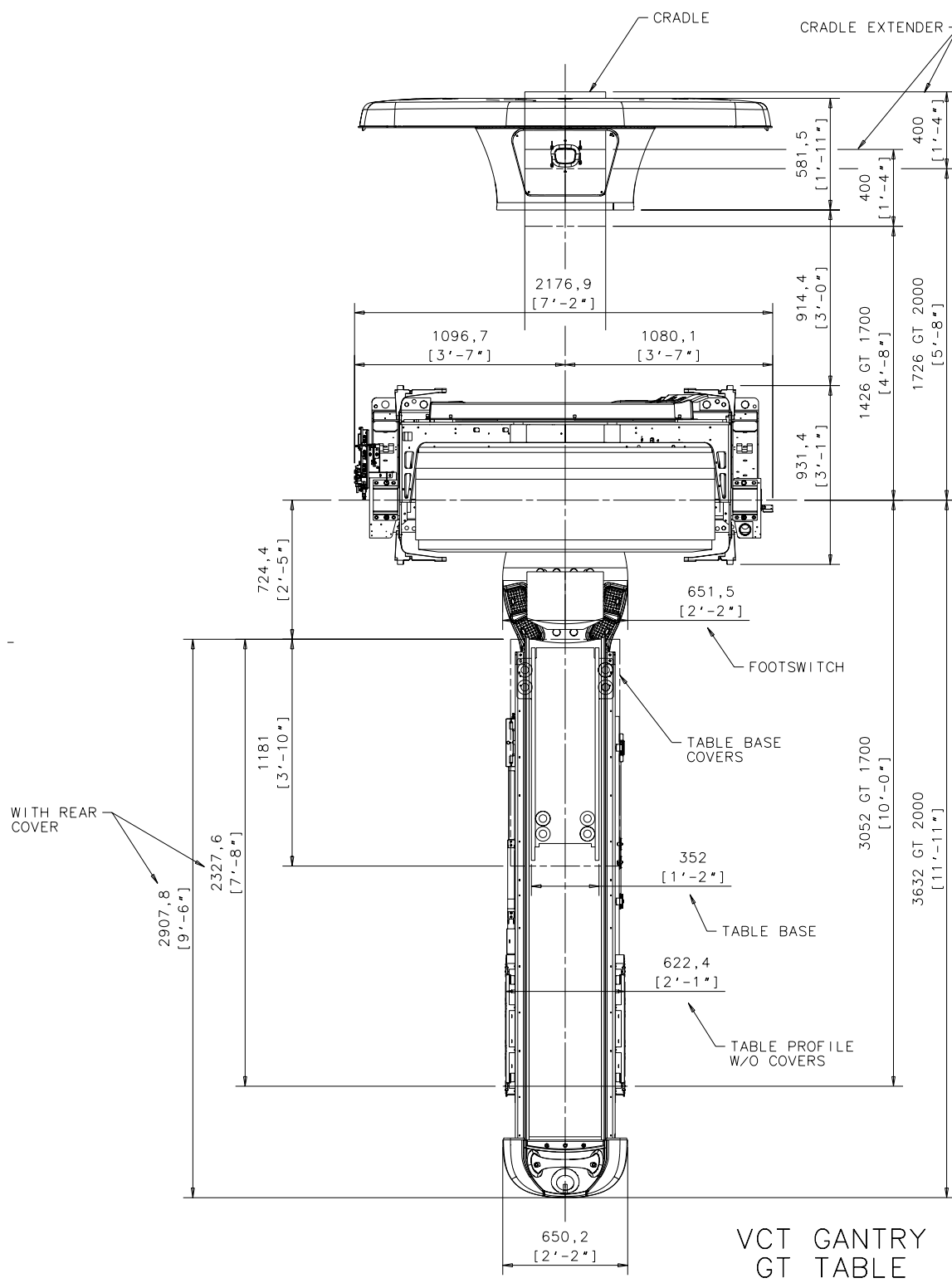


Figure 4-6 LightSpeed VCT Dimensions without Covers

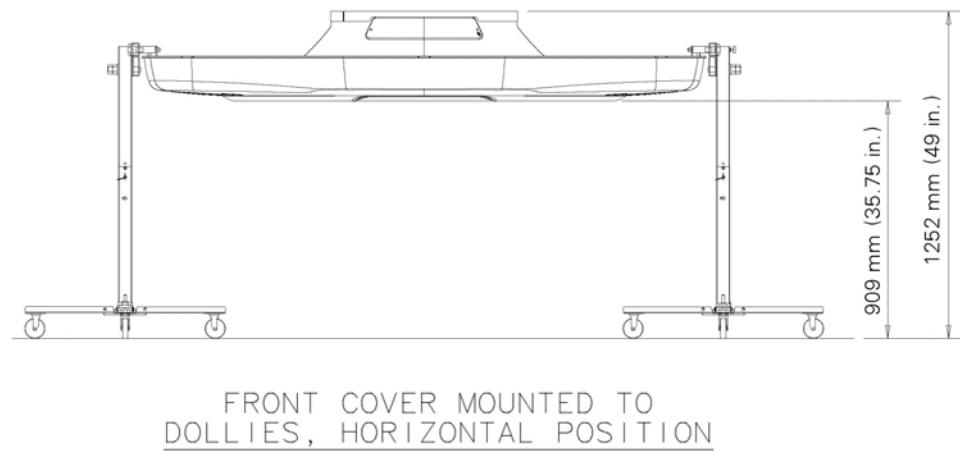
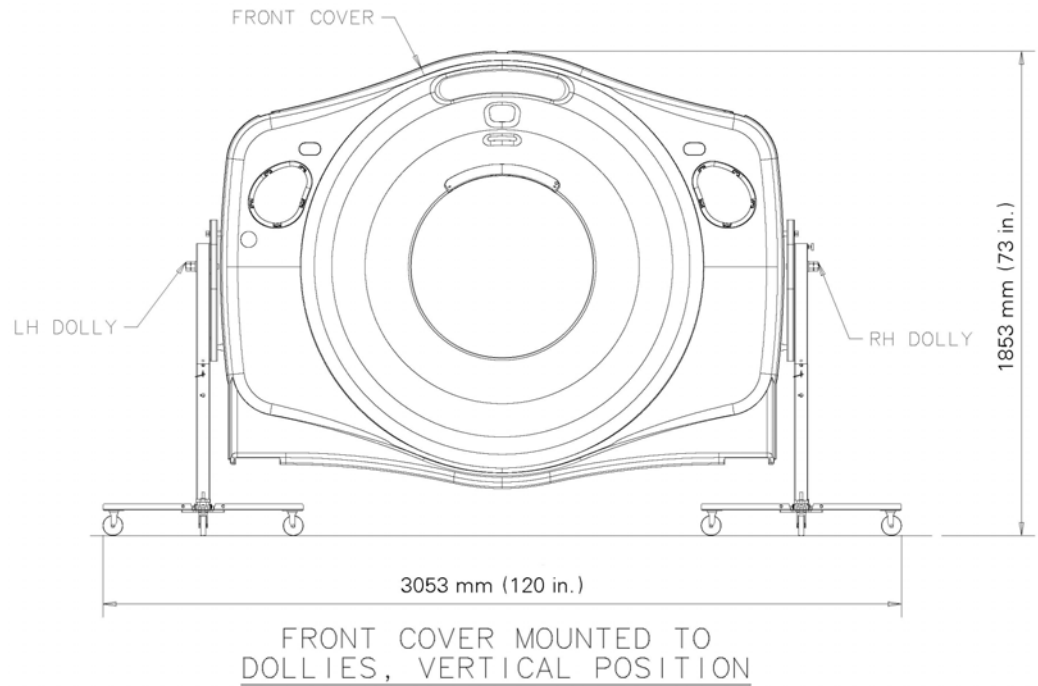


Figure 4-7 Gantry Front Cover with Service Dolly Dimensions

DESCRIPTION	WIDTH		DEPTH		HEIGHT	
	MM	INCH	MM	INCH	MM	INCH
Remote Color Monitor (LCD)	413	16.25	203	8	406	16
Color printer	584	23	457	18	178	7

Table 4-8 Dimensions of Accessories

6.1 Table and Gantry

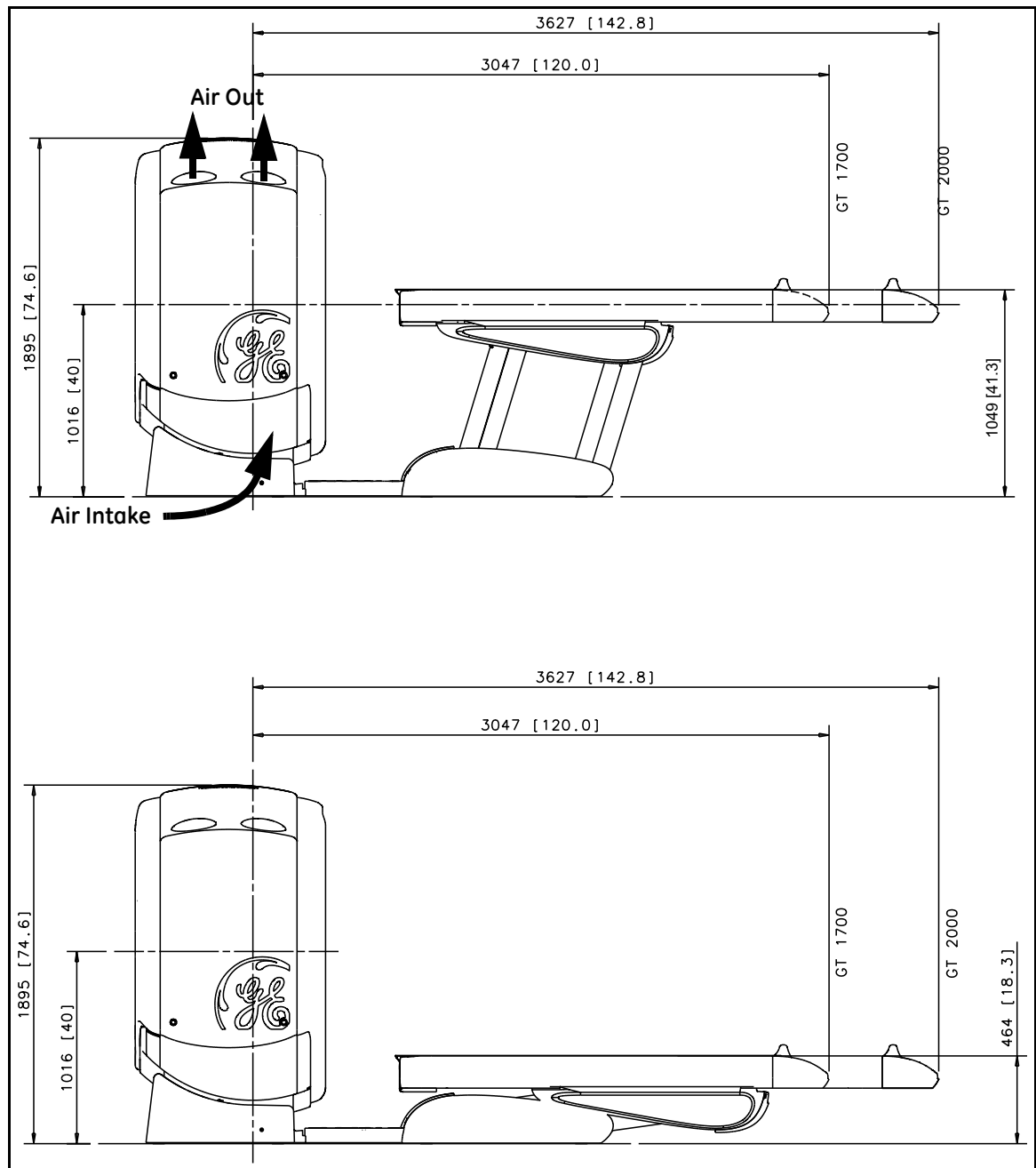


Figure 4-8 GT 1700 and GT 2000 Table and Gantry (Side View)

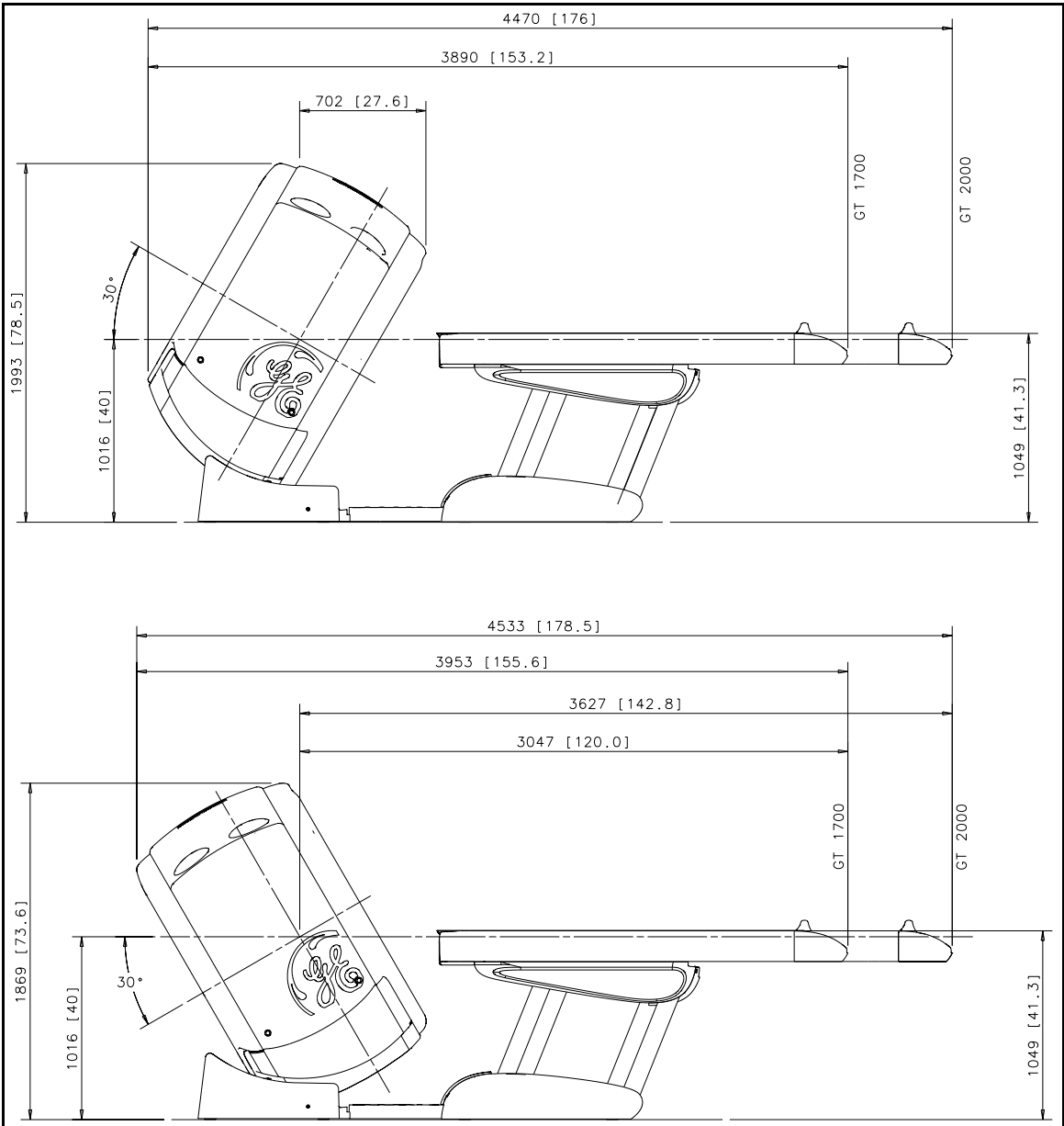


Figure 4-9 Gantry shown tilted +30° (top) and -30° (bottom) - GT 1700 and GT 2000 Table Options

6.2 Power Distribution Unit

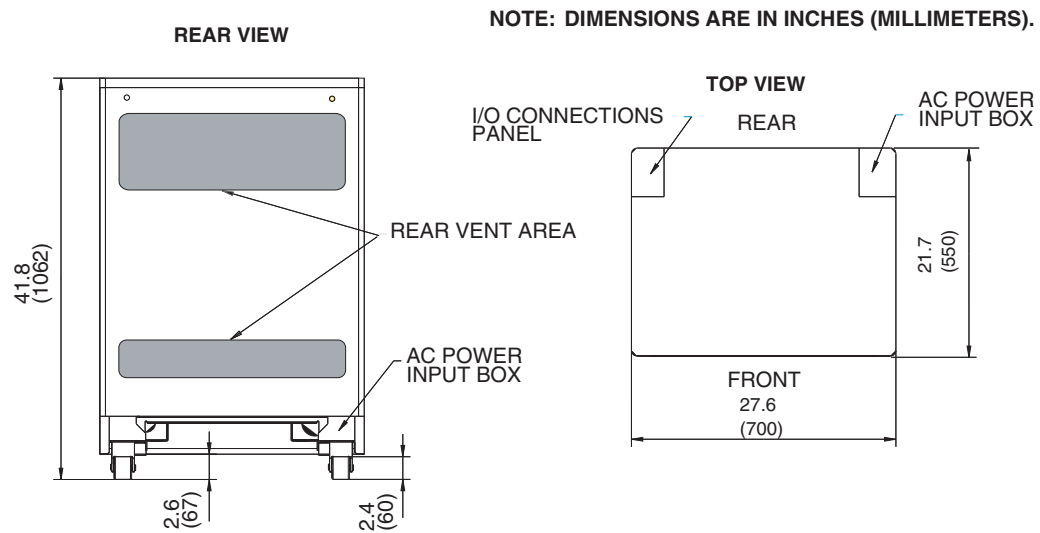


Figure 4-10 Power Distribution Unit (NGPDU-4)

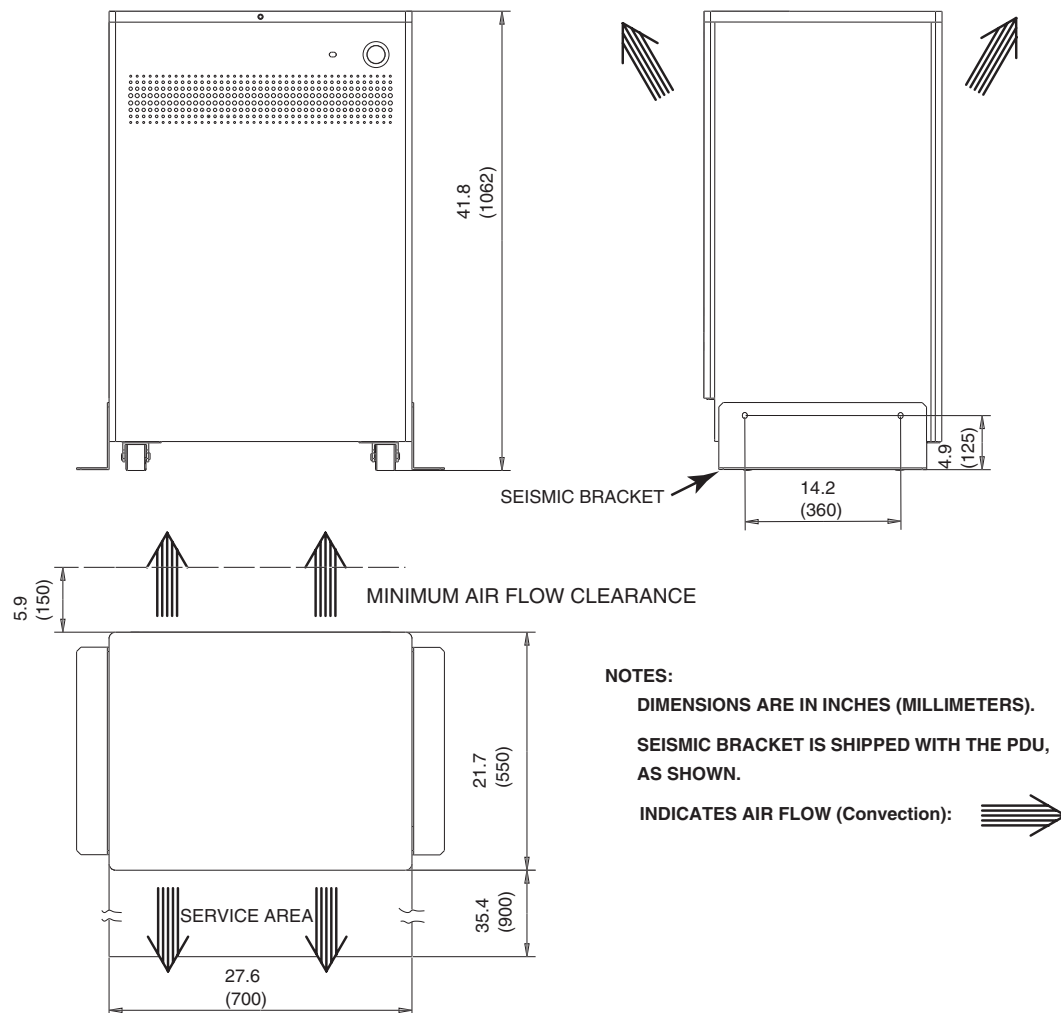


Figure 4-11 Power Distribution Unit (NGPDU-4)

6.3 Operator Console

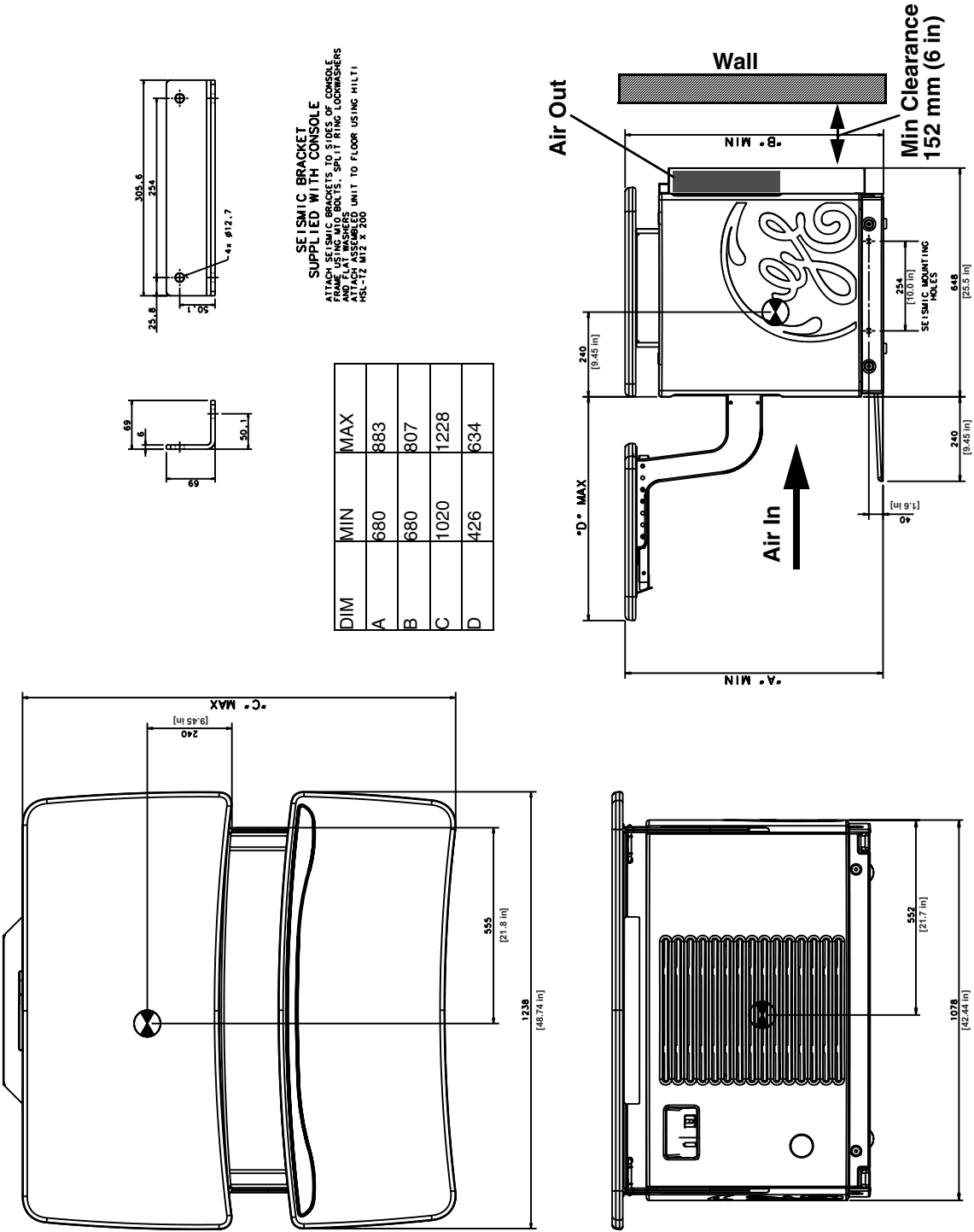


Figure 4-12 Operator's Console

Section 7.0

Minimum Dimensions and Clearances

System Operation	MM	Inches
Ceiling Pedestal mount (optional) (lowest point to floor injector or monitor)	2439 mm	96 in.
Finished ceiling to floor (recommended)	2743 mm	108 in.
Finished ceiling to floor (minimum)	2439 mm	96 in.
Table max extension head end w/extender from Center Line	2030 mm	80 in.
Table extension head end w/extender to obstruction	152 mm	6 in.
Table in lowest pos. w/cradle @home pos. to Center Line	3209 mm	126.5 in.
Back of Console to wall	152 mm	6 in.
Back of PDU to wall	152 mm	6 in.

Table 4-9 Minimum Dimensions and Clearances

7.1 Injector Control

A suitable work area, which is within reach of the operator console, should be provided for placement of the injector control.

Wall mounted, ceiling mounted, and pedestal units need cables to be routed from the gantry area to the operator console area. The supplied cable is 15.24 m (50 ft.) in length. Injectors require AC power that is not supplied by the CT system.

Note: Injector cables should not be routed with the system cables.

Mounts are available in different configurations and lengths. Refer to injector documentation for detailed installation instructions.

Section 8.0

Structural Requirements

8.1 Table and Gantry Mounting Requirements



WARNING **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING
SYSTEM OPERATION.



NOTICE **It is the purchaser's responsibility to provide an approved support structure and mounting method for all floor types other than those listed. General Electric is not responsible for any failure of the support structure or method of anchoring, including seismic requirements.**
GE is not responsible for methods other than those listed.

Table and gantry mounting dimensions are shown in [Figure 6-2](#), [Figure 6-3](#), and [Figure 6-5](#). Refer to [Chapter 6](#) for additional details of floor loadings, component weights, and gantry and table installation and anchoring.

Anchor the gantry and table to the floor by a means that maintains their relative alignment and meets applicable building and other local codes, including seismic structural mounting requirements.

The floor structure must be capable of withstanding the occupied weight of the table and gantry, and the individual contact area loading of these components. Refer to [Figure 4-8](#) for each of the three (3) major components of the LightSpeed 7.X system.

Support areas of the patient table and gantry must rest on solid concrete, not resilient tile or carpeting, which will slowly yield over a period of time and disturb the alignment of the table to the gantry.

Factors that could cause misalignment between the gantry and table due to floor sag should be considered. The cradle can potentially carry a 227 kg (500 lb) patient. The center of gravity changes as the cradle cantilevers.

Take into consideration all other moving weights such as gurneys or personal equipment. Refer to [Chapter 6](#) for gantry and table mounting details.

No part of the floor surface within the table and gantry, nor the two interface areas between the table and gantry, should be higher than the support area for the table and gantry.

8.2 Floor

8.2.1 Anchors

Provided floor anchors are designed for use ONLY on concrete floors that meet the 102 mm (4 in.) concrete floor requirements. At the customer's expense, all other anchoring methods (on floor types other than the 102 mm (4 in.) concrete minimum) must be determined by their structural contractor to meet the stated GE minimum load requirements. The customer's contractor is responsible for the installation of all anchors other than those shipped with the system.

8.2.2 Floor Strength

Concrete floors must have a minimum strength of $f'_c = 2000$ psi (1.4×10^7 Pa) at 28 days for (curing time) mounting floor anchors. It is the responsibility of each customer to have appropriate tests performed to determine and measure concrete strength.

Note: If installing the GE LS scanner on a floor type thinner than a 102 mm (4 in.) concrete floor, the purchaser, at their expense, shall provide acceptable anchoring and mounting methods that meet all structural specifications provided in sections 8.1 through 8.2.4 of this Pre-Installation Manual.

8.2.3 Floor Levelness

The CT Room floor levelness requirement is important for accurate patient positioning. Floor levelness in the Scan Room must not be greater than 3 mm (0.125 in.) between depression and high spots over any 3048 mm (120 in.) distance within the area of the gantry/table template (see the envelope shown in [Figure 2-1, on page 37](#)).

The use of floor shims is not suitable to achieve floor levelness. It is recommended that the concrete be leveled to meet this requirement.

8.2.4 Floor Vibration

The CT equipment may be sensitive to vibration in the frequency range of 0.5 to 20 Hz depending on the amplitude of the vibration. It is the customer's responsibility to contract a vibration consultant or qualified engineer to implement design modifications to meet the specific limits. However, it is ultimately the customer/architect/engineer responsibility to design the site solution.

8.2.4.1 Steady State Vibration

The maximum steady state vibration transmitted through the floor should not exceed 10^{-3} m/s² rms maximum single frequency above ambient baseline from 0.5 to 80 Hz (measured in any 1 hour during a normal operating period).

8.2.4.2 Transient Vibration

The behavioral characteristics must be such that any measurable transient disturbance must also be minimized to less than 0.01 m/s² peak-to-peak.

8.2.4.3 Equipment Location

To minimize the interference, the CT equipment should be placed on a solid floor, located as far as possible from the following vibration sources:

- Parking lots
- Roadways
- Subways
- Trains
- Hallways
- Elevators
- Heliports
- Hospital power plants containing pumps, motors, air handling equipment and air conditioning units

8.2.5 Finished Floor Requirements

Installation requires a finished floor in the scan and control rooms. The scan room must be level by 6 mm (1/4 in) over the table and gantry area to be acceptable. You cannot use shims to level the floor. Eight or more floor covering openings that are 101.6 mm (4 in) in diameter are made to ensure the table and gantry rest on a solid surface. These floor penetrations can be sealed if needed. These requirements apply to all installation types.

FINISHED FLOOR EXCEPTION 1

For sites replacing their scan room floor covering after the table and gantry are installed, the floor can be clean-finished with dust-free concrete. The finish floor in the scan room requires no dust-producing operations when applying final floor covering.

FINISHED FLOOR EXCEPTION 2

Facilities under new construction that have a finished radiology area with a single controlled-access and dust abatement barrier, can have a finished concrete floor in the scan room. The finished concrete floor in the scan room requires no dust-producing operations when applying final floor covering.

8.3 Walls

8.3.1 Scan Window

The recommended patient viewing window dimensions are 1219 mm wide x 1067 mm high (48 in. x 42 in.). The location of the window is dependent on the position of the operator workspace position. Consult [Section 10.0](#) of this chapter and a **qualified radiological health physicist** for radiation protection requirements of the window glass (lead content and thickness).

Note: The operator at the operator workspace must be able to view the patient during a scan.

8.3.2 Finished Walls Requirements

Finished walls inside the scan and control rooms must be painted at the time of installation. This requirement applies to all installation types.

A finished walls exception is made for the following condition:

In new construction and upgraded facilities, a primer coat of paint is acceptable for equipment installation. However, the final coat of paint must be applied using a brush of some type (e.g. roller or bristle). The final coat of paint cannot be applied using a spray method.

8.4 Ceiling Requirements

The minimum ceiling height above the table and gantry shall measure at least 2134 mm (84 in.) or the minimum distance allowed by local laws and codes, whichever is greater, when measured from the floor to the finished ceiling or to the ceiling pedestal mounts of any ceiling-mounted components. The purchaser or their contractor shall complete the installation of all pedestals for ceiling-mounted components. The PMI will provide the necessary bolt hole information upon request.

The support structure for a ceiling-mounted option using a Mavig pedestal, requires a flush ceiling mounting plate. This flush ceiling mounting plate must be designed by a structural engineer and installed by a qualified contractor prior to the CT installation.

8.4.1 Mounting Plate

The pedestal-ceiling mount requires a flush ceiling mounting plate that is structurally supported to handle the weight of the load as shown in [Figure 4-13](#). You can order a Mavig-designed ceiling plate (p/n E8007NZ). This 356 mm x 356 mm (14 in. x 14 in.) plate has pre-drilled 1/2 in. mounting holes in each of the four (4) mounting corners.

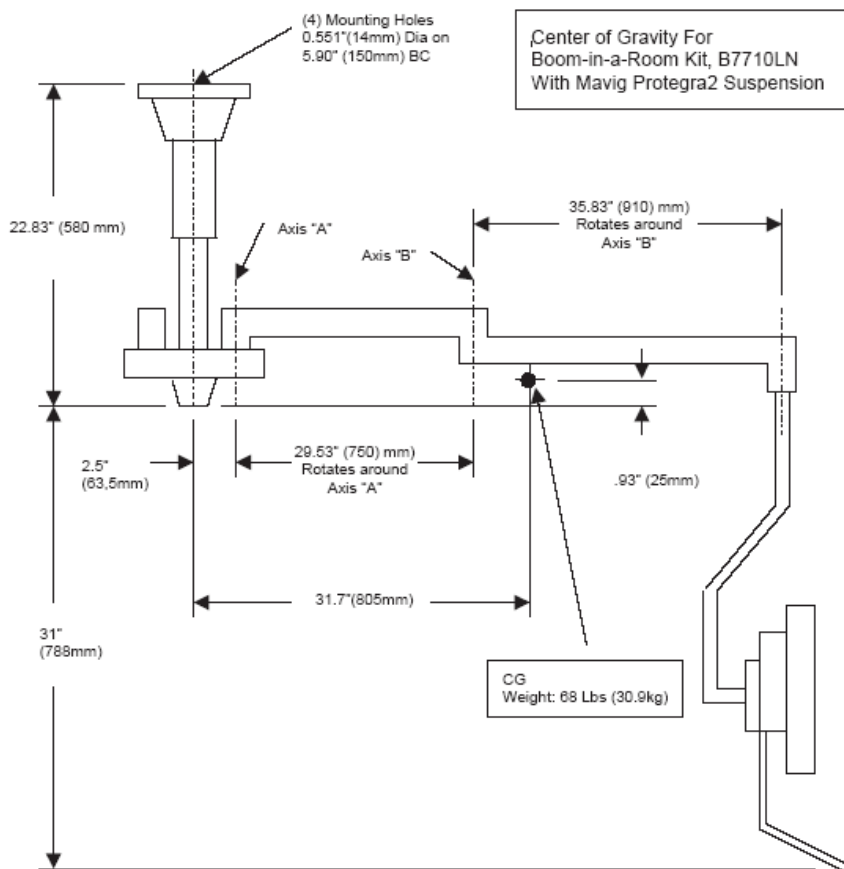


Figure 4-13 Center of Gravity for Boom-in-a-Room Kits, shown on the site print

If a structural contractor designed an equivalent plate, use the supplied template to drill the necessary four 0.39 in. holes in a pattern as shown in [Figure 4-14](#). An additional 0.32 in. hole is strongly recommended for the pedestal-ceiling mount safety chain.

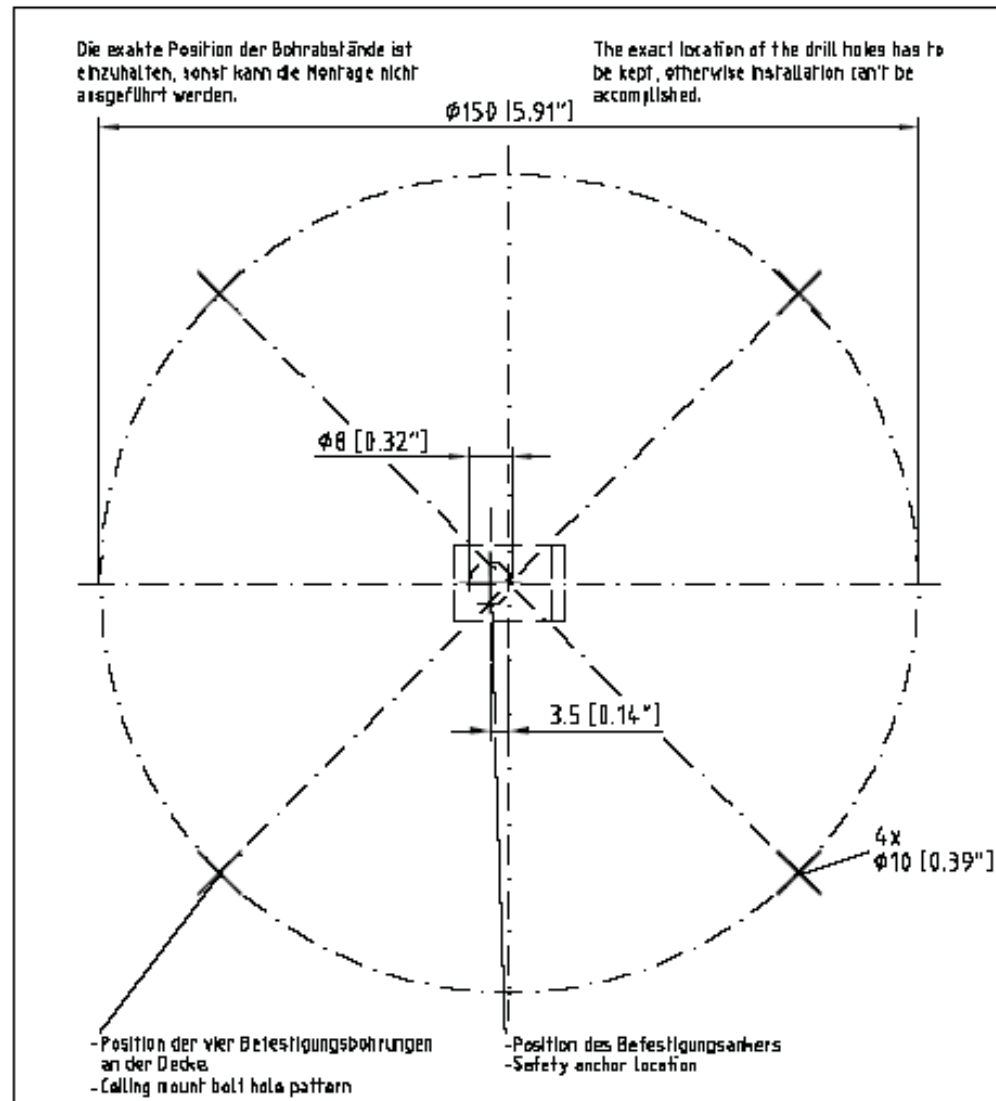


Figure 4-14 Mounting hole pattern for Mavig ceiling pedestal base

8.4.2 Pedestal Length Requirement

The pedestal-ceiling mount assembly must be installed at least 2134 mm (84 in.) from the lowest point to the finished floor.

8.4.3 Electrical Box Requirement

A 152 mm x 152 mm x 102 mm (6 in. x 6 in. x 4 in.) or equivalent ceiling box is strongly recommended to be flush mounted next to the ceiling plate. There should be two (2) conduits exiting into the box and the box grounded to the mounted plate. The electrical box cover plate must be flush mounted to the finished ceiling and with provision to add a 102 mm x 102 mm (4 in. x 4 in.) centered GE-supplied electrical cover plate.

Note:

- Additional mounting information is available on the Mavig website. Refer to the Protegra 2 Installation manual.

Seismic information is also available on the same website.

Section 9.0 Network Connections

Broad-band is the standard network connection and is strongly recommended for all LightSpeed 7.X systems. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the console.

- A patch cable (not to exceed 3.05 m (10 ft.)) should be provided by the customer to connect the operator console to a wall box. (See Notes on [Figure 9-1](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera, and operator console.
- The run from the hospital switch to the CT wall outlet must not exceed 88 m (290 ft.). Bandwidth performance is degraded when the length reaches 91 m (300 ft.) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice-only line.

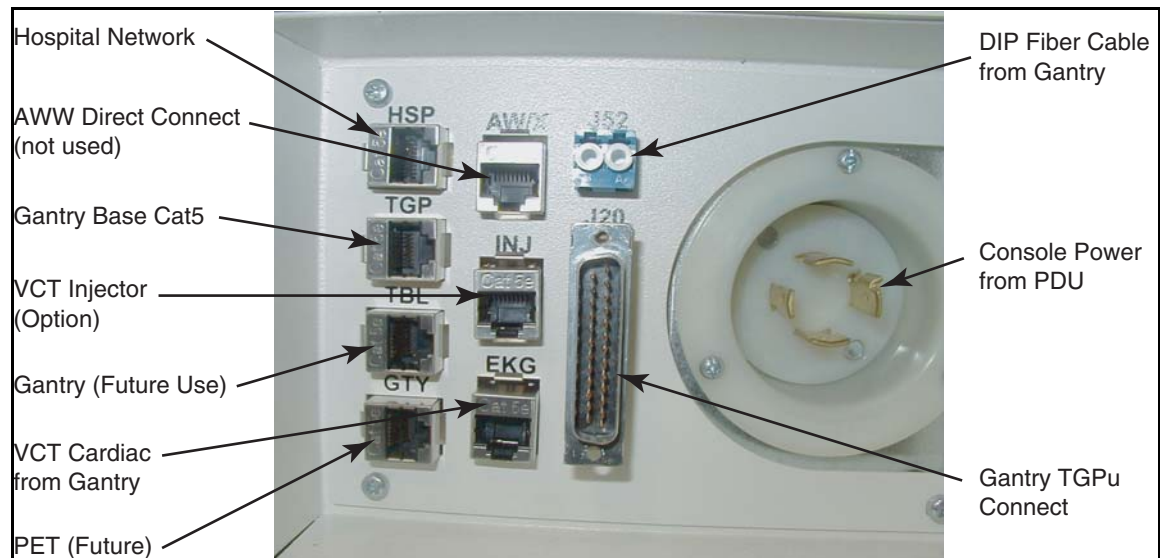


Figure 4-15 Console Rear Bulkhead

9.1 US Broad Band Process Overview

The United States network connectivity requirement for this product is broad-band. The US process relies on the GE PMI to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

9.2 Customer Broad-Band Responsibilities

Provide the GE PMI with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Coordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments.
 - Act as a focal point in assuring site broad-band infrastructure meets GE Healthcare requirements for connection as determined by a mutual assessment with the GE Healthcare Connectivity team.
- IT Contact
 - Complete an equipment assessment with the GE Healthcare Connectivity team to determine site readiness for broad-band.
- Contact your GE PMI, for the name of the zone broad-band specialist to:
 - Work with the Customer Champion to complete any identified infrastructure changes.
 - Provide an IP addresses for new CT equipment.
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption.
 - Utilize an Internet Service Provider that supports static routing.



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ **FE Name:** _____
City, State: _____ **FE Phone:** _____
Date: _____ **FE Email:** _____

- | | Yes | No |
|--|--------------------------|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |

- | | |
|--|---|
| ☐ a) Cisco Pix Firewalls | ☐ g) Symantec (Raptor) firewalls |
| ☐ b) Cisco Routers | ☐ h) Firebox |
| ☐ c) Cisco 3000 Series (Altiga acquisition) | ☐ i) Linux S/WAN |
| ☐ d) Checkpoint Firewalls Software Version 4.1 and higher | ☐ j) Sidewinder |
| ☐ e) Nortel Contivity Software Version 3.2 or higher | ☐ k) Netscreen |
| ☐ f) Redcreek | ☐ l) None |
| | ☐ m) Other _____ |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

- | | | |
|--|--------------------------|--------------------------|
| 5. Does your VPN device support "triple DES" Encryption? | ☐ | ☐ |
| 6. Has approval been given to install this VPN connection? | ☐ | ☐ |

Site Approver's Name _____

7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support	1-262-524-5261
	Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

a) Judy Heyer	judy.hey@med.ge.com
b) Judy Heyer Fax#	414-918-4707

[Use the send button on page 2](#)



MR	Software Version	CT	Software Version
Include all Signa(LX, Ovation, OpenSpeed and CV/I) MR's System Ids and IP Addresses in assessment.		HiSpeed CT/e	Minimum 6.0
		HiSpeed CT/e dual	Minimum 2.06
		HiSpeedNX/I	Minimum 5.5
		LightSpeed QX/I	Minimum 1.3
Prior to completing an InSite checkout systems must be upgraded to Version 9.0 or 9.1 unless shown in the compatible list below.		LightSpeed Plus	Minimum 2.1
		LightSpeed Ultra	Minimum 3
		HiSpeed X/I	
		CT/I	Minimum 6.2
Signa TwinSpeed	Release 9.0	Lxi, Dxi, Fxi, Zxi	Minimum 6.01
Signa	Release 9.1		
Signa	Release 10.x		
Signa CV/I	CNV4		
Signa 3T			
Signa Profile	Minimum 7.66		
Signa Contour	Minimum 7.66		
Magnet Monitor	Minimum 2.3 software		
Network Products	Software Version	Nuclear/Pet	Software Version
AW 4.0	Version 4.0 or Above	Advance	Minimum 5.1
AW 4.0P		eNTEGRA	Minimum 2.03
X-Ray	Software Version		
<u>Cardiac</u>			
INNOVA	All		
<u>Mammography</u>			
Seno2000D	All		
<u>Digital Radiology</u>			
Revolution XQ/I	Version 10.12.5 or above		
Revolution XR/d	Version 18.0		
Multi Vendor			
MR	PSI	CT	PSI
Philips ACS 1.5T	MZP 400	Picker MX-TWIN	CZM 400
Philips NT 2000	MZP 401	Picker MX 8000	CZM 401
Philips NT 3000	MZP 402	Picker PQ	CZM 500
Philips NT ACS	MZP 403	Picker PQ 2000	CZM 501
Picker Outlook 0.23T	MZM 301	Picker PQ 5000	CZM 502
Picker Edge 1.5T	MZM 800	Picker PQ 6000	CZM 503
Picker Vista 1.0T	MZM 801	Picker PQS	CZM 504
Picker Eclipse 1.5T	MZM 900	Siemens AR STAR	CZS 100
Picker Polaris 1.0T	MZM 901	Siemens AR.C	CZS 101
Siemens Harmony 1.0T	MZS 100	Siemens AR.HP	CZS 102
Siemens Impact 1.0T	MZS 200	Siemens AR.SP	CZS 103
Siemens Impact Expert 1.0T	MZS 201	Siemens AR.T	CZS 104
Siemens SP 1.0T	MZS 400	Siemens PLUS-4 EXP	CZS 400
Siemens SP 1.5T	MZS 401	Siemens PLUS-4 POW	CZS 401
Siemens SP4000 1.5T	MZS 403	Siemens PLUS-4 VOL	CZS 402
Siemens Symphony 1.5T	MZS 500		
Siemens Vision 1.5T	MZS 600		

[illegible]☐ Send Now

Section 10.0 Radiation Protection



NOTICE Scanner-room shielding requirements should be reviewed by a qualified radiological health physicist taking into consideration:

- Scatter radiation levels within the scanning room (see [Figure 4-16](#))
- At 40 mm aperture, scan times of typical LightSpeed 7.x patient exams are expected to be two or four times faster than that of LightSpeed 5.x exams with a 20 mm or 10 mm aperture, respectively
- Equipment placement
- Weekly projected work-loads (# patient/day technique (kvp*ma))
- Materials used for construction of walls, floors, ceiling, doors, and windows
- Access to surrounding scan room areas
- Equipment in surrounding scan room areas (e.g., film developer, film storage)

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[Figure 4-16](#) depicts measured radiation levels within the scanning room, while scanning a 32 cm CTDI phantom and using a large filter, with the technique shown. The mAs, kV and aperture scaling factors shown in [Table 4-10](#) can be used to adjust exposure levels to the scan technique used at the site.

For example: The exposure level for a 120 kV, 800 mA, 1 sec. scan at 50 in. away from the scan plane is: 10.4 mGy (from [Figure 4-16](#)) × 0.71 (from [Table 4-10](#)) × 800/100 (from [Table 4-10](#)) = 59.2 mGy.

Note: Actual measurements can vary. Expected deviation equals ±15%, except for the 5 mA and 1 mm techniques, where variation may be greater (up to a factor of 2), due to the inherent deviation in small values. The maximum deviation anticipated for tube output equals ±40%.

CHANGED PARAMETER	MULTIPLICATION FACTOR
mAs	new mAs/100
80 kV	0.24
100 kV	0.45
120 kV	0.71
140 kV	1.00
1mm aperture	0.20
3 mm aperture	0.22
5 mm aperture	0.27
10 mm aperture	0.38
15 mm aperture	0.48
20 mm aperture	0.59
30 mm aperture	0.79
40 mm aperture	1.00

Table 4-10 Shielding Requirements Scaling



NOTICE The units of measure used for radiation levels have been changed in this publication, from mR (millirads) to µGy (micrograys). The conversion factor is: 1 mR = 8.69 µGy

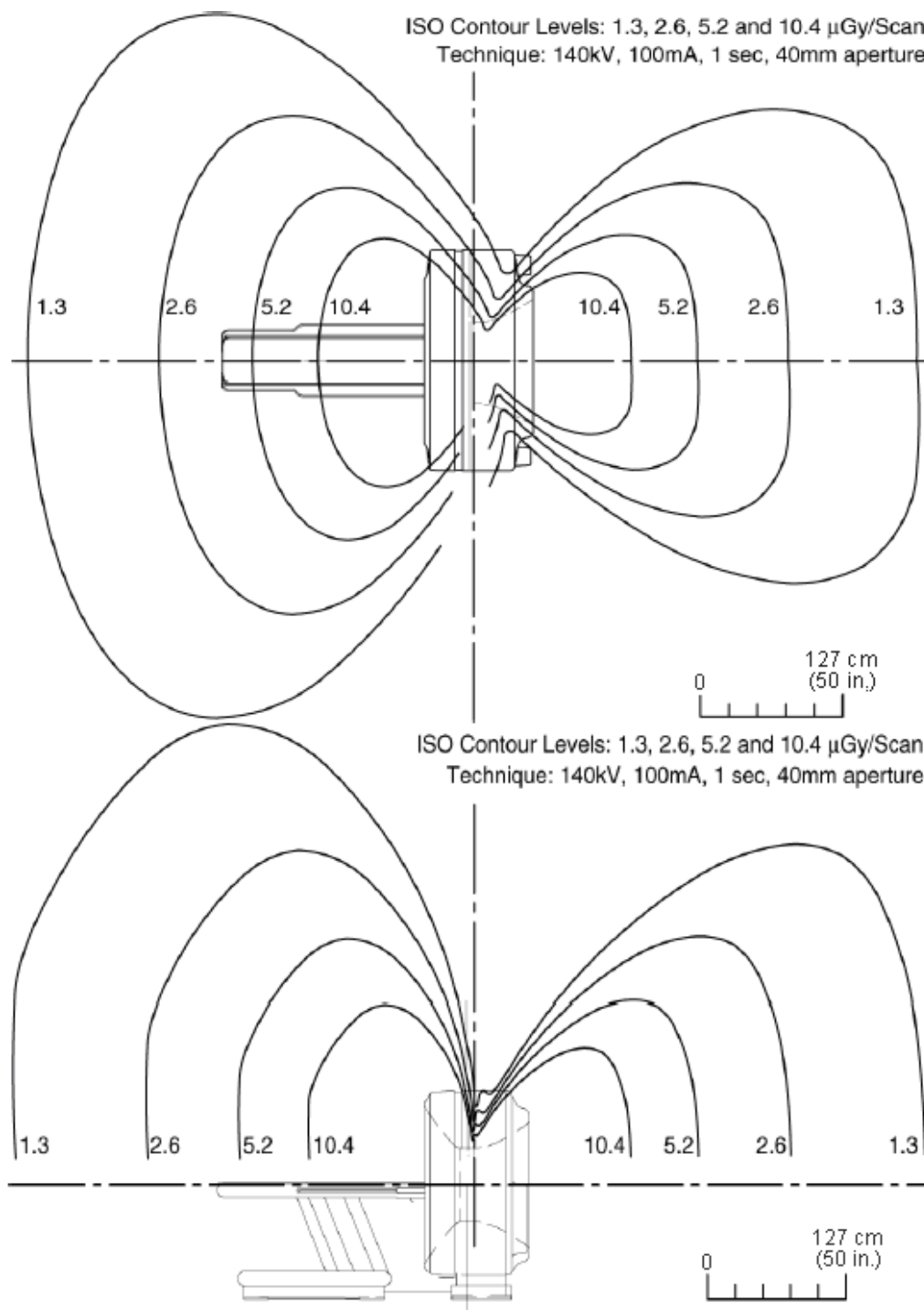


Figure 4-16 Typical Scatter Survey (Large Filter)

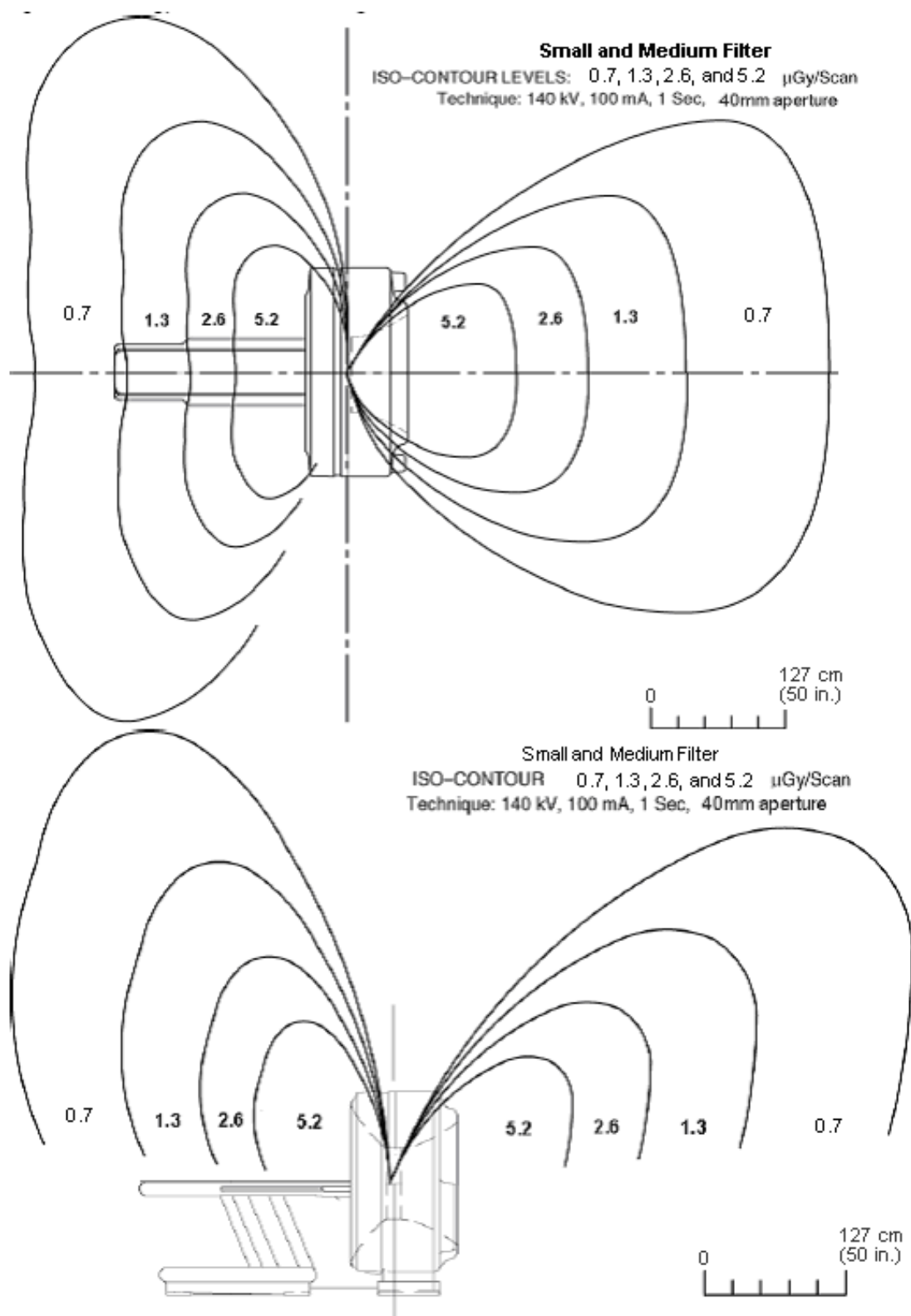


Figure 4-17 Typical Scatter Survey (Small and Medium Filter)

Chapter 5

Environmental Conditions

Ratings and duty cycles of CT subsystems apply if site environment meets the standards of this section. Maintain environmental conditions listed below at all times – including, for example, overnight, weekends and holidays. Shut down the CT system if air conditioning is not working. When system is shut down for major repair, air conditioning may be shut down also.

Section 1.0

Temperature and Humidity Specifications

Environmental specifications apply to the table, gantry, Power Distribution Unit, and operator console.



NOTICE

System operation and image quality may be affected, if environmental specifications are exceeded.

1.1 Temperature (Scan Room & Control Room)

Maximum allowable ambient room temperature:	26° C (79° F)
Recommended ambient room temperature:	22° C (72° F)
Minimum allowable ambient room temperature:	18° C (64° F)
Maximum allowable ambient room temperature rate of change: 3° C per hour (5° F per hour)	

Note: Any cooling equipment cycle control range must be taken into account, such that the maximum and minimum ambient room temperatures shown above are not exceeded during room thermal cycling. For example, if the HVAC is capable of ± 2° C control, then the limits would 20° C - 24° C to maintain absolute limits.

1.2 Humidity (Scan Room & Control Room)

Maximum allowable non-condensing relative humidity:	60%
Minimum allowable non-condensing relative humidity:	30%
Maximum allowable relative humidity rate of change:	5% per hour

1.3 Other Guidelines

- To help determine the hospital room environmental conditions, a temperature and humidity recorder may be temporarily installed close to where the system will be installed. Note readings before and after installation to verify the true temperature and humidity in your environment.
- Consider heating, ventilating, air conditioning (HVAC) needs, and redundancy. An air conditioner with two compressor units, rather than one, may prevent system downtime. A back-up (redundant) air conditioner permits CT system operation during an extended repair of the primary air conditioner.

Section 2.0 Cooling Requirement

Use [Table 5-1](#) to assist in cooling requirements planning. Over half the cooling used by your scanner is needed for gantry operation.

- Gantry air intake is located across the bottom of the gantry.
- Gantry air exhaust is located across the top of the gantry.
- Contact an HVAC specialist to determine optimal placement of all HVAC vents and thermostat.

SYSTEM COMPONENT	BTU/HR	WATT
1. Gantry recommended (See NOTE 1)	43,000	12,600
2. Table	700	200
3. Power Distribution Unit	3400	1000
Recommended Scan Room Subtotal (see notes):	47,100	13,800
4. Operator console/computer with three IGs	7210	2117
LCD Monitor, each (x2)	170 (340)	50 (100)
SCSI Tower	425	125
Injector	425	125
DASM	425	125
Recommended Control Room Subtotal (w/ 3 IGs)	8,825	2,592
System Total (Recommended) - w/ 3 IGs (See NOTE 2)	55,925	16,392
Option: Remote Color Monitor (LCD)	170	50
Option: UPS	2900	850
ROOM TOTAL (SEE NOTE 2)		

NOTE 1: With 75 scan rotations per patient:

Recommended BTU/hr. provides for up to six patients per hour. It is also needed during calibration of the system.

NOTE 2: Cooling requirements do not include cooling for room lighting, personnel, or non-CT equipment.

Table 5-1 Cooling Requirements (Worksheet)

Please refer to [Figure 4-8](#), [Figure 4-11](#), and [Figure 4-12](#) in [Chapter 4, Section 6.0](#), for component air flow requirements.

Recommended placements of HVAC vents (intake and output) and thermostat are shown for the scan room (Figure 5-1) and the control room (Figure 5-2).

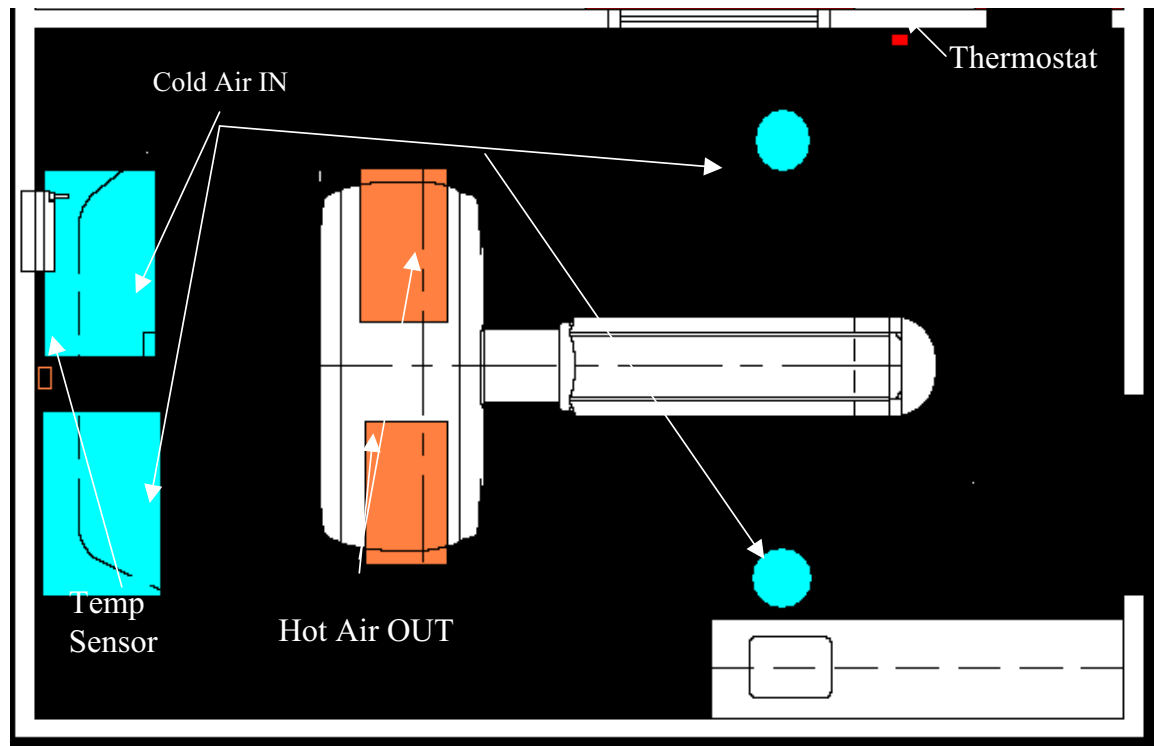


Figure 5-1 HVAC Air Vent Placement - Scan Room

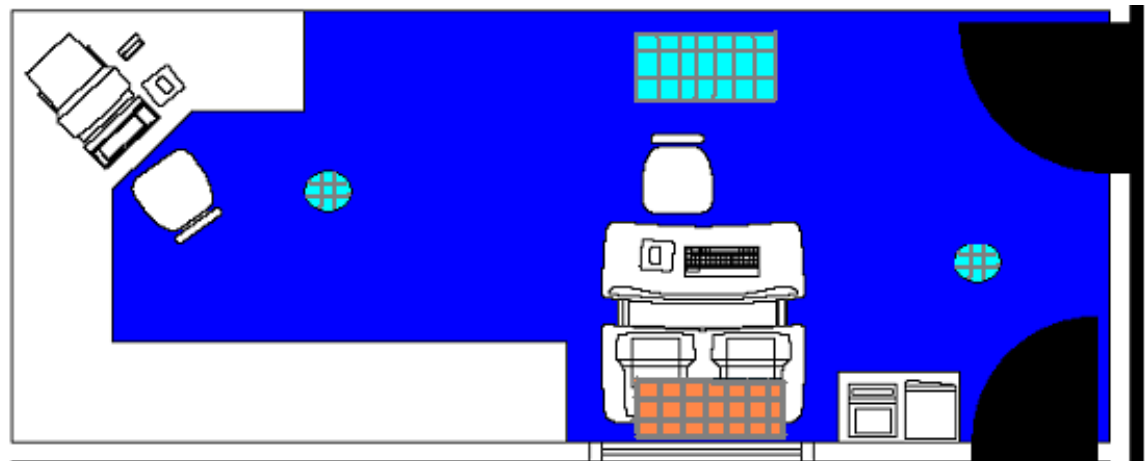


Figure 5-2 HVAC Air Vent Placement - Control Room

Section 3.0 Altitude

System operating altitude is from mean sea level to 3050 m (10,000 ft.).

Section 4.0

Electro-Magnetic Interference (EMI)

4.1 Gantry

Locate the gantry in ambient static magnetic fields of less than 10^{-4} tesla (1,000 milligauss) to guarantee specified imaging performance. Ambient AC magnetic fields must be below 10^{-6} tesla (10 milligauss) peak.

4.2 Color Monitor

Locate color monitors in ambient static magnetic fields of less than 5×10^{-5} tesla (100 milligauss) to guarantee color purity and display geometry. See [Figure 5-3](#).

4.3 Console / Computer Equipment

Locate computer equipment in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss) to guarantee data integrity. See [Figure 5-3](#).

4.4 Magnetic Media

Locate magnetic media in ambient static magnetic fields of less than 10^{-3} tesla (10,000 milligauss).

4.5 PDU

The PDU produces an electromagnetic field that radiates outward from its cabinet in all directions. Do not place sensitive electronics (e.g., operator console or computer equipment - the UPS is not classified as sensitive electronics) within one meter (1 m) (3.28 ft.) of the Power Distribution Unit in any direction (including above or below). See [Figure 5-3](#).

4.6 EMI Reduction

If fields of excessive EMI are known or suspected to be present, consult GE Healthcare Sales & Service for recommendations. Consider the following if you attempt to reduce EMI:

- External field strength decreases rapidly with distance from source of the magnetic field.
- External leakage magnetic field of a three-phase transformer is much less than that of a bank of three single-phase transformers of equivalent power rating.
- Large electric motors are a source of substantial EMI.
- High-powered radio signals are a source of EMI.

Maintain good screening of cables and cabinets.

4.7 UPS

The Uninterruptable Power Supply (UPS) provides a consistent power supply to various electrical components of the system. Also, it continues to provide electrical power to components during a site-wide power outage so components can be safely shut down. The UPS should be kept at least one meter (1 m) (3.28 ft.) away from sensitive electronics (the PDU does not include sensitive electronics).

For UPS interconnect information, please refer to [Section 5.0 on page 117](#).

4.8 Equipment EMI “Envelopes”

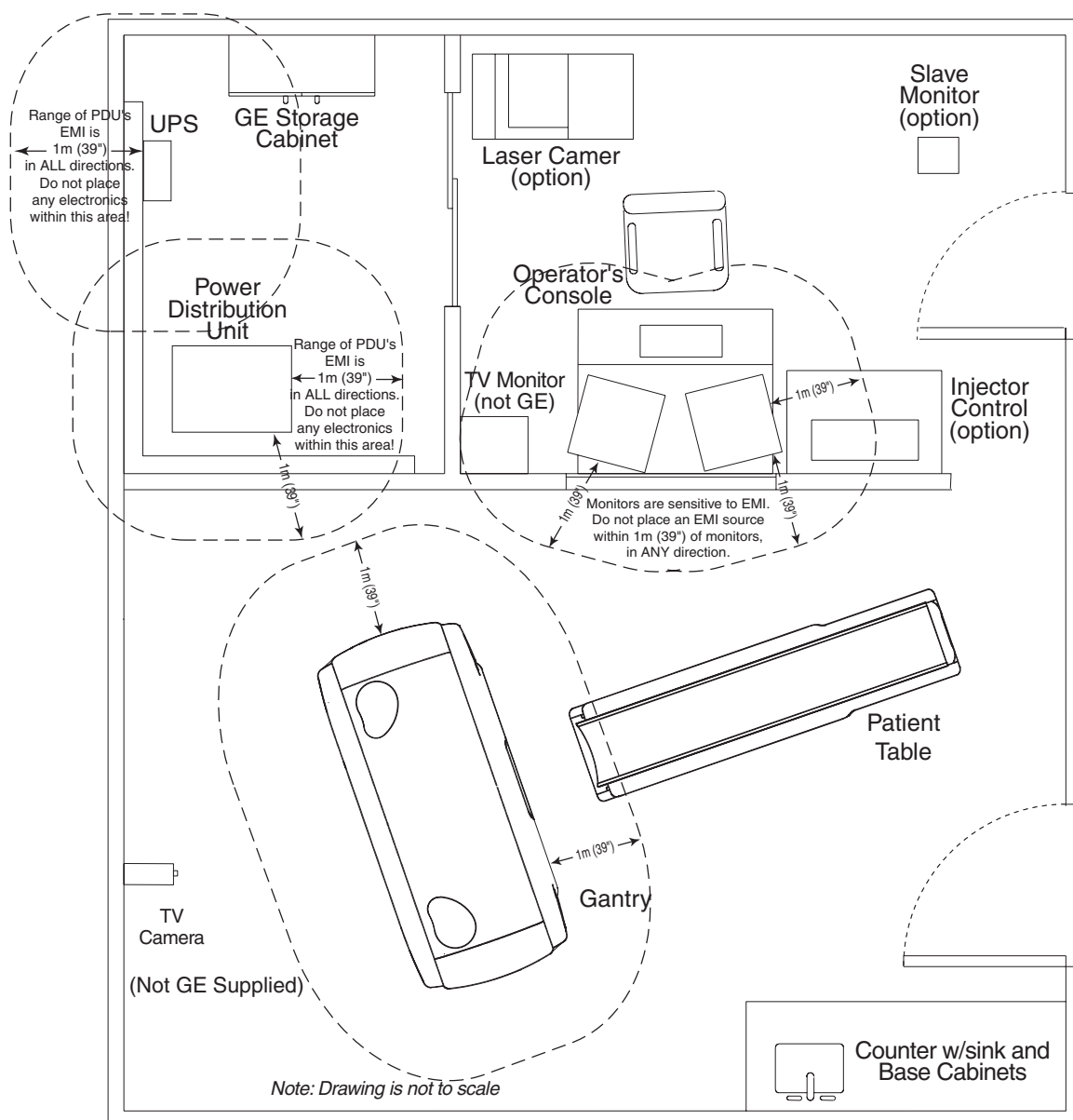


Figure 5-3 Sample Room Layout, showing approximate EMI requirements

4.9 EMC Compliance

4.9.1 General Scope

This equipment complies with IEC60601-1-2 Edition 2 (2001) EMC standard for medical devices.

The LightSpeed 7.X System is suitable to be used in the electromagnetic environment, as per the limits and recommendations described in the tables hereafter:

- Emission Compliance level and limits ([Table 5-2](#))
- Immunity Compliance level and recommendations to maintain equipment clinical utility ([Table 5-3](#) and [Table 5-4](#))

Note: This system complies with above-mentioned EMC standard when used with supplied cables. If different cable lengths are needed, contact a qualified GE service representative for advice.

4.10 Electromagnetic Emission

The LightSpeed 7.X is intended for use in the electromagnetic environment specified below. The customer or the user of the LightSpeed 7.X should assure that it is used in such an environment

EMC Emissions Guidance and Declaration for LightSpeed 7.X		
Emissions Test	Compliance	Electromagnetic Environment Guidance
RF emissions CISPR 11	Group 1	The LightSpeed 7.X uses RF energy only for its internal function. Therefore, its RF emissions are very low and are not likely to cause any interference in nearby electronic equipment.
RF emissions CISPR 11	Class A + 12	WARNING: This equipment is allowed to be installed only in X-ray protected rooms, which provide an attenuation of at least 12 dB for radio disturbances from 20 MHz to 1 GHz.
Harmonic emissions IEC 61000-3-2	Not Applicable	When installed in such a shielded location, the LightSpeed 7.X is suitable for use in all establishments other than domestic and those directly connected to the public low-voltage power supply network that supplies buildings used for domestic purposes.
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Not Applicable	

Table 5-2 Electromagnetic Emissions

4.11 Electromagnetic Immunity

The LightSpeed 7.X is intended for use in the electromagnetic environment specified below. The customer or the user of the LightSpeed 7.X should assure that it is used in such an environment

EMC Emissions Guidance and Declaration for LightSpeed 7.X			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Electrostatic discharge (ESD) IEC 61000-4-2	± 6 kV contact ± 8 kV air	± 6 kV contact ± 8 kV air	Floors should be wood, concrete, or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrical fast transient/burst IEC 61000-4-4	± 2 kV for power supply lines ± 1 kV for input/output lines	± 2 kV for power supply lines ± 1 kV for input/output lines	Mains power quality should be that of a typical commercial or hospital environment
Surge IEC 61000-4-5	± 1 kV differential mode ± 2 kV common mode	± 1 kV differential mode ± 2 kV common mode	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	< 5% U_T (> 95% dip in U_T) for 5 sec	< 5% U_T (> 95% dip in U_T) for 5 sec	Mains power quality should be that of a typical commercial or hospital environment. If the user of the LightSpeed 7.X requires continued operation during power mains interruptions, it is recommended that the LightSpeed 7.X be powered from an uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	3 A/m	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.

Table 5-3 Electromagnetic Immunity

EMC Emissions Guidance and Declaration for LightSpeed 7.X			
Immunity Test	IEC 60601-1-2 Test Level	Compliance Level	Electromagnetic Environment Guidance
Conducted RF IEC 61000-4-6 Radiated RF IEC 61000-4-3 (Alternative method: Full range IEC 61000-4-21 test in lieu of Large, Permanently-Installed Equipment exemption)	3 V_{RMS} 150 kHz to 80 MHz 3 V/m 150 kHz to 80 MHz	3 V 150 kHz to 80 MHz 3 V/m 150 kHz to 80 MHz	Portable and mobile RF communications equipment should be used no closer to any part of the LightSpeed 7.X, including cables, than the recommended separation distance calculated from the equation appropriate for the frequency of the transmitter. Recommended Separation Distance: $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ (See Table 5-4.) $d = \left[\frac{3.5}{3} \right] \sqrt{P}$ 80 MHz to 800 MHz (See Table 5-4.) $d = \left[\frac{7}{3} \right] \sqrt{P}$ 800 MHz to 2,5 GHz (See Table 5-4.) where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site survey,^a should be less than the compliance level in each frequency range.^b Interference may occur in the vicinity of equipment marked with the following symbol: 

- a.) Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast, and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the LightSpeed 7.X is used exceeds the applicable RF compliance level above, the LightSpeed 7.X should be observed to verify normal operation. If abnormal performance is observed, additional measures may be necessary, such as re-orienting or relocating the LightSpeed 7.X.
- b.) Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.

Note: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects, and people

Note: U_T is the AC mains voltage prior to application of the test level.

Table 5-3 Electromagnetic Immunity

The LightSpeed 7.X is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the LightSpeed 7.X can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the LightSpeed 7.X as recommended below, according to the maximum output power of the communications equipment

Recommended Separation Distances Between Portable and Mobile RF Communications Equipment and the LightSpeed 7.X			
Rated Maximum Output Power (P) of Transmitter Watts (W)	Separation Distance (Meters) by Frequency of Transmitter		
	150 kHz to 80 MHz $d = \left[\frac{3.5}{3}\right]\sqrt{P}$	80 MHz to 800 MHz $d = \left[\frac{3.5}{3}\right]\sqrt{P}$	800 MHz to 2.5 GHz $d = \left[\frac{7}{3}\right]\sqrt{P}$
0.01	0.12	0.12	0.23
0.1	0.37	0.37	0.74
1	1.17	1.17	2.33
10	3.69	3.69	7.38
100	11.7	11.7	23.3

For transmitters rated at a maximum output power not listed above, the separation distance can be estimated using the equation in the corresponding column, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.

NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.

NOTE 2: These guidelines may not apply in all situations. electromagnetic propagation is affected by absorption and reflection from structures, objects, and people.

Table 5-4 Recommended Separation Distances

As an example, a 1W mobile phone (800 MHz to 2.5 GHz carrier frequency) shall be separated by at least 2.3 meters from the LightSpeed 7.X (in order to avoid image interference risks.)

LIMITATIONS MANAGEMENT:

Adhering to the distance separation recommended in [Table 5-4](#), between 150 kHz and 2.5 GHz, reduces disturbances recorded at the image level, but may not eliminate all disturbances. However, when installed and operated as specified herein, the system maintains its essential performance by continuing to acquire, display, and store diagnostic quality images safely.

4.12 External Component Use Limitations

The use of accessories, transducers, and cables other than those specified below or supplied with the system may result in degraded ELECTROMAGNETIC COMPATIBILITY of the LightSpeed 7.X system. For additional compatible accessories, consult the *GE Healthcare* Sales and Service representative.

- UPS (P5052PS)
- Enhanced PET Recon Option (P5091RC)
- Cardiac Gating Option Kit for Discovery VCT (P5064PJ)
- Respiratory Option Kit for Discovery VCT (P5064PH)
- SmartStep (P5050SM, P5050SN)
- Laser Camera Interface (B7700L)
- Gammex RMI Laser Tracking System (E8505LF, E8505LK)

4.13 Installation Requirements and Environment Control

In order to minimize interference risks, the following requirements shall apply.

4.13.1 Cable Shielding and Grounding

All interconnect cables to peripheral devices must be shielded and properly grounded. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

The manufacturer is not responsible for any interference caused by using other than recommended interconnect cables or panels, or by unauthorized changes or modifications to this equipment. Unauthorized changes or modifications could void the users' authority to operate the equipment.

All interconnect cables-to-peripheral devices must be shielded and properly grounded, except when technologically prohibited. Use of cables not properly shielded and grounded may result in the equipment causing radio frequency interference.

4.13.2 Environment Control

This product complies with the radiated emission as per CISPR11 Group1 Class A + 12 limits. The LightSpeed 7.X system is predominantly intended for use in non-domestic environments and not directly connected to the Public Mains Network. The system is predominantly intended for use (e.g., in hospitals) with a dedicated supply system, and with an X-ray shielded room. In case of use in a domestic environment (e.g., doctors' offices), in order to avoid interferences, it is recommended to use a separate AC power distribution panel and line, with an X-ray shielded room.

This equipment generates, uses, and can radiate radio frequency energy. The equipment may cause radio frequency interference to other medical and non-medical devices and radio communications. To provide reasonable protection against such interference, this product complies with the radiated emission as per CISPR11 Group1 Class A + 12 limits. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment is found to cause interference (which may be determined by turning the equipment on and off), the user (or qualified service personnel) should attempt to correct the problem by one or more of the following measure(s):

- Reorient or relocate the affected device(s).
- Increase the separation between the equipment and the affected device.
- Power the equipment from a source different from that of the affected device.
- Consult *GE Healthcare* Sales and Service for further suggestions.

4.13.3 Power Supply Distribution for Accessories and Subsystems

All components, accessories, subsystems and systems that are electrically connected to the LightSpeed 7.X must have all AC power supplied by the same power distribution panel and line.

4.13.4 Stacked Components and Equipment

The components/sub-systems of the LightSpeed 7.X should not be used adjacent to, or stacked with, other equipment; if adjacent or stacked use is necessary, observe the LightSpeed 7.X system to verify normal operation in the configuration in which it will be used.

4.13.5 Low Frequency Magnetic Field

To minimize the risk of interference from low-frequency magnetic field sources, any such sources (such as CRT monitors) must be installed at least 1 meter from the LightSpeed 7.X gantry.

4.13.6 Static Magnetic Field Limits

In order to avoid interference on the LightSpeed 7.X system, static field limits from the surrounding environment are specified:

- Less than 1 gauss in examination room, and in the control room.
- Less than 3 gauss in the equipment/technical room.

4.13.7 Electrostatic Discharge Environment and Recommendations

In order to reduce electrostatic discharge interference, install a charge dissipative floor material to avoid electrostatic charge buildup. The dissipative material shall be connected to the system ground reference, if applicable. The relative humidity shall be at least 30%.

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Chapter 6

Floor Loading and Weights

Section 1.0 Floor Loading

LightSpeed 7.X systems have a total floor load of approximately 3102 kg (6839 lbs). About 2521 kg (5557 lbs), including patient, is concentrated in the table-gantry assembly.

Table 6-1 lists CT components with weight, size, floor loading, and normal mounting requirements.

ITEM	NET WEIGHT LB(KG)	OVERALL W X D INCH (MM)	MAX ANCHOR LOAD LB (N)	WEIGHT/ AREA LB/SQ. FT. (KG/M ²)	LOAD PATTERN / IN. (MM)	NORMAL METHOD OF MOUNTING IN. (MM) (GE SUPPLIED) ¹
Gantry (w/covers)	~4050 (~1850)	89.25 X 39.65 (2267 X 1007)	compression 1100 (4895) (assume infinite floor stiffness)	296 (1448)	Four round 64 mm pads in a rectangular pattern. Floor Print: 27.6 x 79.25 (700 x 2013)	Hilti Kwik-Bolt II 1/2in (13 mm) diameter by 8in (203mm) long per P/N 2106573 at four leveling pads into concrete floor.
Dollies (each)	250 (114)					
Top Cover (each)	24.6 (11)					
Side Cover (each)	25 (11)					
Front Cover	95 (43)					
Rear Cover	100 (45)					
Patient Table:						
GT 1700 w/ 225kg patient	880 (400) 1375 (625)	25.6 x 95.9 (650 x 2436)	- 605 (2690)	- 238 (1161)	Four round 64 mm pads: 19.7 x 40.3	Hilti Kwik-Bolt II 1/2in (13 mm) diameter by 8in (203mm) long per P/N 2106573 at the leveling pads into concrete floor.
GT 2000 w/ 227kg patient	1250 (568) 1750 (795)	25.6 x 114.5 (650 x 2910)	- 591 (2630)	- 273 (1337)	Four round 64 mm pads: 19.7 x 40.3	
Footswitch (GT)	33 (15)	-	-	-	-	
Power Distribution Unit	~700 (~336)	28 X 22 (711 X 559)		161 (788)	Four Casters support area of 28 X 22 (711 X 559).	Casters are for positioning and service. Set on floor. May be anchored to floor using angle brackets ² in seismic zones.
Console w/HP & w/o monitors	450 (204)	48 X 39 (1219x991)		140 (681)	Four Casters or Leveling Feet support area of 46 X 19 (1168 X 483).	Casters are for positioning. Set on floor. Console may be anchored to floor using angle brackets ³
Monitor - LCD (ea)	22 (10)					

Notes:

- 1.) Use the GE Supplied mounting hardware ONLY IF APPROVED by qualified personnel.
[See statements in [8.1 - Table and Gantry Mounting Requirements](#), on [page 63](#).]
- 2.) Seismic brackets are available in the OPTIONAL Seismic Kit (R4390JC)
- 3.) Angle brackets are included on the shipping skid, and are also available in the OPTIONAL Seismic Kit (R4390JC).

Table 6-1 LightSpeed 7.X System Floor Loading

Section 2.0

Mounting Data, Including Seismic

 **WARNING** **POTENTIAL FOR PATIENT INJURY.**
IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.
PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

The following pages show center-of-gravity information for system components:

- Gantry: [Figure 6-2](#)
- Table: [Figure 6-5](#)
- Power Distribution Unit: [Figure 6-6](#)
- Operator's Console/Computer: [Figure 6-8](#)

Floor mounting hole locations for components that don't have templates are also in this section.

An optional Seismic Kit (R4390JC) is available, for facilities in areas that require one.

The customer is responsible for seismic mounting. Refer to all applicable codes in your area.

GE-provided floor anchors ([Figure 6-1](#)) are designed to be used ONLY on concrete floors that meet the concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 76 mm (3 in.) at each anchor point. Any anchors having more than 25 mm (1 in.) of thread showing above the nut, when torqued to 75 N-m (55 ft.-lbs), shall have a second anchor installed in the closest adjacent hole. the second anchor shall be installed to the standard depth and torque specifications.

MOUNTING REQUIREMENTS	GANTRY	TABLE
Minimum Floor Thickness:	102 mm (4 in.)	102 mm (4 in.)
Recommended Drilling Depth:	95 mm (3.75 in.)	95 mm (3.75 in.)
Average Anchor Embedment:	89 mm (3.5 in.)	89 mm (3.5 in.)
Minimum Anchor Embedment:	76 mm (3 in.)	76 mm (3 in.)
Available Alternate Anchor Locations:	Yes	Yes
Shipped Anchor Size:	203 mm (8 in.)	203 mm (8 in.)
Alternate Anchoring Methods:	Yes (see notes, below)	Yes (see notes, below)
Floor Levelness Requirement:	8 mm (0.3125 in.) over 3 m (10 ft.)	8 mm (0.3125 in.) over 3 m (10 ft.)

Table 6-2 Gantry and Table Mounting Requirements

Note: If the Installers cannot set all four anchors for the GT1700 or GT2000 tables, the installer must inform the customer that the minimum anchoring cannot be met, and a structural engineering contractor is strongly recommended to determine the anchoring method and certify that their anchoring meets the stated GE minimum load requirement and torque specifications.

Note: All other anchoring methods on floor types other than the concrete minimum must be determined at the customer's expense by a structural engineering contractor, and anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

Note: If installing the GE LS scanner on a floor type other than a 102 mm (4 in.) concrete floor, all structural specifications in this document must be reviewed and met.

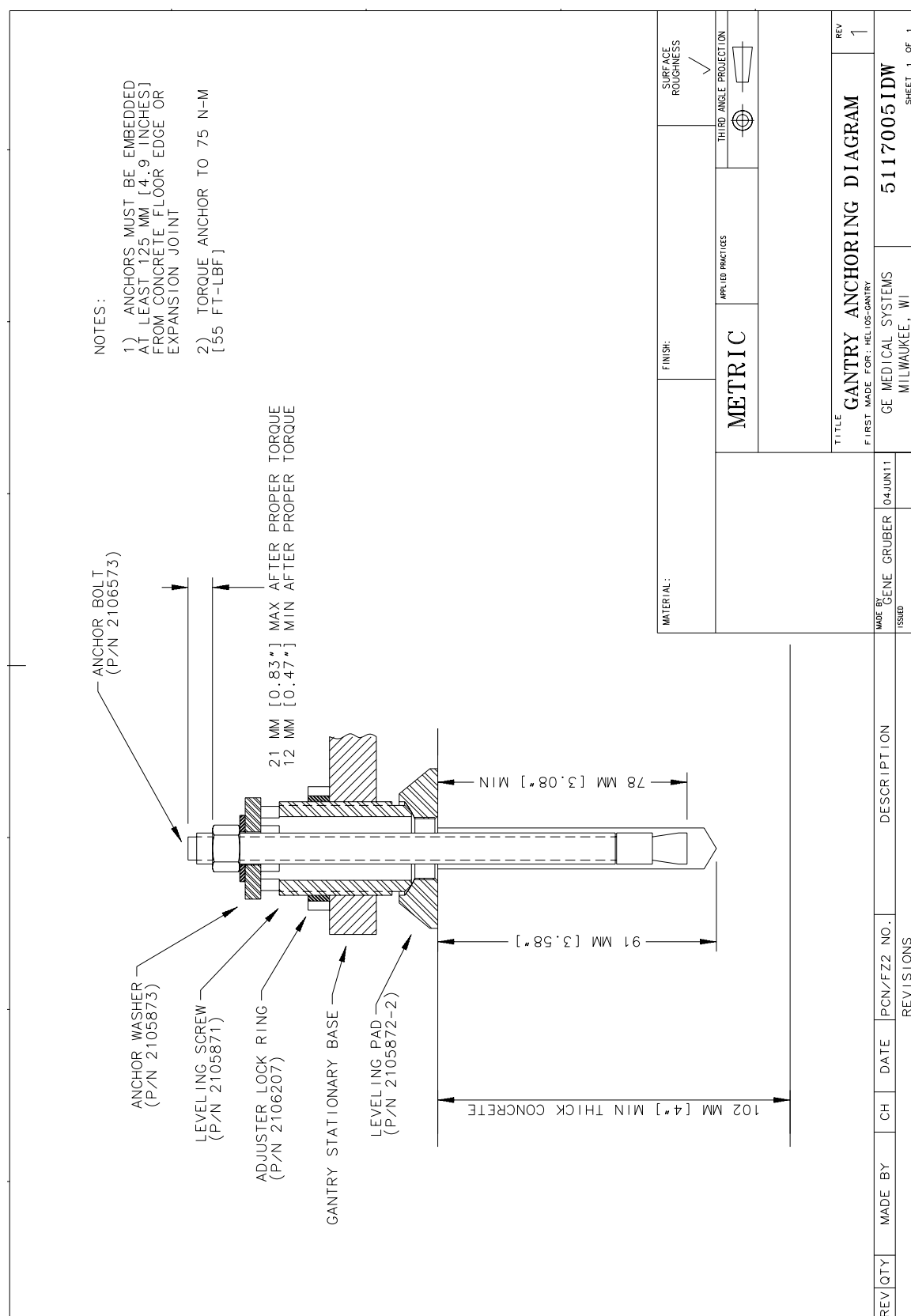


Figure 6-1 Typical Floor Anchor, Gantry and Table

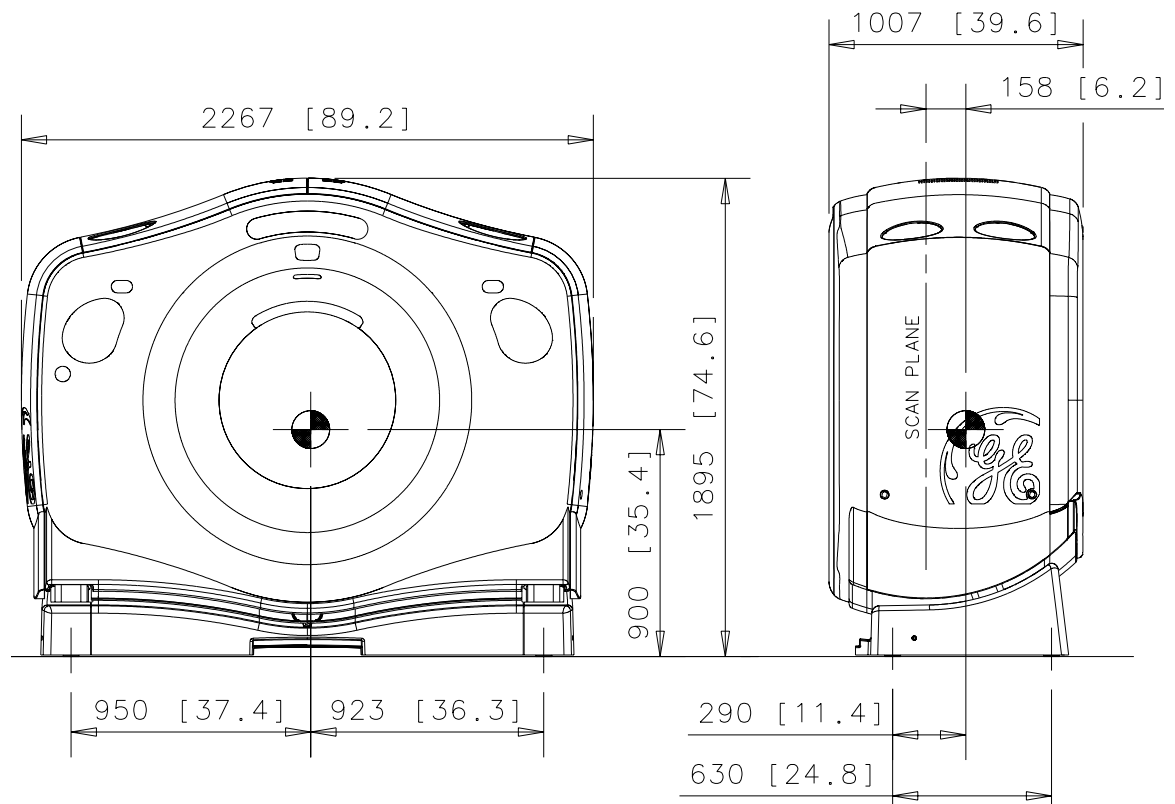


Figure 6-2 Gantry Center of Gravity

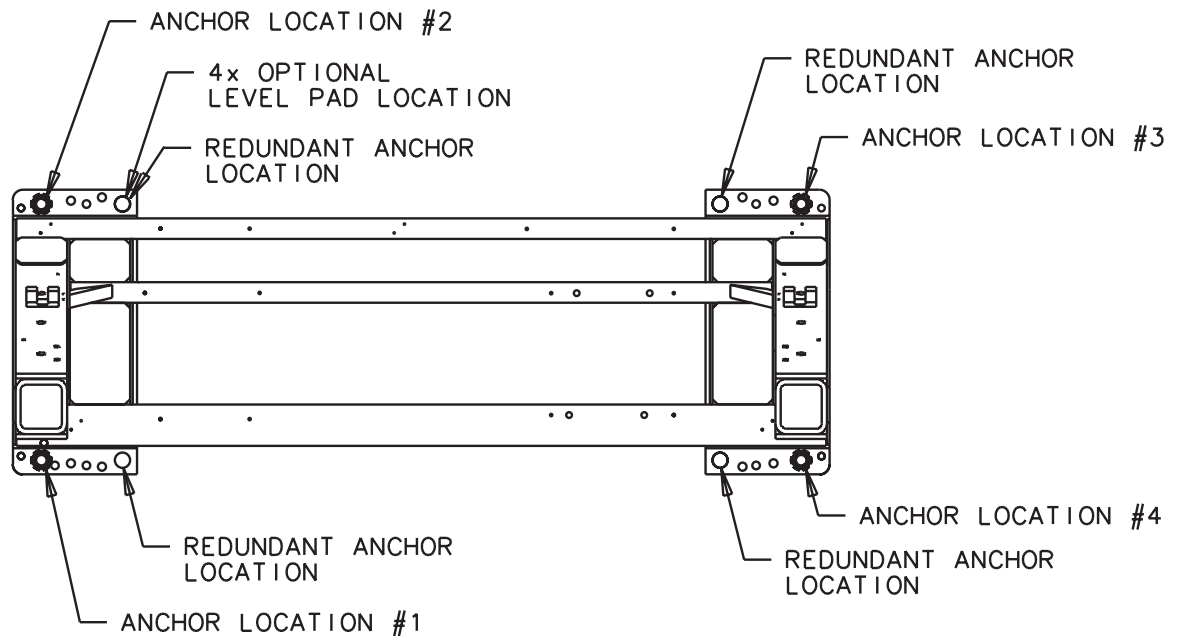


Figure 6-3 Gantry Anchor Locations



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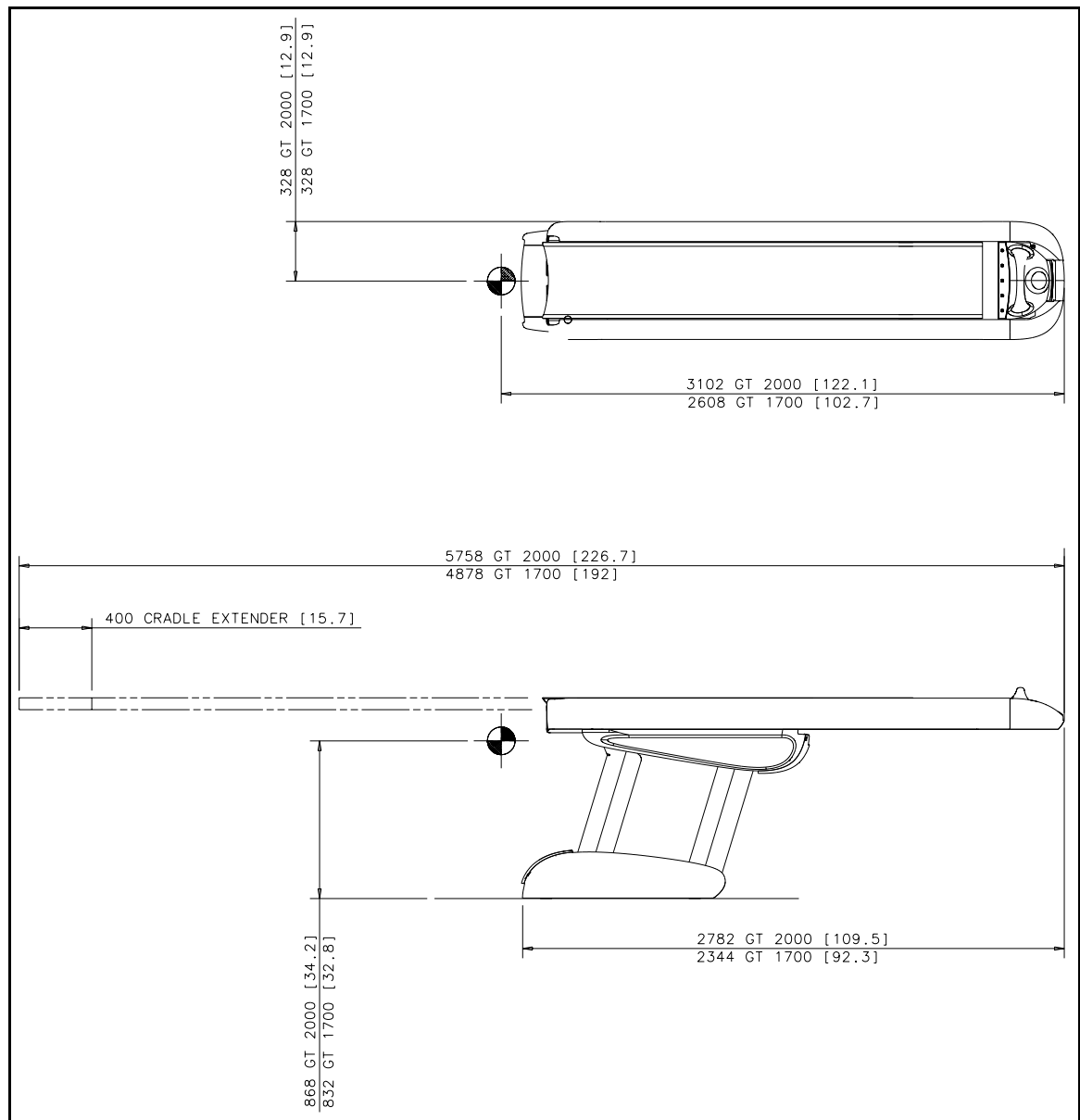


Figure 6-5 Patient Table (CT1) - GT 1700 & GT 2000 (Cradle *Mechanical* Limits shown)

Note: See [Figure 4-8](#), on [page 58](#), for cradle *operational* limits.

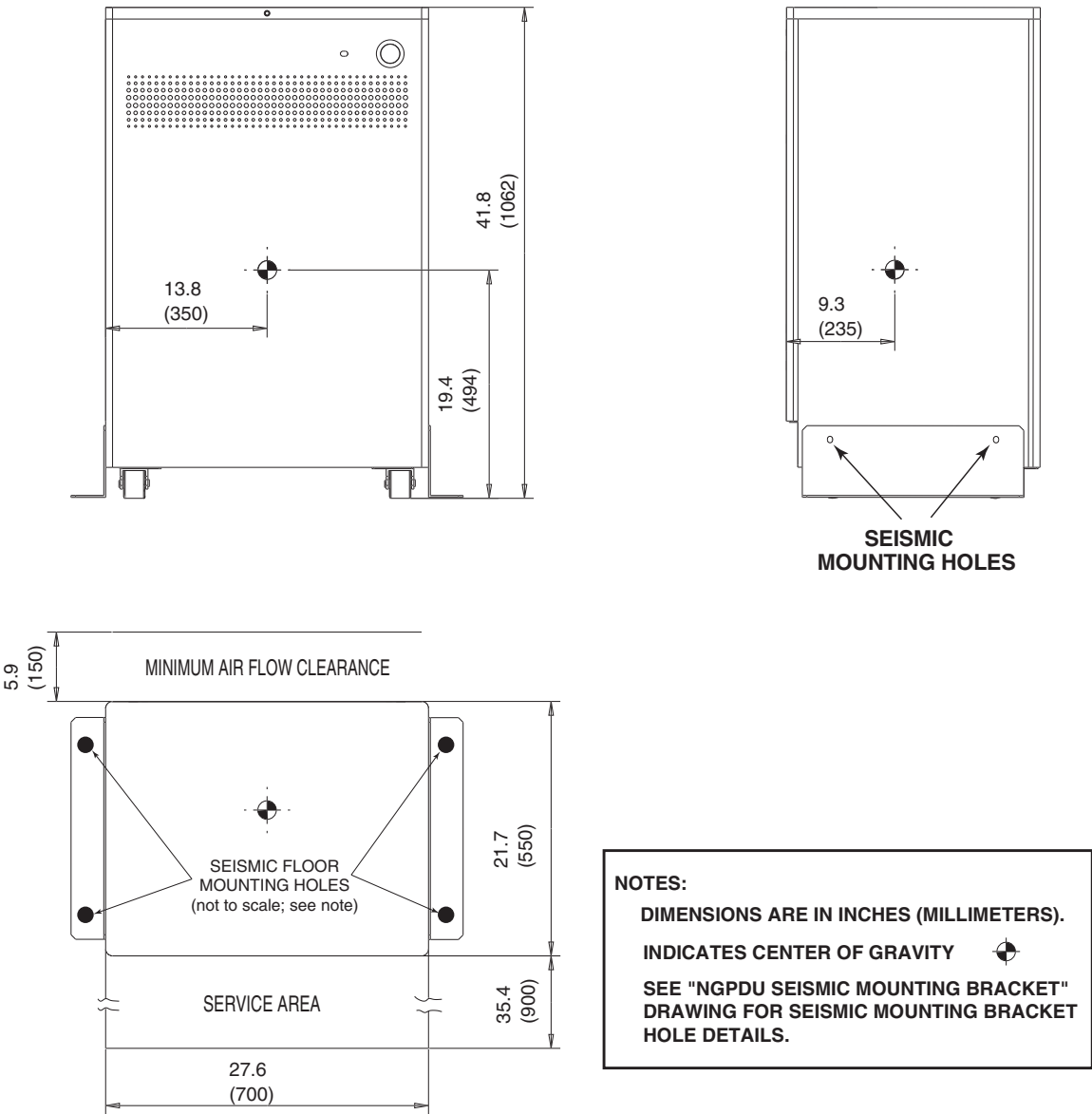


Figure 6-6 Power Distribution Unit (NGPDU-4)

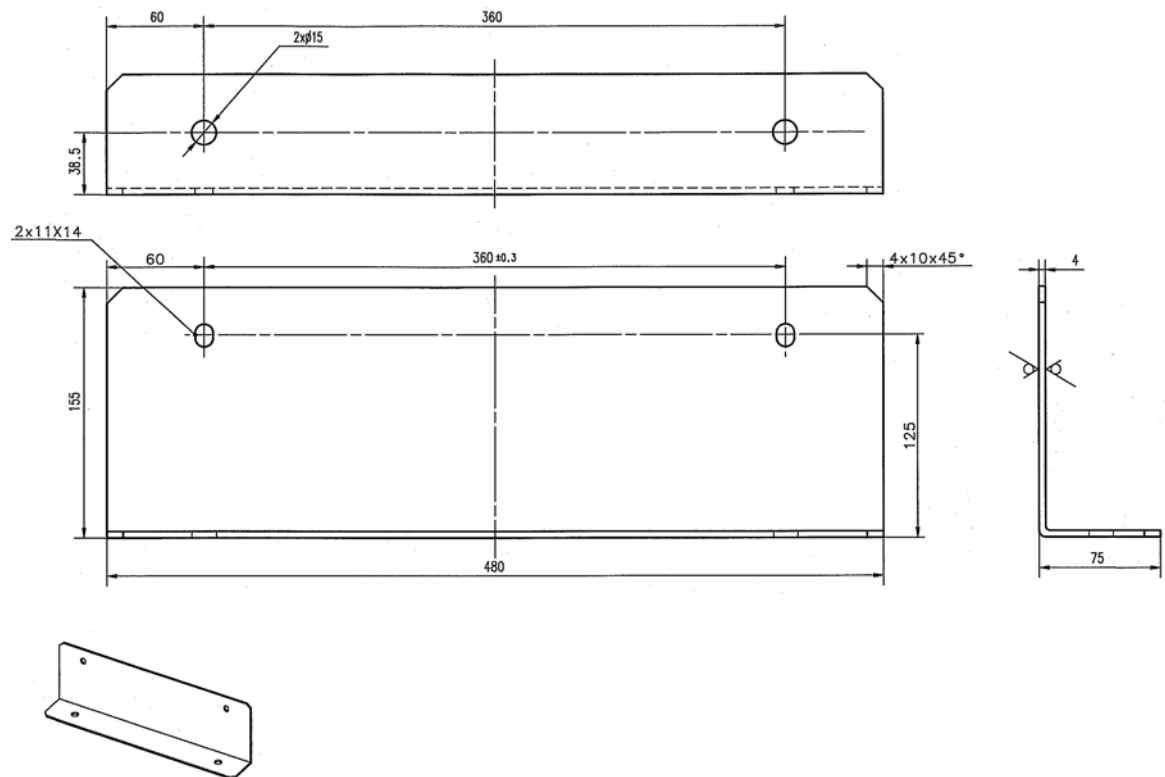


Figure 6-7 NGPDU Seismic Mounting Bracket (dimensions are in mm)

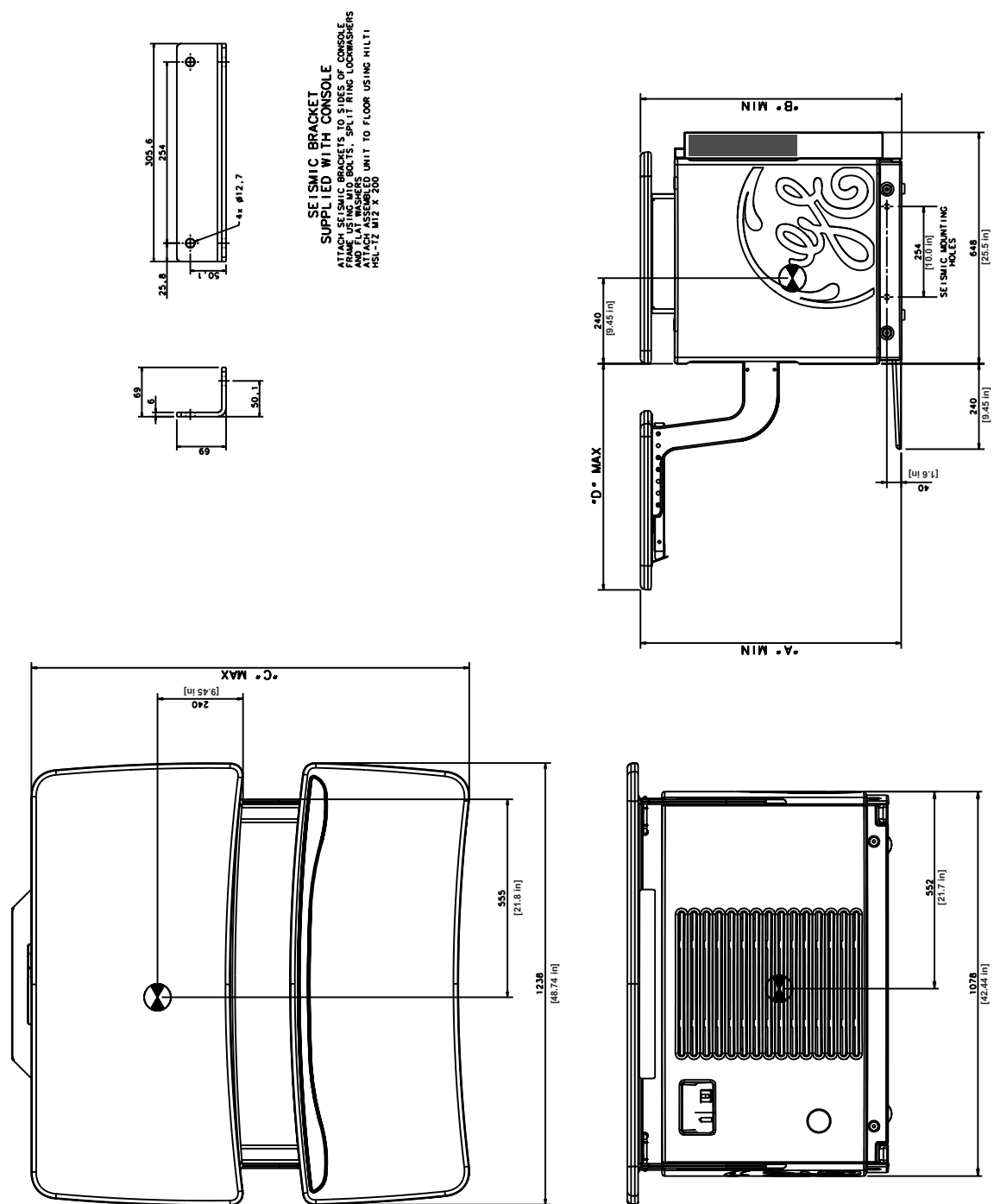


Figure 6-8 Operator Console

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Chapter 7

Delivery Data



WARNING SOME ASSEMBLIES ARE TOP-HEAVY. BE CAREFUL NOT TO TIP.

Section 1.0

Van Delivery

The CT system is packed for van shipment with minimum tear-down of components. It consists of approximately 20 shipping containers, which include dollies, skids, and boxes without skids.

Section 2.0

Delivery/Shipping Requirements & Considerations

The LightSpeed 7.X System is not designed to tolerate excessive mishandling, including dropping, shock, vibration, tipping, or hoisting.

The gantry, console, table and PDU must NEVER be dropped. A drop from a height greater than 13 mm (0.5 in.) may induce structural damage to the frame or other major components. Damage resulting from a drop (e.g., bent frame, misalignment) may not be obvious until after system installation is complete.

To avoid dropping the gantry, dock-to-dock shipment is recommended. Other methods are acceptable, provided that the system is not dropped or otherwise mishandled. For example, the system may be moved via flat-bed wrecker or by rolling it across SMOOTH sidewalks or other paved surfaces.

When moving the gantry off of a flat-bed wrecker, attach the straps to the lowest point possible on the dolly. Lower the gantry at the slowest reasonable rate. See [Figure 7-1](#).

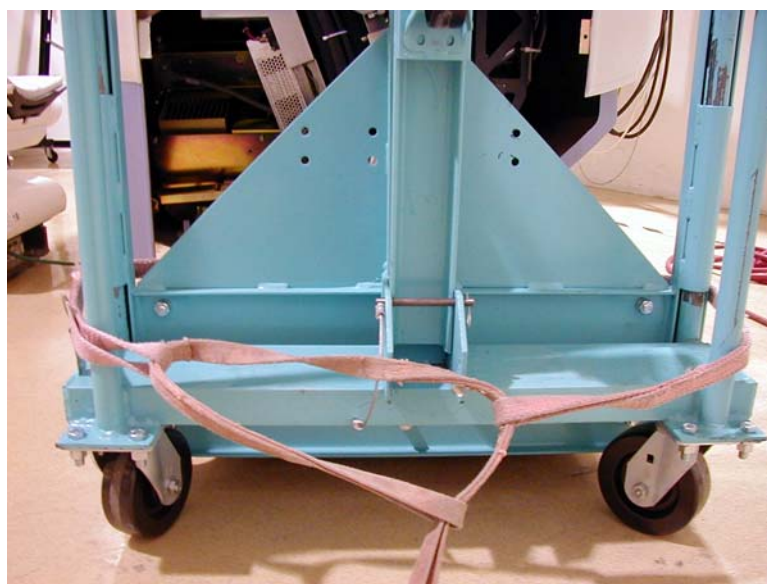


Figure 7-1 Gantry Strap Location

The LightSpeed 7.X System, including the gantry, console, table and PDU, is not designed to tolerate the excessive shock or vibration that may occur during unloading. For example, rolling the console across a “washboard” style ramp may vibrate components to the extent of loosened or broken connections, etc. Damage resulting from shock or vibration (e.g., monitor, CD-ROM, hard-drive, or octane failure) may not be evident until after system installation is complete.

All system components must remain upright at all times, and must not be tipped. The gantry should not be hoisted; it should be moved by rolling it on its dollies only. Movement through hallways, doorways, elevators, etc., must be done without tipping or lifting the gantry.

Protection for flooring along the move path (from dock to scan room) is advised.



NOTICE Lifting a gantry requires engineering approval for each occurrence. Your GE PMI should contact CT Engineering for all special lifting requirements.

- Do Not lift the gantry with a forklift.
- Unauthorized gantry lifting will cause gantry bearing damage.

Section 3.0

Site Environmental Considerations

3.1 Dust/Dirt Contamination

LightSpeed 7.X systems (consisting of: console, PDU, table, and gantry) are highly susceptible to airborne contaminants, especially concrete and drywall dust. Due to the possibility of contamination, these systems should NEVER be installed in a construction site. Any site with unfinished floors, walls or ceilings is considered a construction site, and is not suitable for system installation.



NOTICE The act of installing a GE CT scanner in a construction (i.e., unfinished) site will likely result in the following adverse effects:

- Increased installation time
- Decreased installation quality
- Increased scanner downtime, due to increased service calls

3.2 Chemical Contamination

Wet film processors must never be installed in the same room as the scanner, due to the possibility of chemical contamination of LightSpeed 7.X CT scanner components. Such chemicals can contribute to increased equipment failures, increased system downtime, and decreased reliability. Film processor equipment installation must meet the manufacturer's requirements (e.g. ventilation specifications) and all applicable national and local codes. Also, consideration should be given to the location of this equipment and chemical fumes relative to human contact as it relates to locating this equipment and chemicals in the control room.

Section 4.0

Storage Requirements



NOTICE Failure to adhere to storage requirements can result in equipment damage.

4.1 Short-term Storage (Less than Six Months)

If storing the CT system before installation for less than six months, store it in a temperature- and humidity-controlled warehouse. Protect it from weather, dirt, and dust.

Meeting the following requirements prevents rust and corrosion from forming on bearing surfaces due to condensation:

- Storage temperature should not exceed 4° to 27° C (40° to 80° F).
- Maintain relative humidity (non-condensing) between 20% and 60%.
- Maximum rate of relative humidity change measures 5%/hr.
- Maximum rate of temperature change measures 3° C/hr. (5° F/hr.)
- Storage longer than 6 months is not recommended



NOTICE Between delivery qualifies as short-term storage. Van storage must meet the same specifications listed above.

4.2 Construction-Site Storage

When storing the CT system at a construction site be sure to adhere to the following storage requirements:

- Do not damage or puncture the shipping crate.
- Do not remove packaging until all construction is completed at the site and all dust created by the construction is removed.
- Maintain a storage temperature within the range of 10° to 32° C (50° to 90° F).



NOTICE Between delivery is considered short-term storage. Van storage must meet the same specifications as above.

Section 5.0

Extreme Temperature Transportation and Deliveries



NOTICE Failure to adhere to extreme temperature requirements during delivery and storage can result in equipment damage.

Avoid extreme temperatures during system transportation and delivery.

Extreme temperatures consist of temperatures below -18° C (0° F), or above 49° C (120 ° F), without humidity control.

When transporting the CT system, prevent extended exposure of the system to temperatures or humidity outside of the following specifications:

- Temperature: -40° to +70° C (-40° to +158° F)
- Humidity: 5% to 95%



NOTICE Component freezing occurs when exposing the CT system to temperatures below -18° C (0° F) for a period longer than two (2) days. Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature prior to installation.

Section 6.0

Gantry Considerations

The gantry is shipped with most covers installed. The assembly is mounted between two dollies. See [Figure 7-2](#). Two side rails are bolted to the dollies to stabilize the dollies and protect the gantry. Use dolly elevating casters to lift the gantry off its base and roll it into position.



Figure 7-2 Gantry with Shipping Dollies and Side Rails

Door Openings. Clear door openings for moving equipment into building must be 1067 mm X 2083 mm (42 in. X 82 in.) minimum; 2439 mm (8 ft.) corridor width is helpful.

Elevator requirements. To move the gantry from the receiving location to the scanning room, consider elevator capacity and size. By removing side rails and one dolly after the gantry is placed in the elevator, the gantry width/length and elevator depth requirements are reduced. All weights are average due to gantry component weight differences. This change can be ± 18.14 kg (± 40 lb). Contact a representative of the elevator manufacturer if the gantry weight (see [Table 7-1](#)) exceeds elevator capacity.

CONFIGURATION	LENGTH	WIDTH	HEIGHT	WEIGHT
Dollies On, Side Rails On	2900 mm (114 in.)	1290 mm (51 in.)	2000 mm (79 in.)	1932 kg (4260 lb)
Dollies On, Side Rails Removed	2900 mm (114 in.)	1000 mm (39.4 in.)	2000 mm (79 in.)	1910 kg (4210 lb)
Dollies Off, Covers Off	2060 mm (81 in.)	860 mm (34 in.)	1850 mm (73 in.)	1864 kg (4110 lb)

Table 7-1 Size of Gantry & Dollies, with and without Side Rails

Minimum hallway and door size for a gantry with covers and dollies attached, but side rails removed, is 1016 mm (40 in.).

For alternative lifting arrangements and instructions, contact GE Installation Support Services.

Dollies: Typically, dollies are used on the gantry, table, and operator console for domestic shipments. Once the gantry, table and operator console are installed at the site, return the dollies to GE using the shipping document located in Box #1.

Dollies can be purchased for international shipments (B7850LD) to use at the customer site. After the system is removed from the crates, dollies shipped with international shipments **only** are not to be shipped back to GE in Milwaukee, WI, USA, but to be sent to the local GE office or warehouse. Zero clearance dollies are available from UMI. For information on ordering zero clearance dollies, please go to: <http://www.umi-dollyshop.com>

Section 7.0

Table Considerations

The table is shipped without side covers installed. Covers are shipped in four separate boxes. The table is mounted between two dollies.

Table shipping dimensions are 3200 mm (126 in.) long, 650 mm (25.6 in.) wide, and 1200 mm (47.2 in.) high.

	LENGTH		WIDTH		HEIGHT	
	MM	IN	MM	IN	MM	IN
GT2000 w/Dollies	2997	118	762	30	1143	45
GT1700 w/Dollies	2489	98	762	30	1143	45
GT2000 w/o Dollies	2997	118	660	26	889	35
GT1700 w/o Dollies	2489	98	660	26	835	35

Table 7-2 GT Table Dimensions -- with and without dollies

	LENGTH		WIDTH		HEIGHT		WEIGHT	
	MM	IN	MM	IN	MM	IN	KG	LB
GT2000 Dollies	2210	87	762	30	127	5	132	290
GT1700 Dollies	210	87	762	30	127	5	132	290
GT2000 Tilting	762	30	762	30	889	35	136	300

Table 7-3 GT Table Dolly Dimensions

	LENGTH		WIDTH		HEIGHT		WEIGHT	
	MM	IN	MM	IN	MM	IN	KG	LB
GT2000	2997	118	762	30	1143	45	632	1390
GT1700	2489	98	762	30	1143	45	602	1325
GT2000 - Tilting (approx. dimensions)	2489 - 2921	98 - 115	660	26	1778 - 2032	70 - 80	636	1400

Table 7-4 GT Table Elevator Delivery Dimensions

Section 8.0

Tilt Dollies

A very limited number of tilt dollies are available for U.S. deliveries. For information on ordering tilt dollies, please go to: <http://www.umi-dollyshop.com>.

If the system is being stored prior to installation, avoid ordering tilt dollies.

If tilt dollies are not available for delivery, riggers are strongly recommended in their place.

Section 9.0

Console Considerations

The operator console is shipped without the keyboard table installed. The keyboard table is shipped with the operator console.

- The operator console is shipped on a skid equipped with ramps for unloading.
- Do not remove the operator console from the shipping skid until it is in the CT equipment room.
- Console shipping dimensions (on the shipping skid) are 1016 mm (40 in.) deep, 1346 mm (53 in.) wide, and 1041 mm (41 in.) high.

Note: The dimensions of the operator console alone (as shipped) are 888 mm (34.95 in.) deep, 1238 mm (48.74 in.) wide, and 750 mm (29.5 in.) high.



NOTICE Do not lift the operator console by the monitor table top to remove it from the shipping skid.



Figure 7-3 Console ready to be unloaded from shipping skid

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Chapter 8

Power Requirements

Section 1.0

Introduction

The Power Distribution Unit (PDU) supplied with the LightSpeed 7.X system transforms and distributes power to all system components. The PDU is the only power entry point necessary to operate the system.

To minimize voltage regulation effects, power wiring between the facility main distribution panel and the PDU should be kept as short as possible.

When routing the power wiring, all three-phase wires and ground must be run in the same conduit or raceway duct. Power wires should be routed separately from the system control and signal cables using a separate conduit or trough in a raceway duct.

A metallic conduit, floor duct, or surface raceway may be used for running cables, depending upon local codes and practices. However, cable passageways should be large enough to install any cable with all other cables already installed. The use of non-metallic conduit is not recommended.

Section 2.0

System Input Power

2.1 Power Source Configuration

The LightSpeed 7.X CT Scanner is designed to operate on a three-phase, four-wire wye power source. A solidly grounded wye source is preferred. The neutral wire does not need to be run to the system, i.e., four-wire connection. If a neutral wire is run, then it should be terminated in the A1 box. A dedicated feeder from the nearest Main Distribution Panel (MDP) should supply power to the scanner. In accordance with the National Electric Code (U.S.), and similar applicable national and local codes, a protective disconnect device must be provided in the power line supplying the PDU. It must be located within 10 m (32 ft.) of the PDU, visible to PDU service personnel, and must have lockout /tagout provisions. This disconnect device is identified as “A1” in the interconnection schematic diagrams.

2.2 Rating

The LightSpeed 7.X system operates on three-phase power that meets the following specifications.

Voltage	380 to 480 VAC
Capacity	150 kVA
Frequency	50 or 60 Hz $\pm$ 3 Hz

- Maximum power demand = 150 kVA @ 0.85 PF at a selected technique of 140 kV, 715 mA.
- Average (continuous) power demand at maximum duty cycle = 25 kVA.

The A1 disconnect device referenced above must provide overcurrent protection for the system and facilitate multi-point remote “Emergency Off” control of system power. A disconnect utilizing undervoltage release control is preferred over shunt trip devices. The rating of the A1 disconnect device depends on the nominal line voltage at the site. Refer to [Section 3.0: “Recommended Power Distribution System”](#), for minimum rating requirements and suggested disconnect devices.

2.3 Regulation

Total load regulation as measured at the PDU input terminals must not exceed 6%. The capacity of the facility transformer and size & length of feeder wires directly affect the load regulation presented to the system. Refer to [Section 3.0: “Recommended Power Distribution System”](#), for recommended single-unit installation specifics.

2.4 Phase Imbalance

The difference between the highest line-to-line voltage and lowest line-to-line voltage must not exceed 2% of the lowest line-to-line voltage.

2.5 Sags, Surges & Transients

Sags and surges of the power line must not exceed the absolute range limits shown in 8-1. Maximum transient voltages should be limited to 1500 V peak.

2.6 Grounding

Metal conduit, raceway, or the armor of armored cable used to power the system, should be bonded to the PDU cabinet. However, in addition to such mechanical grounding, a dedicated 1/0 (55 mm²) or larger insulated copper ground wire must be run with the phase wires from the main distribution panel to the PDU.

Note: The shield or armor of armored cable is not sufficient for this purpose.

The ground wire should be bonded to intermediate distribution panels through which it passes in accordance with local codes. The resistance between the PDU ground and the facility earth ground must not exceed 0.5 ohm. In addition, the total resistance between the PDU ground and earth must not exceed 2 ohms.

Section 3.0

Recommended Power Distribution System

A dedicated feeder run from the facility main isolation transformer is recommended to power the LightSpeed 7.X CT scanner. If the scanner must be powered from an existing distribution transformer and secondary feeder, such as the equipment distribution panel of an X-ray department, installation with other X-Ray equipment that use rapid film changers should be avoided. These changers use a large number of high powered, closely-spaced exposures, which may coincide with the CT scan and produce image artifacts.

If a dedicated distribution transformer is provided for the scanner, the minimum recommended transformer size is 225 kVA, rated 2.4% regulation at unity power factor. For this configuration, the minimum recommended feeder size and overcurrent protection device based on line voltage is shown in 8-2 Minimum Feeder Wire Size.

In all cases, qualified personnel must verify that the transformer and feeder, at point of take-off, plus the run to the LightSpeed 7.X CT scanner, meet all the requirements stated in this document.

SYSTEM CHARACTERISTICS:

- Maximum power demand = 150kVA @ 0.85 PF: at a Selected Technique of 140 kV, 715 mA.
- Continuous (average) power demand at maximum duty cycle = 25kVA.
- Maximum allowable total source regulation is 6%.
- Minimum recommended transformer size: 225 kVA, with 2.4% rated regulation at unity power factor. Resultant maximum allowable feeder regulation is 3.4%.

The nominal line voltage must fall within one of the ranges listed below

Nominal Line Voltage	380	400	420	440	460	480
Hi-Line Limit, +10%	418	440	462	484	506	528
Lo-Line Limit, -10%	342	360	378	396	414	432
Continuous Line Current	38	36	34	33	31	30
Momentary Line Current	228	217	206	197	188	180
Maximum Line Current	253	241	229	219	209	200
Minimum Recommended Circuit Protection Rating	150	150	150	125	125	125

Table 8-1 Nominal Line Voltage

FEEDER LENGTH (MDA TO A1)	MINIMUM FEEDER WIRE SIZE, AWG OR MCM (SQ. MM)/ VAC					
	380 VAC	400 VAC	420 VAC	440 VAC	460 VAC	480 VAC
15 m (50 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
30 m (100 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
46 m (150 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
61 m (200 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)
76 m (250 ft)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)	1 (45)	1 (45)
91 m (300 ft)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)	1/0 (55)
107 m (350 ft)	4/0 (100)	3/0 (85)	3/0 (85)	2/0 (70)	2/0 (70)	1/0 (55)
122 m (400 ft)	250 (125)	4/0 (100)	3/0 (85)	3/0 (85)	3/0 (85)	2/0 (70)

Table 8-2 Minimum Feeder Wire Size

SUB-FEEDER LENGTH (A1 TO PM)	MINIMUM SUB-FEEDER WIRE, AWG OR MCM (SQ. MM)					
	380 VAC	400 VAC	420 VAC	440VAC	460VAC	480VAC
9.7536 m (32 ft)	1/0 (55)	1/0 (55)	1/0 (55)	1 (45)	1 (45)	1 (45)

Table 8-3 Minimum Sub-Feeder Wire Size

- 1.) 8-1, 8-2, and 8-3 (above) are based on the use of copper wire, rated 75 C and run in steel conduit. Ampacity is determined in accordance with the National Electrical Code (NFPA 70), Table 310-16 (2002).
- 2.) The minimum feeder size is determined by the ampacity of the circuit protection device listed above, except where a larger size is necessary to meet total source regulation limits.
- 3.) A 1/0 (55 sq. mm) ground wire is recommended in all cases.

Section 4.0 Ground System

The LightSpeed 7.X CT scanner has been designed to use an equal potential grounding system. The strongly recommended ground system is shown in [Figure 7-1](#). There are three primary grounding points:

- A system power ground point located in the PDU.
- A reference ground point located between gantry and table base.
- A patient ground point located at the front of the table base.

All exposed metal surfaces in the patient vicinity are grounded to the reference ground point.

For additional information, refer to Electrical Safety Equipment, Direction 46-014505.

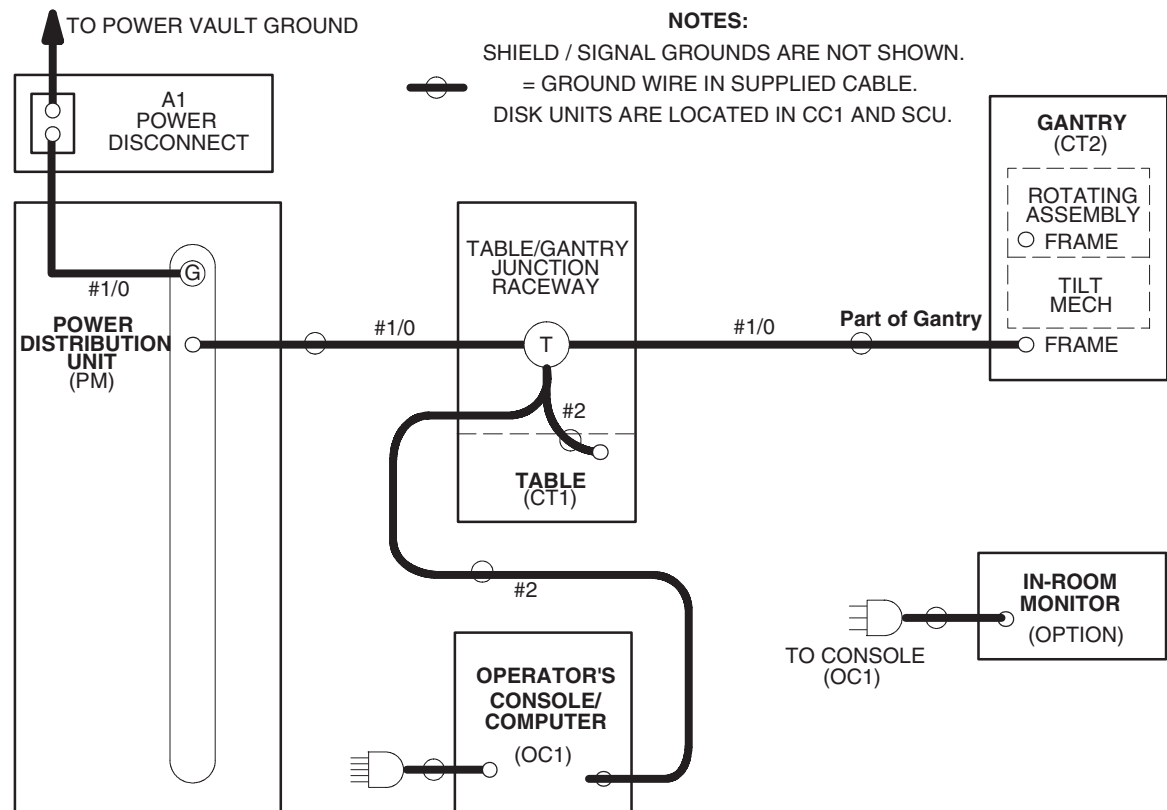


Figure 8-1 System Ground Map

Chapter 9

Interconnection Data

Section 1.0 Introduction

Figure 9-1 shows interconnection runs for a 50/60 Hz system.

Table 9-1 shows component designators for supplied equipment and options and wall power outlets.

Table 9-4 lists customer-installed wiring and supplied cables. The actual length of each run is less than the length of supplied cables to allow for routing inside the equipment. Cable diameters and sizes of connectors are provided to aid in sizing conduit and access plates.

Table 9-2 and Table 9-3 list details for connection to LightSpeed series equipment using standard (short) length and non-standard (long) length cables, respectively. Details are listed for the following types of runs as appropriate:

- Flush-floor duct
- Computer floor
- Through-wall bushing
- Junction box
- Surface floor duct
- Through-floor duct
- Wall duct
- Conduit

The need for additional junction boxes is minimized by using either a cable raceway system or a raised computer floor. LightSpeed series systems use prefabricated cables with large plugs. Therefore, conduit or pipe is not recommended for cable runs.

Section 2.0 Component Designators

DESIGNATOR	APPLIES TO	SOURCE
A1	Primary power disconnect	Contractor supplied
CT1	Patient table	System
CT2	Gantry	System
ITL	InSite telephone lines	Contractor supplied
OC1	Operator console/computer	System
PDU	Power Distribution Unit	System
SEO	System emergency off	Contractor supplied
SM	Slave monitor	Option
WL	"X-ray on" warning light	Contractor supplied
DS	Door Interlock Switch	Contractor supplied
BBNC	Broad-band network connection	Contractor supplied

Table 9-1 Component Designators

Section 3.0 Interconnect Runs, Wiring and Cables

3.1 GE Healthcare Supplied (Standard Length)

RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	28 (20)	9 (6)	2343529-2	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 0(.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	28 (20)	9 (6)	2343530-2	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	11 (0.44) Dia
052	28 (20)	9 (6)	2343528-2	LVAC, PDU to Gantry	2587	FT4	600	120/208Y	90	14 0(.542)	5	8	56 (2.22) Dia
053	65 (60)	20 (18)	5121809-2	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (0.483)	4	10	56 (2.22) Dia
054			n/a	LVAC, Gantry to Table	1015		600	120VAC			3	14	
055	43 (35)	13 (11)	2371450-2	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 0(.608)	1	1/0	16 (0.62) Dia
056	71 (60)	22 (18.3)	2371450-4	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (0.467)	1	2	12 (0.48) Dia
100	32.5 (20)	10 (6)	5120646-2	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	71 (60)	22 (18)	5120645-2	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	68 (60)	21 (18)	2373436-3	Signal (LAN), Gantry to OC			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
103	68 (60)	21 (18)		Fiber Optic, Gantry to OC			N/A	N/A			1	N/A	10 (0.39) Dia
104			n/a	Signal, Gantry to Table		FT-4	300		80		25	22	

Table 9-2 GE Healthcare Supplied Cables (Standard Run) - UL Information

3.2 GE Healthcare Supplied (Optional, Long Run)

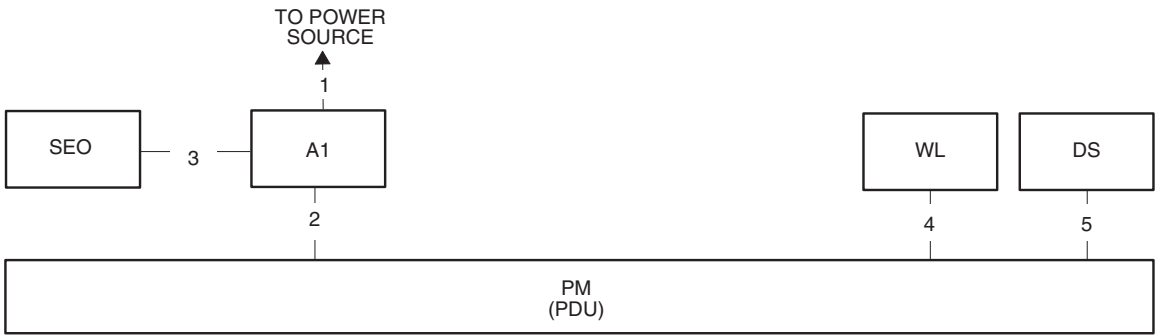
RUN #	LENGTH, ACTUAL (USABLE)		PART #	DESCRIPTION	UL CABLE INFORMATION								PULL SIZE MM (INCHES)
	ft	m			UL Style	Flam. Rating	Voltage Rating	Voltage Actual	Temp. Rating (C)	Dia. mm (inch)	# of Cond	Size AWG	
050	63 (55)	19 (17)	2343529	HVDC, PDU to Gantry	2587	FT4	600	+ & - 350VDC	90	19 (0.751)	3	(2) 4 (1) 8	22 (0.87) Dia
051	62.5 (55)	19 (17)	2343530	HVAC, PDU to Gantry	2587	FT4	600	440Y/254	90	15 (0.604)	4	14	11 (0.44) Dia
052	60 (55)	19 (17)	2343528	LVAC, PDU to Gantry	2587	FT4	600	208Y/120	90	14 (0.542)	5	8	56 (2.22) Dia
053	80 (75)	25 (23)	5121809	LVAC, PDU to Console	2587	FT4	600	120VAC	90	12 (0.483)	4	10	56 (2.22) Dia
055	63 (55)	19 (17)	2371450	Ground, PDU to Raceway	1284	VW-1 (FT-1)	600	0	105	16 (0.608)	1	1/0	16 (0.62) Dia
056	83 (75)	26 (23)	2371450-3	Ground, Raceway to Console	1283	VW-1 (FT-1)	600	0	105	12 (0.467)	1	2	12 (0.48) Dia
100	63 (55)	19 (17)	5120646	Signal, Gantry MSUB to PDU		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
101	83 (75)	26 (23)	5120645	Signal, Gantry MSUB to OC		FT-4	300	<30VDC	80	11 (0.440)	25	22	17 x 58 (0.68 x 2.30) 19 x 51 (0.75 x 2.01)
102	80 (75)	24 (23)	2373436-2	Signal (LAN), Gantry to OC			1900	<30VDC		6 (0.234)	8	24	15 (0.59) Dia
103	80 (75)	24 (23)		Fiber Optic, Gantry to OC			N/A	N/A			2	N/A	10 (0.39) Dia

Table 9-3 GE Healthcare Supplied Cables (Optional, Long Run) - UL Information

3.3 Contractor (Customer) Supplied

CUSTOMER INSTALLED WIRING		DESCRIPTION	CABLES SUPPLIED			PLUG PULLING DIMENSIONS		WIRE & CABLE PIGTAILS FT. (M.)	
QTY	SIZE AWG (MM ²)		PART NO	LENGTH FT. (M.)	DIA. IN (MM)	FROM	TO	FROM	TO
RUN NO. 1 FROM PRIMARY POWER SOURCE TO FACILITY DISCONNECT (POWER SOURCE - A1)									
Maximum Run Length *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
RUN NO. 2 FROM FACILITY DISCONNECT TO POWER MODULE (A1 - PM) MAXIMUM RUN LENGTH *									
3	*	POWER						3 (1)	3(1)
1	1/0 (50)	GROUND						3 (1)	3 (1)
-	-	NEUTRAL -- Not Required						3 (1)	3 (1)
RUN NO. 3 FROM FACILITY DISCONNECT TO SYSTEM EMERGENCY OFF (A1 - SEO)									
2	14 (2)	POWER						6 (2)	6 (2)
1	14 (2)	GROUND						6 (2)	6 (2)
RUN NO. 4 POWER MODULE TO WARNING LIGHT CONTROL (PM - WL)									
2	14 (2)	WARNING LIGHT 24 VOLT CONTROL TS6 1, 2, 3, 4, 5, 6, 7, 8							
RUN NO. 5 POWER MODULE TO SCAN ROOM DOOR INTERLOCK (PM - DOOR SWITCH)									
2	14 (2)	SCAN ROOM DOOR INTER LOCK TS6 9, 10							
*	REFER TO Table 8-2 on page 109 FOR AWG (MM2) WIRE SIZES.								
RUN NO. n/a BBNC									
1	customer determined	Hospital Broad-Band Network Connection (Wall Jack)							

Table 9-4 Runs 1, 2, 3, 4 and 5 Connections



NOTES:

- 1) Used for remote diagnostics - Option
- 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
- 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m
- 4) In order to avoid any violation of each National Regulation (NEC in USA, CCC in China, etc.), use of the compliant cable/wire is recommended. For China market, China end-user shall purchase the power supply cable that has the CCC mark.

Only one phone connection is required for the system.

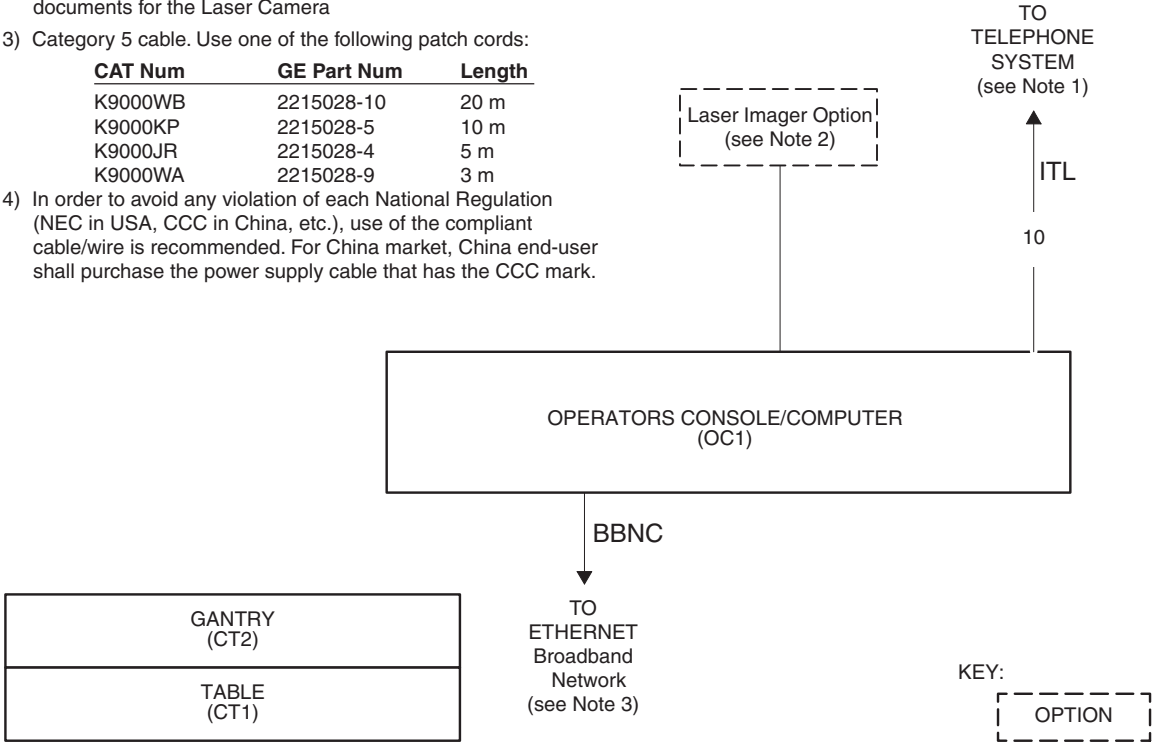


Figure 9-1 Interconnection Runs

Section 4.0

Contractor Supplied Components

REFERENCE	ASSOCIATED EQUIPMENT	MATERIAL/LABOR SUPPLIED BY CUSTOMER CONTRACTOR	USA VENDOR / CAT NO. GE CATALOG
A1 380 - 480V 50/60 Hz	Circuit Breaker with Magnetic Contactor	3 Pole, 380V - 480V, Combination breaker with magnetic contactor. Includes control transformer, optional UPS interface, On/Off controls and auto-restart feature	Recommend*: - E4502AE (125A) - E4502AF (150A) Optional remote operator control available from GE Supply, Cat # GESCTR0CS1
BBNC (strongly recommended)	Broad-Band Network Connection	Broad-Band network connection wall jack, located within 1m (39inches) of Operator Console location, for internal hospital networking and InSite Broad-Band connectivity. Cabling to conform to facility's IT standards.	
ITL (optional)	In-suite Telephone Lines	Supply 2 voice-grade telephone lines. One line must be a direct number from outside the facility – do not route this line through a telephone switchboard. Telephone line operating charges are paid by customer.	
	System Components	Reference the system installation drawings supplied by Installation Support Services within your geographic area. Room Warning Light Controller	E4500AM

*Refer to Table 9-6 on page 117.

Table 9-5 Contractor-Supplied Components

Section 5.0

UPS Interconnect



Figure 9-2 Typical UPS

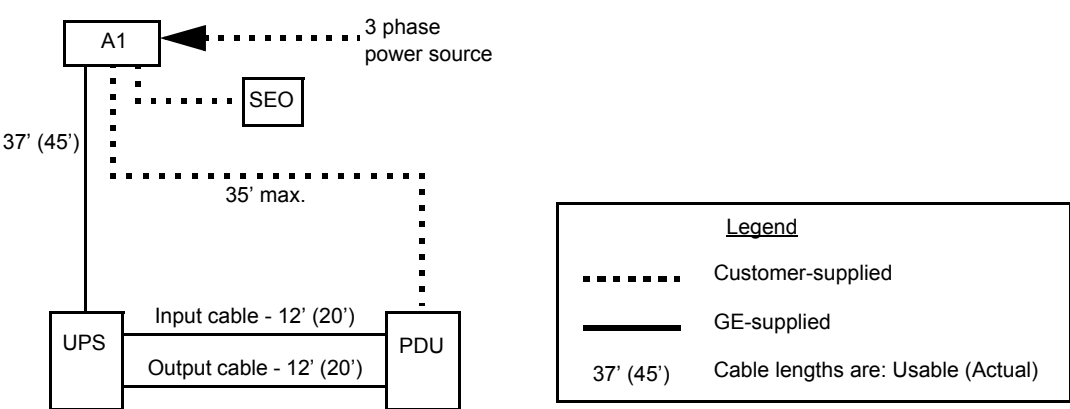


Figure 9-3 Typical UPS Interconnect

The UPS Kit catalog number is B7864PP. See table below, for details.



NOTICE **UPS Kit B7864PP REQUIRES** installation of one of the A1 panels listed below.

PDU Type & Model #	Max. Mom. kVA Rating	Strongly Recommended Main Disconnect (A1) Cat #		Optional Partial UPS Kit Cat #
		Europe & Asia (380-400V or 420V)	North America (440V or 460-480V)	
NGPDU-4 2326492-4	150kVA	E4502AF (150A) (incl. Auto Restart & Integrated UPS Control) (Additional A1 Disconnects are available for Europe through the European Sales Team.)	E4502AE (125A) (incl. Auto Restart & Integrated UPS Control)	B7864PP PowerWare 9330 (15kVa - 40A) REQUIRES one of the A1 Panels shown at left, or equal.

Table 9-6 LightSpeed 7.X Partial UPS Back-up Options

Note: Conduit is strongly recommended between the:

- A1 and UPS
- UPS and PDU
- PDU and A1

Section 6.0

Typical Customer Supplied Wiring - U.S.

6.1 Primary Power Disconnect



Figure 9-4 Primary Power Disconnect (A1)

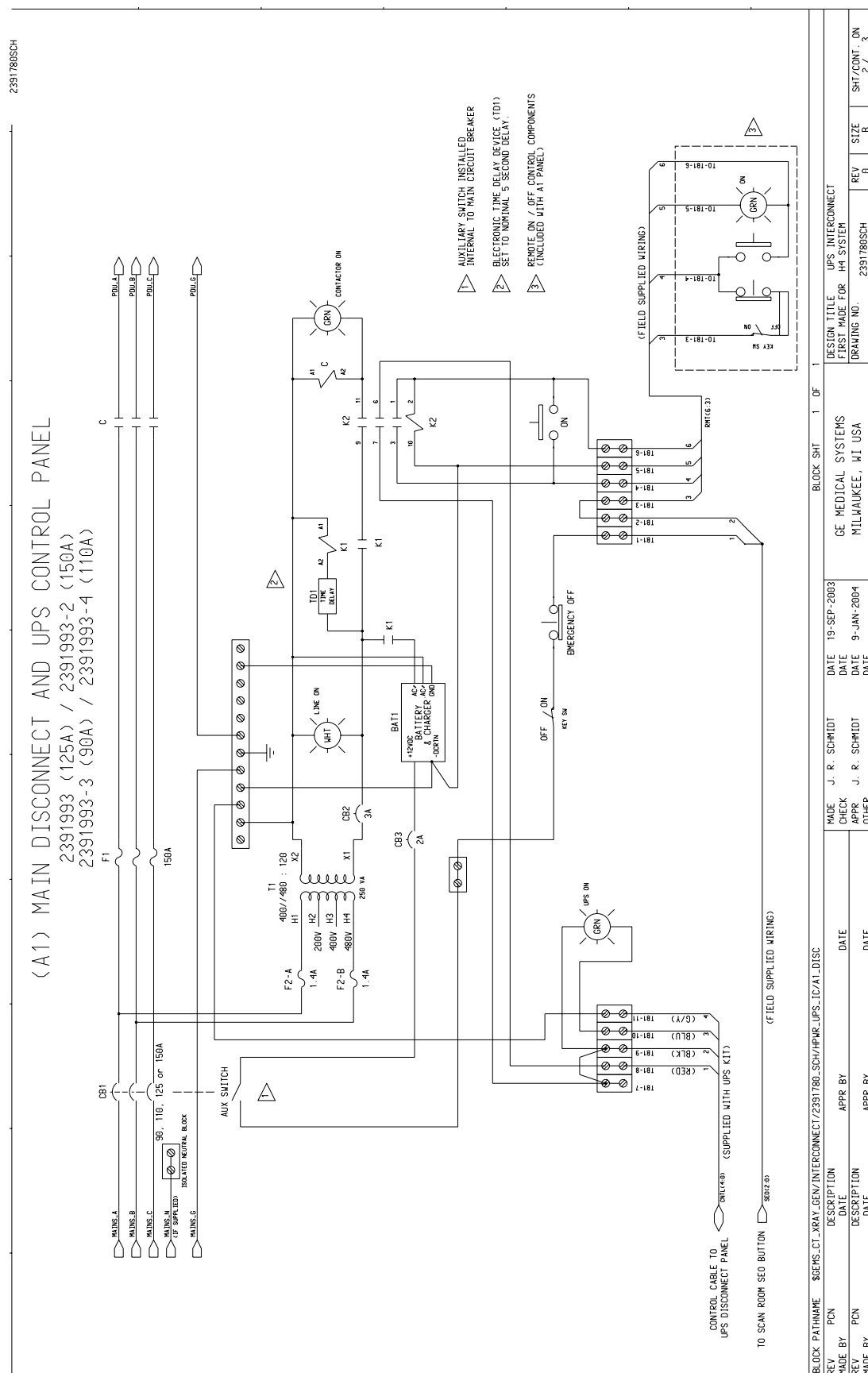


Figure 9-5 Primary Power Disconnect (A1) – Fusible Disconnect and Magnetic Contactor

6.2 Scan Room Warning Light & Door Interlock

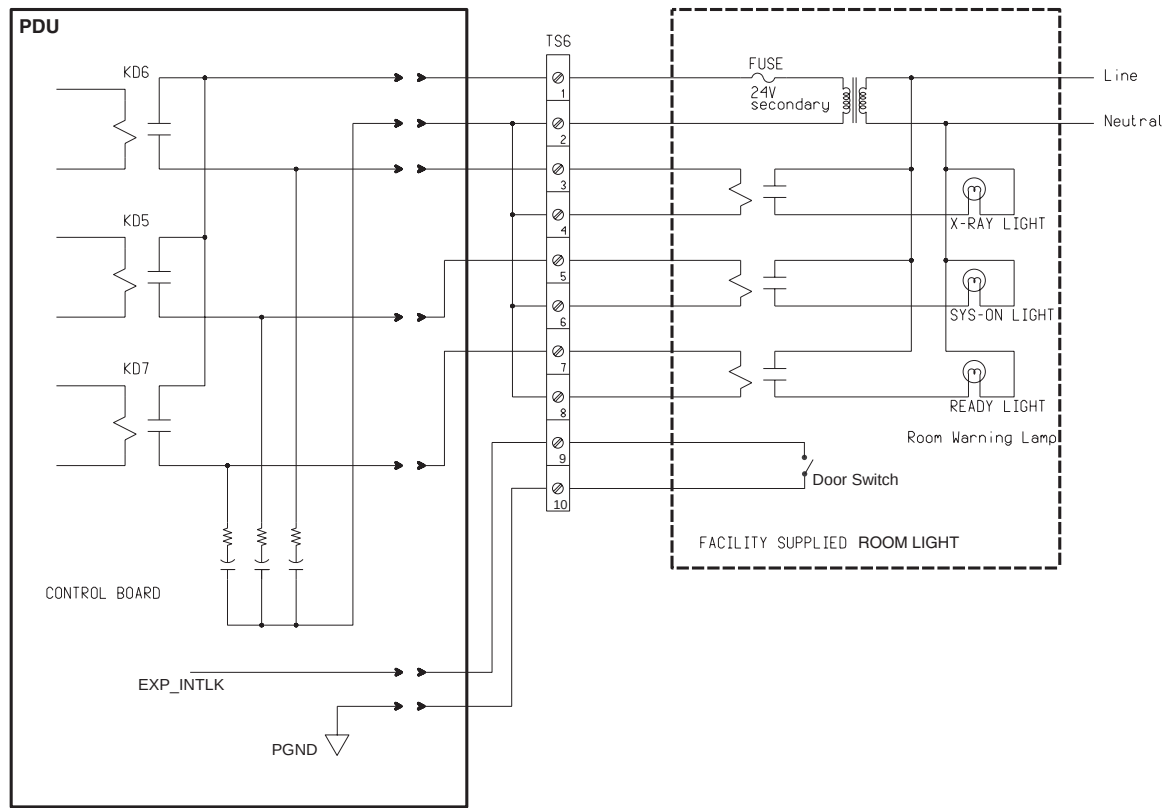


Figure 9-6 TS6 X-Ray Warning Light Connections

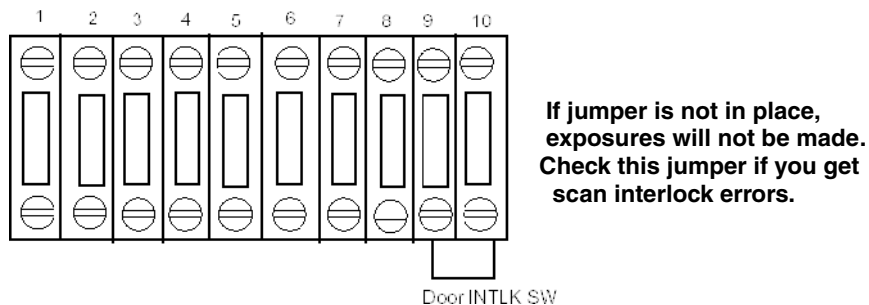


Figure 9-7 TS6 Room Door Interlock Connections - Without a Door Interlock

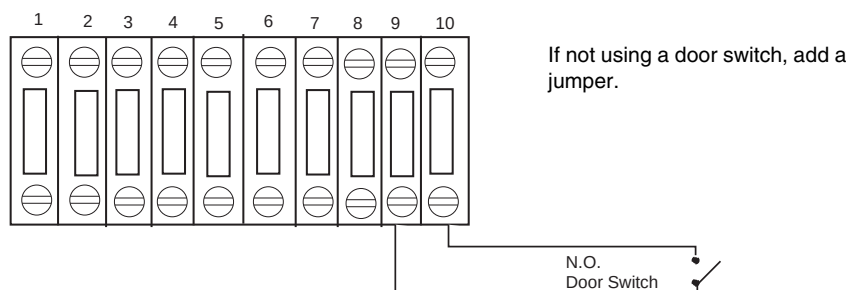


Figure 9-8 TS6 Room Door Interlock Connections - With a Door Interlock

Appendix

VCT & FCT Installation Site Ready Form

This appendix contains the FCT Installation Site Ready Form, which is required for the Pre-Installation process.

Complete and submit the electronic (MS Word) version of this form, which can be found on the LightSpeed 7.X Pre-Installation CD (included with the Pre-Installation Kit, p/n 5136343).

FCT Products Installation Site Ready Inspection Form

Required Site Information

Customer Name Install Date Site Visit Date
Order Number System ID: GEE Technician

IS Present: Select Customer Present: Select GE final site prints on site Select

This site ready visit must be completed at least 3 business days prior to the installation date

These items must be on site.

SRV Items

- 1) Actual room size must match the site print, including cabinets, sinks and accessories
- 2) Do Not continue this inspection without an approved final site print.

a) GE site prints Project/Rev # Product type Select

Check if present

- b) Quick start kit or PIM on site ☐ **required for this task**
c) Pre-install video reviewed ☐ **required for this task**
d) Floor template ☐ Optional order if required

Customer Checks

Customer Awareness Items (review only)

Radiation Shielding Was shielding reviewed by a health physics for the product to be installed ☐
Structural Floor Vibration Checked ☐ Site EMI Reviewed ☐
Operational Clearances Will a bed fit in room ☐ Work space around table gantry reviewed ☐

Site Delivery Information

Delivery route inspection Doors and hallways check will Select tilt, or zero clearance dollies needed

Select One Select One Riggers needed Select Elevator size & load limits checked ☐

Site Information

System ID created Select One Network ID's address sheet completed Select One
Camera setup information available Select One Equipment Order reviewed Select One
Customer Contact Phone # Identify customer to accept system
De-Installation Required /planned Date:

Requirements

Site Requirements Escalations required if specification are not met.

Network Communication Broad Band installed & Active Select One see page 2

Scan Room Requirements Met [90" min. ceiling ☐

Room size [w/GT1700 table min. 13'-6" x 19'-21"] ☐ [w/GT2000 table min. 13'-6" x 22'-0"] ☐

Structural Requirements Met

4" Floor thickness ☐ Meets floor level specification (+/- 1/8") ☐ Anchoring Select one

HVAC Requirements Met

Can maintain (70 degree F) minimum ☐ [Temp = 64° to 79°F, max. 5°F change per hr]

Can maintain minimum Humidity spec ☐ 30% minimum

Electrical Requirements Met 480 VAC 125 Amps ☐ 380 VAC 150 Amps ☐ LOTO Compatible Disconnect ☐

Power & Quality Audit Done Preliminary inspection must be completed Select One

All Regulatory Clearances Confirmed Select One **Note:** Must meet ALL **Regulatory requirements to pass**

Site condition:

Finished Ceiling ☐, Floor & Cabinets Installed ☐ Walls Painted ☐ ALL Construction in the area Completed? ☐

Note: The Area must be in move in condition. IF **NO** Reschedule

FCT Products Installation Site Ready Inspection Form

Site ready Report

GE Energy Site Ready Report:

Site Issues: (Pro's & Con's) List all observed site issues that would present installation problems

GE PMI - Delivery Details:

Check if a short term storage location is required. Will floor protection be required for the delivery

Broadband:

Has the BB Specialist completed the BB installation GE setup and setup? Notify the zone BB specialist if there are questions. Broadband is expected to be complete at install.

Inspection Results

The site ready visit will result in a report to the project manager indicating one of the following:

☐ **Pass** - All required items are present, completed and the site is ready for installation. Must meet all regulatory requirements, have a final site print that accurately matches the room to **pass**.

☐ **Conditional Pass** - is issued when 80% of all of the required tasks are completed and all parties agree that the 20% will be completed by the installation delivery date.

If a "Conditional Pass" is granted on the inspection date, the project manager must present conclusive evidence that unfinished tasks are completed and that the site will be ready one business day prior to delivery. Must meet all regulatory requirements, have a final site print that accurately matches the room.

☐ **Fail** - is issued when less than 80% of the tasks are completed, area is under construction and all parties cannot agree that the remaining work will be completed by the requested installation delivery date. Failed sites shall be rescheduled when all of the above items are met and the room is completed.

Re-inspection Date Are pictures attached?

Inspection Attendance

Zone LCT

Comments:

All US installations - E-mail to: brian.rendina@ge.com

All US site escalation should be sent to:

brian.rendina@ge.com; general site ready issues

bradley.bassler@med.ge.com or SRV, site prints, construction and delivery items

george.farrington@med.ge.com or charles.brukl@med.ge.com regulatory and clearances issues

Note: Regulatory requirements can not be changes or deviated.

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